Calculus in Context

The Five College Calculus Project

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Note on the PreTeXt Edition

This edition of *Calculus in Context* has been converted from its original LaTeX source into PreTeXt. The conversion was carried out by Aaron Weinberg as part of an effort to make the text more accessible, adaptable, and compatible with modern publishing formats.

While every effort has been made to preserve the content and intent of the original, the conversion process is necessarily imperfect. Some errors may have been introduced in translating mathematical notation, figures, and formatting. In addition, aspects of the original LaTeX source that relied on stylistic or presentational conventions have required reinterpretation into the semantic structures required by PreTeXt.

This edition should therefore be regarded as a work in progress. Readers are encouraged to report errors, inconsistencies, or unclear passages so that future revisions can improve the quality and fidelity of the text.

The long-term goal of this project is not only to reproduce the original text, but also to refine its structure, improve clarity, and take advantage of the capabilities of PreTeXt for interactive and accessible mathematical exposition.

Advisory Committee

Preface to the 2008 Edition

We are publishing this edition of *Calculus in Context* online to make it freely available to all users. It is essentially unchanged from the 1994 edition.

The continuing support of Five Colleges, Inc., and especially of the Five College Coordinator, Lorna Peterson, has been crucial in paving the way for this new edition. We also wish to thank the many colleagues who have shared with us their experiences in using the book over the last twenty years—and have provided us with corrections to the text.

Preface to the 1994 Edition

Our point of view. We believe that calculus can be for our students what it was for Euler and the Bernoullis: A language and a tool for exploring the whole fabric of science. We also believe that much of the mathematical depth and vitality of calculus lies in these connections to the other sciences. The mathematical questions that arise are compelling in part because the answers matter to other disciplines as well.

The calculus curriculum that this book represents started with a "clean slate;" we made no presumptive commitment to any aspect of the traditional course. In developing the curriculum, we found it helpful to spell out our **starting points**, our **curricular goals**, our **functional goals**, and our view of the **impact of technology**. Our starting points are a summary of what calculus is really about. Our curricular goals are what we aim to convey about the subject in the course. Our functional goals describe the attitudes and behaviors we hope our students will adopt in using calculus to approach scientific and mathematical questions. We emphasize that what is missing from these lists is as significant as what appears. In particular, we did *not* not begin by asking what parts of the traditional course to include or discard.

Starting Points.

- Calculus is fundamentally a way of dealing with functional relationships that occur in scientific and mathematical contexts. The techniques of calculus must be subordinate to an overall view of the underlying questions.

- Technology radically enlarges the range of questions we can explore and the ways we can answer them. Computers and graphing calculators are much more than tools for teaching the traditional calculus.

- The concept of a dynamical system is central to science Therefore, differential equations belong at the center of calculus, and technology makes this possible *at the introductory level.*

- The process of successive approximation is a key tool of calculus, even when the outcome of the process—the limit—cannot be explicitly given in closed form.

Curricular Goals.

- Develop calculus in the context of scientific and mathematical questions.

- Treat systems of differential equations as fundamental objects of study.

- Construct and analyze mathematical models.
- Use the method of successive approximations to define and solve problems.
- Develop geometric visualization with hand-drawn and computer graphics.
- Give numerical methods a more central role.

Functional Goals.

- Encourage collaborative work.
- Empower students to use calculus as a language and a tool.
- Make students comfortable tackling large, messy, ill-defined problems.
- Foster an experimental attitude towards mathematics.
- Help students appreciate the value of approximate solutions.
- Develop the sense that understanding concepts arises out of working on problems, not simply from reading the text and imitating its techniques.

Impact of Technology.

- Differential equations can now be solved numerically, so they can take their rightful place in the introductory calculus course.
- The ability to handle data and perform many computations allows us to explore examples containing more of the messiness of real problems.
- As a consequence, we can now deal with credible models, and the role of modelling becomes much more central to our subject.
- In particular, introductory calculus (and linear algebra) now have something more substantial to offer to life and social scientists, as well as to physical scientists, engineers and mathematicians.
- The distinction between pure and applied mathematics becomes even less clear (or useful) than it may have been.

By studying the text you can see, quite explicitly, how we have pursued the curricular goals. In particular, every one of those goals is addressed within the very first chapter. It begins with questions about describing and analyzing the spread of a contagious disease. A model is built, and the model is a system of coupled non-linear differential equations. We then begin a numerical assault on those equations, and the door is opened to a solution by successive approximations.

Our implementation of the functional goals is less obvious, but it is still evident. For instance, the text has many more words than the traditional calculus book—it is a book to be read. Also, the exercises make unusual demands on students. Most exercises are not just variants of examples that have been worked in the text. In fact, the text has rather few simple "template" examples.

Shifts in Emphasis. It will also become apparent to you that the text reflects substantial shifts in emphasis in comparison to the traditional course. Here are some of the most striking:

Table 0.0.1

How the emphasis shifts:

increase	decrease
concepts	techniques
geometry	algebra
graphs	formulas
brute force	elegance
numerical solutions	closed-form solutions

Euler's method is a good example of what we mean by "brute force." It is a general method of wide applicability. Of course when we use it to solve a differential equation like $y'(t) = t$, we are using a sledgehammer to crack a peanut. But at least the sledgehammer *does* work. Moreover, it works with coconuts (like $y' = y(1-y/10)$), and it will just as happily knock down a house (like $y' = \cos^2(t)$). Of course, students also see the elegant special methods that can be invoked to solve $y' = t$ and $y' = y(1-y/10)$ (separation of variables and partial fractions are discussed in chapter 11), but they understand that they are fortunate indeed when a real problem will succumb to these special methods.

Audience. Our curriculum is not aimed at a special clientele. On the contrary, we think that calculus is one of the great bonds that unifies science, and all students should have an opportunity to see how the language and tools of calculus help forge that bond. We emphasize, though, that this is not a "service" course or calculus "with applications," but rather a course rich in mathematical ideas that will serve all students well, including mathematics majors. The student population in the first semester course is especially diverse. In fact, since many students take only one semester, we have aimed to make the first six chapters stand alone as a reasonably complete course. In particular, we have tried to present contexts that would be more or less broadly accessible. The emphasis on the physical sciences is clearly greater in the later chapters; this is deliberate. By the second semester, our students have gained skill and insight that allows them to tackle this added complexity.

Handbook for Instructors. Working toward our curricular and functional goals has stretched us as well as our students. Teaching in this style is substantially different from the calculus courses most of us have learned from and taught in the past. Therefore we have prepared a handbook based on our experiences and those of colleagues at other schools. We urge prospective instructors to consult it.

Origins. The Five College Calculus Project has a singular history. It begins almost thirty years ago, when the Five Colleges were only Four: Amherst, Mount Holyoke, Smith, and the large Amherst campus of the University of Massachusetts. These four resolved to create a new institution which would be a site for educational innovation at the undergraduate level; by 1970, Hampshire College was enrolling students and enlisting faculty.

Early in their academic careers, Hampshire students grapple with primary sources in all fields—in economics and ecology, as well as in history and literature. And journal articles don't shelter their readers from home truths: if a mathematical argument is needed, it is used. In this way, students in the life and social sciences found, sometimes to their surprise and dismay, that they needed to know calculus if they were to master their chosen fields. However, the calculus they needed was not, by and large, the calculus that was actually being taught. The journal articles dealt directly with the relation between quantities and their rates of change—in other words, with differential equations.

Confronted with a clear need, those students asked for help. By the mid-1970s, Michael Sutherland and Kenneth Hoffman were teaching a course for those students. The core of the course was calculus, but calculus as it is *used* in contemporary science. Mathematical ideas and techniques grew out of scientific questions. Given a process, students had to recast it as a model; most often, the model was a set of differential equations. To solve the differential equations, they used numerical methods implemented on a computer.

The course evolved and prospered quietly at Hampshire. More than a decade passed before several of us at the other four institutions paid some attention to it. We liked its fundamental premise, that differential equations belong at the center of calculus. What astounded us, though, was the revelation that differential equations could really *be* at the center—thanks to the use of computers.

This book is the result of our efforts to translate the Hampshire course for a wider audience. The typical student in calculus has not been driven to study calculus in order to come to grips with his or her own scientific questions—as those pioneering students had. If calculus is to emerge organically in the minds of the larger student population, a way must be found to involve that population in a spectrum of scientific and mathematical questions. Hence, calculus *in context*. Moreover, those contexts must be understandable to students with no special scientific training, and the mathematical issues they raise must lead to the central ideas of the calculus—to differential equations, in fact.

Coincidentally, the country turned its attention to the undergraduate science curriculum, and it focused on the calculus course. The National Science Foundation created a program to support calculus curriculum development. To carry out our plans we requested funds for a five-year project; we were fortunate to receive the only multi-year curriculum development grant awarded in the first year of the NSF program. This text is the outcome of our effort.

To the Students

In a typical high school math text, each section has a "technique" which you practice in a series of exercises very like the examples in the text. This book is different. In this course you will be learning to use calculus both as a tool and as a language in which you can think coherently about the problems you will be studying. As with any other language, a certain amount of time will need to be spent learning and practicing the formal rules. For instance, the conjugation of *être* must be almost second nature to you if you are to be able to read a novel—or even a newspaper—in French. In calculus, too, there are a number of manipulations which must become automatic so that you can focus clearly on the content of what is being said. It is important to realize, however, that becoming good at these manipulations is not the goal of learning calculus any more than becoming good at declensions and conjugations is the goal of learning French.

Up to now, most of the problems you have met in math classes have had definite answers such as "17," or "the circle with radius 1.75 and center at (2,3)." Such definite answers are satisfying (and even comforting). However, many interesting and important questions, like "How far is it to the planet Pluto," or "How many people are there with sickle-cell anemia," or "What are the solutions to the equation $x^5 + x + 1 = 0$" can't be answered exactly. Instead, we have ways to **approximate** the answers, and the more time and/or money we are willing to expend, the better our approximations may be. While many calculus problems do have exact answers, such problems often tend to be special or atypical in some way. Therefore, while you will be learning how to deal with these "nice" problems, you will also be developing ways of making good approximations to the solutions of the less well-behaved (and more common!) problems.

The computer or the graphing calculator is a tool that that you will need for this course, along with a clear head and a willing hand. We don't assume that you know anything about this technology ahead of time. Everything necessary is covered completely as we go along.

You can't learn mathematics simply by reading or watching others. The only way you can internalize the material is to work on problems yourself. It is by grappling with the problems that you will come to see what it is you do understand, and to see where your understanding is incomplete or fuzzy.

One of the most important intellectual skills you can develop is that of exploring questions on your own. Don't simply shut your mind down when you come to the end of an assigned problem. These problems have been designed not so much to capture the essence of calculus as to prod your thinking, to get you wondering about the concepts being explored. See if you can think up and answer variations on the problem. Does the problem suggest other questions? The ability to ask good questions of your own is at least as important as being able to answer questions posed by others.

We encourage you to work with others on the exercises. Two or three of you of roughly equal ability working on a problem will often accomplish much more than would any of you working alone. You will stimulate one another's imaginations, combine differing insights into a greater whole, and keep up each other's spirits in the frustrating times. This is particularly effective if you first spend time individually working on the material. Many students find it helpful to schedule a regular time to get together to work on problems.

Above all, take time to pause and admire the beauty and power of what you are learning. Aside from its utility, calculus is one of the most elegant and richly structured creations of the human mind and deserves to be profoundly admired on those grounds alone. Enjoy!

Acknowledgements

Certainly this book would have been possible without the support of the National Science Foundation and of Five Colleges, Inc. We particularly want to thank Louise Raphael who, as the first director of the calculus program at the National Science Foundation, had faith in us and recognized the value of what had already been accomplished at Hampshire College when we began our work. Five College Coordinators Conn Nugent and Lorna Peterson supported and encouraged our efforts, and Five College treasurer and business manager Jean Stabell has assisted us in countless ways throughout the project.

We are very grateful to the members of our Advisory Board: to Peter Lax, for his faith in us and his early help in organizing and chairing the Board; to Solomon Garfunkel, for his advice on politics and publishing; to Barry Simon, for using our text and giving us his thoughtful and imaginative suggestions for improving it; to Gilbert Strang, for his support of a radical venture; to John Truxal, for his detailed commentaries and insights into the world of engineering.

Among our colleagues, James Henle of Smith College deserves special thanks. Besides his many contributions to our discussions of curriculum and pedagogy, he developed the computer programs that have been so valuable for our teaching: Graph, Slinky, Superslinky, and Tint. Jeff Gelbard and Fred Henle ably extended Jim's programs to the MacIntosh and to DOS Windows and X Windows. All of this software is available on anonymous ftp at emmy.smith.edu. Mark Peterson, Robert Weaver, and David Cox also developed software that has been used by our students.

Several of our colleagues made substantial contributions to our frequent editorial conferences and helped with the writing of early drafts. We offer thanks to David Cohen, Robert Currier and James Henle at Smith; David Kelly at Hampshire; and Frank Wattenberg at the University of Massachusetts. Mary Beck, who is now at the University of Virginia, gave heaps of encouragement and good advice as a co-teacher of the earliest version of the course at Smith. Anne Kaufmann, an Ada Comstock Scholar at Smith, assisted us with extensive editorial reviews from the student perspective.

Two of the most significant new contributions to this edition are the appendix for graphing calculators and a complete set of solutions to all the exercises. From the time he first became aware of our project, Benjamin Levy has been telling us how easy and natural it would be to adapt our Basic programs for graphing calculators. He has always used them when he taught *Calculus in Context*, and he created the appendix which contains translations of our programs for most of the graphing calculators in common use today. Lisa Hodsdon, Diane Jamrog, and Marcia Lazo have worked long hours over an entire summer to solve all the exercises and to prepare the results as LaTeX documents for inclusion in the Handbook for Instructors. We think both these contributions do much to make the course more useful to a wider audience.

We appreciate the contributions of our colleagues who participated in numerous debriefing sessions at semester's end and gave us comments on the evolving text. We thank George Cobb, Giuliana Davidoff, Alan Durfee, Janice Gifford, Mark Peterson , Margaret Robinson and Robert Weaver at Mount Holyoke; Michael Albertson, Ruth Haas, Mary Murphy, Marjorie Senechal, Patricia Sipe, and Gerard Vinel at Smith. We learned, too, from the reactions of our colleagues in other disciplines who participated in faculty workshops on Calculus in Context.

We profited a great deal from the comments and reactions of early users of the text. We extend our thanks to Marian Barry at Aquinas College, Peter Dolan and Mark Halsey at Bard College, Donald Goldberg and his colleagues at Occidental College, Benjamin Levy at Beverly High School, Joan Reinthaler at Sidwell Friends School, Keith Stroyan at the University of Iowa, and Paul Zorn at St. Olaf College. Later users who have helped us are Judith Grabiner and Jim Hoste at Pitzer College; Allen Killpatrick, Mary Scherer, and Janet Beery at the University of Redlands; and Barry Simon at Caltech.

Dissemination grants from the NSF have funded regional workshops for faculty planning to adopt Calculus in Context. We are grateful to Donald Goldberg, Marian Barry, Janet Beery, and to Henry Warchall of the University of North Texas for coordinating workshops.

We owe a special debt to our students over the years, especially those who assisted us in teaching, but also those who gave us the benefit of their thoughtful reactions to the course and the text. Seeing what they were learning encouraged us at every step.

We continue to find it remarkable that our text is to be published the way we want it, not softened or ground down under the pressure of anonymous reviewers seeking a return to the mean. All of this is due to the bold and generous stance of W. H. Freeman. Robert Biewen, its president, understands—more than we could ever hope—what we are trying to do, and he has given us his unstinting support. Our aquisitions editors, Jeremiah Lyons and Holly Hodder, have inspired us with their passionate conviction that our book has something new and valuable to offer science education. Christine Hastings, our production editor, has shown heroic patience and grace in shaping the book itself against our often contrary views. We thank them all.

Contents

Chapter 1

A Context for Calculus

Calculus gives us a language to describe how quantities are related to one another, and it gives us a set of computational and visual tools for exploring those relationships. Usually, we want to understand how quantities are related in the context of a particular problem—it might be in chemistry, or public policy, or mathematics itself. In this chapter we take a single context—an infectious disease spreading through a population—to see how calculus emerges and how it is used.

1.1 The Spread of Disease

1.1.1 Making a Model

Many human diseases are contagious: you "catch" them from someone who is already infected. Contagious diseases are of many kinds. Smallpox, polio, and plague are severe and even fatal,

Some properties of contagious diseases

while the common cold and the childhood illnesses of measles, mumps, and rubella are usually relatively mild. Moreover, you can catch a cold over and over again, but you get measles only once. A disease like measles is said to "confer immunity" on someone who recovers from it. Some diseases have the potential to affect large segments of a population; they are called *epidemics* (from the Greek words *epi*, upon + *demos*, the people.) *Epidemiology* is the scientific study of these diseases.

An epidemic is a complicated matter, but the dangers posed by contagion—and especially by the appearance of new and uncontrollable diseases—compel us to learn as much as we can about the nature of epidemics. Mathematics offers a very special kind of help. First, we can try to draw out of the situation its essential features and describe them mathematically. This is calculus as *language*. We substitute an "ideal" mathematical world for the real one.

The idea of a mathematical model

This is calculus as *tool*. Any conclusion we reach about the model can then be interpreted to tell us something about the reality.

To give you an idea how this process works, we'll build a model of an epidemic. Its basic purpose is to help us understand the way a contagious disease spreads through a population—to the point where we can even predict what fraction falls ill, and when. Let's suppose the disease we want to model is like measles. In particular,

- it is mild, so anyone who falls ill eventually recovers;
- it confers permanent immunity on every recovered victim.

In addition, we will assume that the affected population is large but fixed in size and confined to a geographically well-defined region. To have a concrete image, you can imagine the elementary school population of a big city.

At any time, that population can be divided into three distinct classes:

- **Susceptible**: those who have never had the illness and can catch it.
- **Infected**: those who currently have the illness and are contagious.
- **Recovered**: those who have already had the illness and are immune.

Suppose we let S, I, and R

The quantities that our model analyzes

denote the number of people in each of these three classes, respectively. Of course, the classes are all mixed together throughout the population: on a given day, we may find persons who are susceptible, infected, and recovered in the same family. For the purpose of organizing our thinking, though, we'll represent the whole population as separated into three "compartments" as in the following diagram:

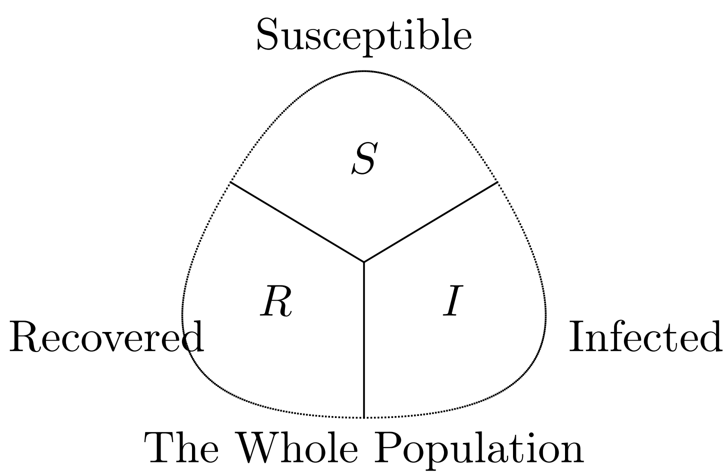


Figure 1.1.1

The goal of our model is to determine what happens to the numbers S, I, and R over the course of time. Let's first see what our knowledge and experience of childhood diseases might lead us to expect. When we say there is a "measles outbreak," we mean that there is a relatively sudden increase in the number of cases, and then a gradual decline. After some weeks or months, the illness virtually disappears. In other words, the number I is a **variable**; its value changes over time. One possible way that I might vary is shown in the following graph.

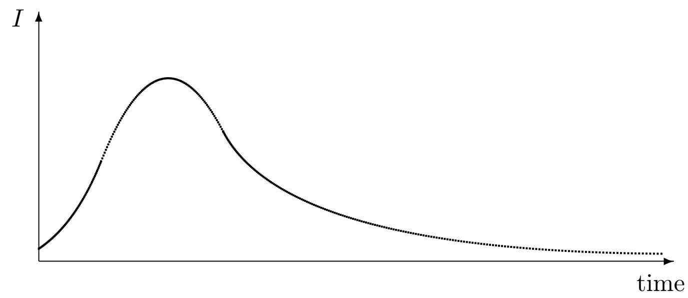


Figure 1.1.2

During the course of the epidemic, susceptibles are constantly falling ill. Thus we would expect the number S to show a steady decline. Unless we know more about the illness, we cannot decide whether everyone eventually catches it. In graphical terms, this means we don't know whether the graph of S levels off at zero or at a value above zero. Finally, we would expect more and more people in the recovered group as time passes. The graph of R should therefore climb from left to right. The graphs of S and R might take the following forms:

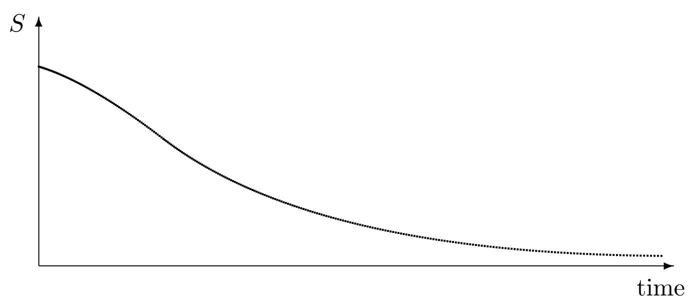


Figure 1.1.3

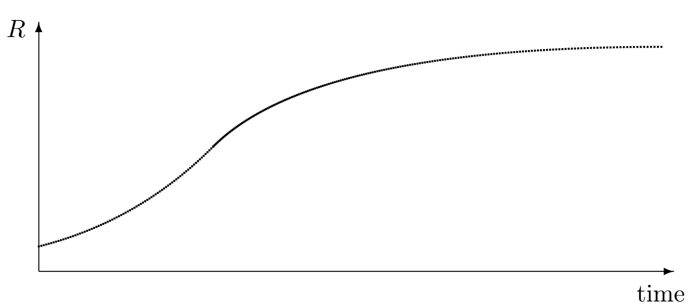


Figure 1.1.4

While these graphs give us an idea about what might happen, they raise some new questions, too. For example, because there are no scales marked along the axes,

Some quantitative questions that the graphs raise

the first graph does not tell us how large I becomes when the infection reaches its peak, nor when that peak occurs. Likewise, the second and third graphs do not say how rapidly the population either falls ill or recovers. A good model of the epidemic should give us graphs like these and it should also answer the quantitative questions we have already raised—for example: When does the infection hit its peak? How many susceptibles eventually fall ill?

1.1.2 A Simple Model for Predicting Change

Suppose we know the values of S, I, and R today; can we figure out what they will be tomorrow, or the next day, or a week or a month from now? Basically, this is a problem of predicting the future. One way to deal with it is to get an idea how S, I, and R are *changing*. To start with a very simple example, suppose the city's Board of Health reports that the measles infection has been spreading at the rate of 470 new cases per day for the last several days. If that rate continues to hold steady and we start with 20,000 susceptible children, then we can expect 470 fewer susceptibles with each passing day. The immediate future would then look like this:

Table 1.1.5 Infections Over Time

Days after today	Accumulated number of new infections	Remaining number of susceptibles
0	0	20 000
1	470	19 530
2	940	19 060
3	1410	18 590

Of course, these numbers will be correct only if the infection continues to spread at its present rate of 470 persons per day.

Knowing rates, we can predict future values

If we want to follow S, I, and R into the future, our example suggests that we should pay attention to the **rates** at which these quantities change. To make it easier to refer to them, let's denote the three rates by S', I', and R'. For example, in the illustration above, S is changing at the rate $S' = -470$ persons per day. We use a minus sign here because S is *decreasing* over time. If S' stays fixed we can express the value of S after t days by the following formula:

$$S = 20000 + S' \cdot t = 20000 - 470\,t \text{ persons.}$$

Check that this gives the values of S found in the table when $t = 0$, 1, 2, or 3. How many susceptibles does it say are left after 10 days?

Our assumption that $S' = -470$ persons per day amounts to a mathematical characterization

The equation $S' = -470$ is a model

of the susceptible population—in other words, a model! Of course it is quite simple, but it led to a formula that told us what value we could expect S to have at any time t.

The model will even take us backwards in time. For example, two days ago the value of t was -2; according to the model, there were

$$S = 20000 - 470 \times -2 = 20940$$

susceptible children then. There is an obvious difference between going backwards in time and going forwards: we already know the past. Therefore, by letting t be negative we can generate values for S that can be checked against health records. If the model gives good agreement with *known* values of S we become more confident in using it to predict *future* values.

To predict the value of S using the rate S' we clearly need to have a starting point—a known value of S

Predictions depend on the initial value, too

from which we can measure changes. In our case that starting point is $S = 20000$. This is called the **initial value** of S, because it is given to us at the "initial time" $t = 0$. To construct the formula $S = 20000 - 470\,t$, we needed to have an initial value as well as a rate of change for S.

In the following pages we will develop a more complex model for all three population groups that has the same general design as this simple one. Specifically, the model will give us information about the rates S', I', and R', and with that information we will be able to predict the values of S, I, and R at any time t.

1.1.3 The Rate of Recovery

Our first task will be to model the recovery rate R'. We look at the process of recovering first, because it's simpler to analyze. An individual caught in the epidemic first falls ill and then recovers—recovery is just a matter of time. In particular, someone who catches measles has the infection for about fourteen days. So if we look at the entire infected population today, we can expect to find some who have been infected less than one day, some who have been infected between one and two days, and so on, up to fourteen days. Those in the last group will recover today. In the absence of any definite information about the fourteen groups, let's assume they are the same size. Then 1/14-th of the infected population will recover today:

$$\text{today the change in the recovered population} = \frac{I \text{ persons}}{14 \text{ days}}.$$

There is nothing special about today, though; I has a value at any time. Thus we can make the same argument about any other day:

$$\text{every day the change in the recovered population} = \frac{I \text{ persons}}{14 \text{ days}}.$$

This equation is telling us about R', the rate at which R is changing. We can write it more simply in the form

The first piece of the S-I-R model

$$R' = \frac{1}{14} I \text{ persons per day.}$$

We call this a **rate equation**. Like any equation, it links different quantities together. In this case, it links R' to I. The rate equation for R is the first part of our model of the measles epidemic.

Are you uneasy about our claim that 1/14-th of the infected population recovers every day? You have good reason to be. After all, during the first few days of the epidemic almost no one has had measles the full fourteen days, so the recovery rate will be much less than $I/14$ persons per day. About a week before the infection disappears altogether there will be no one in the early stages of the illness. The recovery rate will then be much greater than $I/14$ persons per day. Evidently our model is not a perfect mirror of reality!

Don't be particularly surprised or dismayed by this. Our goal is to gain insight into the workings of an epidemic and to suggest how we might intervene to reduce its effects. So we start off with a model which, while imperfect, still captures some of the workings. The simplifications in the model will be justified if we are led to inferences which help us understand how an epidemic works and how we can deal with it. If we wish, we can then refine the model, replacing the simple expressions with others that mirror the reality more fully.

Notice that the rate equation for R' does indeed give us a tool to predict future values of R. For suppose today 2100 people are infected and 2500 have already recovered. Can we say how large the recovered population will be tomorrow or the next day? Since $I = 2100$,

$$R' = \frac{1}{14} \times 2100 = 150 \text{ persons per day}\,.$$

Thus 150 people will recover in a single day, and twice as many, or 300, will recover in two. At this rate the recovered population will number 2650 tomorrow and 2800 the next day.

These calculations assume that the rate R' holds steady at 150 persons per day for the entire two days. Since $R' = I/14$, this is the same as assuming that I holds steady at 2100 persons. If instead I varies during the two days we would have to adjust the value of R' and, ultimately, the future values of R as well. In fact, I *does* vary over time. We shall see this when we analyze how the infection is transmitted. Then, in chapter 2, we'll see how to make the adjustments in the values of R' that will permit us to predict the value of R in the model with as much accuracy as we wish.

Other diseases. What can we say about the recovery rate for a contagious disease other than measles? If the period of infection of the new illness is k days, instead of 14, and if we assume that $1/k$ of the infected people recover each day, then the new recovery rate is

$$R' = \frac{I \text{ persons}}{k \text{ days}} = \frac{1}{k} I \text{ persons per day.}$$

If we set $b = 1/k$ we can express the recovery rate equation in the form

$$R' = bI \text{ persons per day.}$$

The constant b is called the **recovery coefficient** in this context.

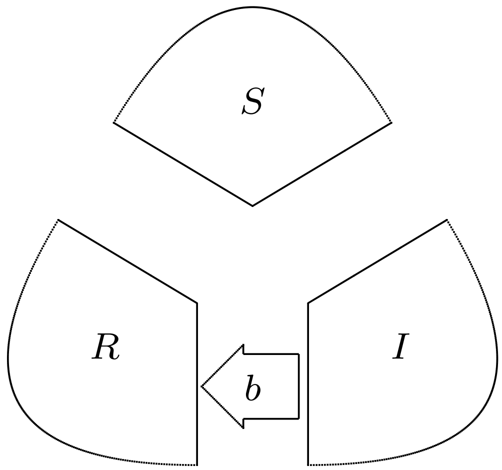


Figure 1.1.6

Let's incorporate our understanding of recovery into the compartment diagram. For the sake of illustration, we'll separate the three compartments. As time passes, people "flow" from the infected compartment to the recovered. We represent this flow by an arrow from I to R. We label the arrow with the recovery coefficient b to indicate that the flow is governed by the rate equation $R' = bI$.

1.1.4 The Rate of Transmission

Since susceptibles become infected, the compartment diagram above should also have an arrow that goes from S to I and a rate equation for S' to show how S changes as the infection spreads. While R' depends only on I, because recovery involves only waiting for people to leave the infected population, S' will depend on both S and I, because transmission involves contact between susceptible and infected persons.

Here's a way to model the transmission rate. First, consider a single susceptible person on a single day. On average, this person will contact only a small fraction, p, of the infected population. For example, suppose there are 5000 infected children, so $I = 5000$. We might expect only a couple of them—let's say 2—will be in the same classroom with our "average" susceptible. So the fraction of contacts is $p = 2/I = 2/5000 = .0004$. The 2 contacts themselves can be expressed as $2 = (2/I) \cdot I = pI$ contacts per day per susceptible.

To find out how many daily contacts the *whole*

Contacts are proportional to both S and I

susceptible population will have, we can just multiply the average number of contacts per susceptible person by the number of susceptibles: this is $pI \cdot S = pSI$.

Not all contacts lead to new infections; only a certain fraction q do. The more contagious the disease, the larger q is. Since the number of daily contacts is pSI, we can expect $q \cdot pSI$ new infections per day (i.e., to convert contacts to infections, multiply by q). This becomes aSI if we define a to be the product qp.

Recall, the value of the recovery coefficient b depends only on the illness involved. It is the same for all populations. By contrast, the value of a depends on the general health of a population and the level of social interaction between its members. Thus, when two different populations experience the same illness, the values of a could be different. One strategy for dealing with an epidemic is to alter the value of a. Quarantine does this, for instance; see the exercises.

Since each new infection decreases the number of susceptibles, we have the rate equation for S:

Here is the second piece of the S-I-R model

$$S' = -aSI \text{ persons per day}\,.$$

The minus sign here tells us that S is decreasing (since S and I are positive). We call a the **transmission coefficient**.

Just as people flow from the infected to the recovered compartment when they recover, they flow from the susceptible to the infected when they fall ill. To indicate the second flow let's add another arrow to the compartment diagram. Because this flow is due to the transmission of the illness, we will label the arrow with the transmission coefficient a. The compartment diagram now reflects all aspects of our model.

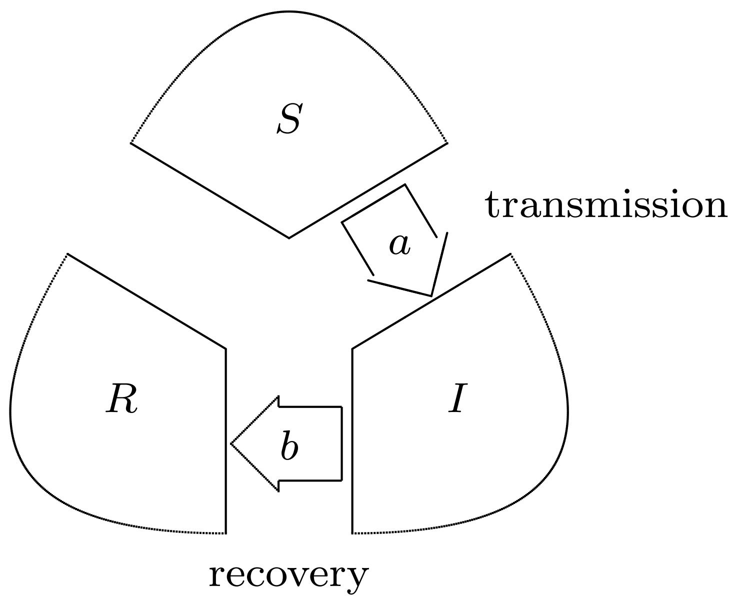


Figure 1.1.7

We haven't talked about the units in which to measure a and b. They must be chosen so that any equation in which a or b appears will balance. Thus, in $R' = bI$ the units on the left are persons/day; since the units for I are persons, the units for b must be 1/(days). The units in $S' = -aSI$ will balance only if a is measured in 1/(person-day).

The reciprocals have more natural interpretations. First of all, $1/b$ is the number of days a person needs to recover. Next, note that $1/a$ is measured in person-days (i.e., persons $\times$ days), which are the natural units in which to measure exposure. Here is why. Suppose you contact 3 infected persons for each of 4 days. That gives you the same exposure to the illness that you get from 6 infected persons in 2 days—both give 12 "person-days" of exposure. Thus, we can interpret $1/a$ as the level of exposure of a typical susceptible person.

1.1.5 Completing the Model

The final rate equation we need—the one for I'—reflects what is already clear from the compartment diagram: every loss in I is due to a gain in R, while every gain in I is due to a loss in S.

Here is the complete S-I-R model

$$\begin{aligned} S' &= -aSI \\ I' &= aSI \qquad\qquad\qquad - bI \end{aligned}$$

$$R' = bI$$

If you add up these three rates you should get the overall rate of change of the whole population. The sum is zero. Do you see why?

You should not draw the conclusion that the only use of rate equations is to model an epidemic. Rate equations have a long history, and they have been put to many uses. Isaac Newton (1642–1727) introduced them to model the motion of a planet around the sun. He showed that the same rate equations could model the motion of the moon around the earth and the motion of an object falling to the ground. Newton created calculus as a tool to analyze these equations. He did the work while he was still an undergraduate—on an extended vacation, curiously enough, because a plague epidemic was raging at Cambridge!

Today we use Newton's rate equations to control the motion of earth satellites and the spacecraft that have visited the moon and the planets. We use other rate equations to model radioactive decay, chemical reactions, the growth and decline of populations, the flow of electricity in a circuit, the change in air pressure with altitude—just to give a few examples. You will have an opportunity in the following chapters to see how they arise in many different contexts, and how they can be analyzed using the tools of calculus.

The following diagram summarizes, in a schematic way, the relation between our model and the reality it seeks to portray.

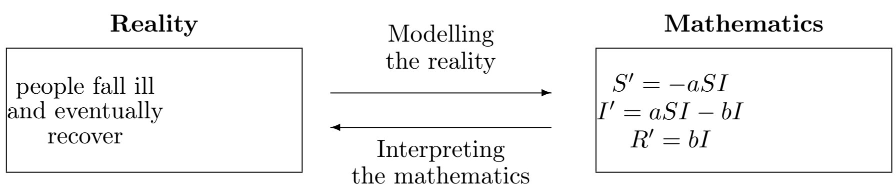


Figure 1.1.8

The diagram calls attention to several facts.

The model is part of mathematics; it only approximates reality

First, the model is a part of mathematics. It is distinct from the reality being modelled. Second, the model is based on a simplified interpretation of the epidemic. As such, it will not match the reality exactly; it will be only an **approximation**. Thus, we cannot expect the values of S, I, and R that we calculate from the rate equations to give us the exact sizes of the susceptible, infected, and recovered populations. Third, the connection between reality and mathematics is a two-way street. We have already travelled one way by constructing a mathematical object that reflects some aspects of the epidemic. This is model-building. Presently we will travel the other way. First we need to get mathematical answers to mathematical questions; then we will see what those answers tell us about the epidemic. This is interpretation of the model. Before we begin the interpretation, we must do some mathematics.

1.1.6 Analyzing the Model

Now that we have a model we shall analyze it as a mathematical object. We will set aside, at least for the moment, the connection between the mathematics and the reality. Thus, for example, you should not be concerned when our calculations produce a value for S of 44,446.6 persons at a certain time—a value that will never be attained in reality. In the following analysis S is just a numerical quantity that varies with t, another numerical quantity. Using only mathematical tools we must now extract from the rate equations the information that will tell us just how S and I and R vary with t.

We already took the first steps in that direction when we used the rate equation $R' = I/14$ to predict the value of R two days into the future (see page). We assumed that I remained fixed at 2100 during those two days, so the rate $R' = 2100/14 = 150$ was also fixed. We concluded that if $R = 2500$ today, it will be 2650 tomorrow and 2800 the next day.

A glance at the full S-I-R model tells us those first steps have to be modified. The assumption

Rates are continually changing; this affects the calculations

we made—that I remains fixed—is not justified, because I (like S and R) is continually changing. As we shall see, I actually increases over those two days. Hence, over the same two days, R' is not fixed at 150, but is continually increasing also. That means that R' becomes larger than 150 during the first day, so R will be larger than 2650 tomorrow.

The fact that the rates are continually changing complicates the mathematical work we need to do to find S, I, and R. In chapter 2 we will develop tools and concepts that will overcome this problem. For the present we'll assume that the rates S', I', and R' stay fixed for the course of an entire day. This will still allow us to produce reasonable estimates for the values of S, I, and R. With these estimates we will get our first glimpse of the predictive power of the S-I-R model. We will also use the estimates as the starting point for the work in chapter 2 that will give us precise values. Let's look at the details of a specific problem.

1.1.6.1 The Problem.

Consider a measles epidemic in a school population of 50,000 children. The recovery coefficient is $b = 1/14$. For the transmission coefficient we choose $a = .00001$, a number within the range used in epidemic studies. We suppose that 2100 people are currently infected and 2500 have already recovered. Since the total population is 50,000, there must be 45,400 susceptibles. Here is a summary of the problem in mathematical terms:

Rate equations:

$$\begin{aligned} S' &= -.00001SI, \\ I' &= .00001SI - I/14, \\ R' &= I/14. \end{aligned}$$

Initial values: when $t = 0$,
$S = 45400, \qquad I = 2100, \qquad R = 2500.$

Figure 1.1.9

1.1.6.2 Tomorrow

From our earlier discussion, $R' = 2100/14 = 150$ persons per day, giving us an estimated value of $R = 2650$ persons for tomorrow. To estimate S we use

$$S' = -.00001\,SI = -.00001 \times 45400 \times 2100 = -953.4 \text{ persons/day.}$$

Hence we estimate that tomorrow

$$S = 45400 - 953.4 = 44446.6 \text{ persons.}$$

Since $S + I + R = 50000$ and we have $S + R = 47096.6$ tomorrow, a final subtraction gives us $I = 2903.4$ persons. (Alternatively, we could have used the rate equation for I' to estimate I.)

The fractional values in the estimates for S and I remind us that the S-I-R model describes the behavior of the epidemic only approximately.

1.1.6.3 Several days hence

According to the model, we estimate that tomorrow $S = 44446.6$, $I = 2903.4$, and $R = 2650$. Therefore, from the new I we get a new approximation for the value of R' tomorrow; it is

$$R' = \frac{1}{14}I = \frac{1}{14} \times 2903.4 = 207.4 \text{ persons/day.}$$

Hence, two days from now we estimate that R will have the new value $2650 + 207.4 = 2857.4$. Now follow this pattern to get new approximations for S' and I', and then use those to estimate the values of S and I two days from now.

The pattern of steps that just carried you from the first day to the second will work just as well to carry you from the second to the third.

Stop and do the calculations

Pause now and do all these calculations yourself. See exercises 15 and 16 on page . If you round your calculated values of S, I, and R to the nearest tenth, they should agree with those in the following table.

Estimates for the first three days

Table 1.1.10

t	S	I	R	S'	I'	R'
0	45 400.0	2100.0	2500.0	−953.4	803.4	150.0
1	44 446.6	2903.4	2650.0	−1290.5	1083.1	207.4
2	43 156.1	3986.5	2857.4	−1720.4	1435.7	284.7
3	41 435.7	5422.1	3142.1			

1.1.6.4 Yesterday

We already pointed out, on page , that we can use our models to go *backwards* in time, too. This is a valuable way to see how well the model fits reality, because we can compare estimates that the model generates with health records for the days in the recent past.

To find how S, I, and R change when we go one day into the future we multiplied the rates S', I', and R' by a time step of $+1$. To find how they change when we go one day into the past we do the same thing, except that we must now use a time step of -1. According to the table above, the rates at time $t = 0$ (i.e., today) are

$$S' = -953.4, \qquad I' = 803.4, \qquad R' = 150.0.$$

Therefore we estimate that, one day ago,

$$\begin{aligned} S &= 45400 + (-953.4 \times -1) &&= 45400 + 953.4 = 46353.4, \\ I &= 2100 + (803.4 \times -1) &&= \ \ 2100 - 803.4 = 1296.6, \\ R &= 2500 + (150.0 \times -1) &&= \ \ 2500 - 150.0 = 2350.0. \end{aligned}$$

Just as we would expect with a spreading infection, there are more susceptibles yesterday than today, but fewer infected or recovered. In the exercises for section 3 you will have an opportunity to continue this process, tracing the epidemic many days into the past. For example, you will be able to go back and see when the infection started—that is, find the day when the value of I was only about 1.

1.1.6.5 There and back again

Go forward a day and then back again

What happens when we start with tomorrow's values and use tomorrow's rates to go back one day—back to today? We should get $S = 45400$, $I = 2100$, and $R = 2500$ once again, shouldn't we? Tomorrow's values are

$$\begin{aligned} S &= 44446.6, & I &= 2903.4, & R &= 2650.0, \\ S' &= -1290.5, & I' &= 1083.1, & R' &= 207.4. \end{aligned}$$

To go backwards one day we must use a time step of -1. The predicted values are thus

$$\begin{aligned} S &= 44446.6 + (-1290.6 \times -1) &&= 45737.2, \\ I &= 2903.4 + (1083.1 \times -1) &&= 1820.3, \\ R &= 2650.0 + (207.4 \times -1) &&= 2442.6. \end{aligned}$$

These are *not* the values that we had at the start, when $t = 0$. In fact, it's worth noting the difference between the original values and those produced by "going there and back again."

Table 1.1.11

	original value	there and back again	difference
S	45400	45737.1	337.1
I	2100	1820.3	–279.7
R	2500	2442.6	–57.4

Do you see why there are differences? We went forward in time using the rates that were current at $t = 0$, but when we returned we used the rates that were current at $t = 1$. Because these rates were different, we didn't get back where we started. These differences do not point to a flaw in the model; the problem lies with the way we are trying to extract information from the model. As we have been making estimates,

The differences measure how rough an estimate is

we have assumed that the rates don't change over the course of a whole day. We already know that's not true, so the values that we have been getting are not exact. What this test adds to our knowledge is a way to measure just *how* inexact those values are—as we do in the table above.

In chapter 2 we will solve the problem of rough estimates by recalculating all the quantities ten times a day, a hundred times a day, or even more. When we do the computations with shorter and shorter time steps we will be able to see how the estimates improve. We will even be able to see how to get values that are mathematically exact!

1.1.6.6 Delta notation

This work has given us some insights about the way our model predicts future values of S, I, and R. The basic idea is very simple: *determine how S, I, and R change*. Because these changes play such an important role in what we do, it is worth having a simple way to refer to them. Here is the notation that we will use:

Δx stands for a *change* in the quantity x

The symbol "Δ" is the Greek capital letter *delta*; it corresponds to the Roman letter "D" and stands for **difference**.

Delta notation gives us a way to refer to changes of all sorts. For example, in the table on page , between day 1 and day 3 the quantities t and S change by

$$Deltat = 2 \text{ days},$$
$$\Delta S = -3010.9 \text{ persons}.$$

We sometimes refer to a change as a **step**.

Δ stands for a change, a difference, or a step

For instance, in this example we can say there is a "t step" of 2 days, and an "S step" of -3010.9 persons. In the calculations that produced the table on page we "stepped into" the future, a day at a time. Finally, delta notation gives us a concise and vivid way to describe the relation between rates and changes. For example, if S changes at the constant rate S', then under a t step of Δt, the value of S changes by

$$\Delta S = S' \cdot \Delta t.$$

1.1.6.7 Using the computer as a tool.

Suppose we wanted to find out what happens to S, I, and R after a month, or even a year. We need only repeat—30 times, or 365 times—the three rounds of calculations we used to go three days into the future. The computations would take time, though. The same is true if we wanted to do ten or one hundred rounds of calculations per day—which is the approach we'll take in chapter 2 to get more accurate values. To save our time and effort we will soon begin to use a computer to do the repetitive calculations.

A computer does calculations by following a set of instructions called a **program**. Of course, if we had to give a million instructions to make the computer carry out a million steps, there would be no savings in labor. The trick is to write a program with just a few instructions that can be repeated over and over again to do all the calculations we want. The usual way to do this is to arrange the instructions in a **loop**. To give you an idea what a loop is, we'll look at the S-I-R calculations. They form a loop. We can see the loop by making a flow chart.

1.1.6.8 The flow chart

We'll start by writing down the three steps that take us from one day to the next:

Algorithm 1.1.12 Updating the SIR Model One Day at a Time.

1. *Given the current values of S, I, and R, we get current S', I', and R' by using the rate equations*

$$\begin{aligned} S' &= -aSI \\ I' &= aSI \qquad\qquad\qquad\qquad - bI \\ R' &= \qquad\qquad\qquad\qquad\qquad\quad\;\; bI \end{aligned}$$

2. *Given the current values of S', I', and R', we find the changes ΔS, ΔI, and ΔR over the course of a day by using the equations*

$$\begin{aligned} \Delta S \text{ persons} &= S' \ \frac{\text{persons}}{\text{day}} \times 1 \text{ day}, \\ \Delta I \text{ persons} &= I' \ \frac{\text{persons}}{\text{day}} \times 1 \text{ day}, \\ \Delta R \text{ persons} &= R' \ \frac{\text{persons}}{\text{day}} \times 1 \text{ day}. \end{aligned}$$

3. *Given the current values of ΔS, ΔI, and ΔR, we find the new values of S, I, and R a day later by using the equations*

$$\begin{aligned} \text{new } S &= \text{current } S + \Delta S, \\ \text{new } I &= \text{current } I + \Delta I, \\ \text{new } R &= \text{current } R + \Delta R. \end{aligned}$$

Each step takes three numbers as **input**, and produces three new numbers as **output**. Note that the output of each step is the input of the next, so the steps flow together. The diagram below, called a **flow chart**,

The flow chart forms a loop

shows us how the steps are connected.

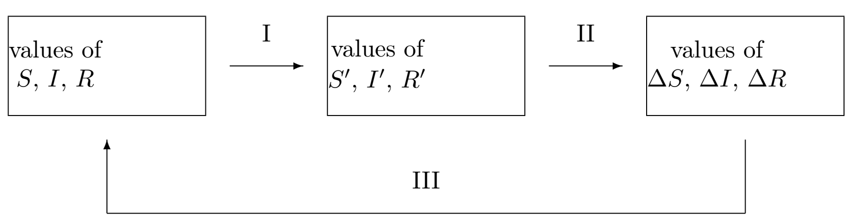


Figure 1.1.13

The calculations form a **loop**, because the output of step III is the input for step I. If we go once around the loop, the output of step III gives us the values of S, I, and R *on the following day*. The steps do indeed carry us into the future.

Each step involves calculating three numbers. If we count each calculation as a single instruction, then it takes nine instructions to carry the values of S, I, and R one day into the future. To go a million days into the future, we need add only one more instruction: “Go around the loop a million times.” In this way, a computer program with only ten instructions can carry out a million rounds of calculations!

Later in this chapter (section 3) you will find a real computer program that lists these instructions (for three days instead of a million, though).

A computer program will carry out the three steps

Study the program to see which instructions accomplish which steps. In particular, see how it makes a loop. Then run the program to check that the computer reproduces the values you already computed by hand. Once you see how the program works, you can modify it to get further information—for example, you can find out what happens to S, I, and R thirty days into the future. You will even be able to plot the graphs of S, I, and R.

Rate equations have always been at the heart of calculus, and they have been analyzed using mechanical and electronic computers for as long as those tools have been available. Now that small powerful computers have begun to appear in the classroom, it is possible for beginning calculus students to explore interesting and complex problems that are modelled by rate equations. Computers are changing how mathematics is done and how it is learned.

1.1.6.9 Analysis without a computer

A computer is a powerful tool for exploring the S-I-R model, but there are many things we can learn about the model without using a computer. Here is an example.

According to the model, the rate at which the infected population grows is given by the equation

$$I' = .00001\,SI - I/14 \quad \text{persons/day.}$$

In our example, $I' = 803.4$ at the outset. This is a positive number, so I increases initially. In fact, I will continue to increase as long as I' is positive. If I' ever becomes negative, then I decreases. So let's ask the question: when is I' positive, when is it negative, and when is it zero? By factoring out I in the last equation we obtain

$$I' = I\left(.00001\,S - \frac{1}{14}\right) \text{ persons/day.}$$

Consequently $I' = 0$ if either

$$I = 0 \qquad \text{or} \qquad .00001\,S - \frac{1}{14} = 0.$$

The first possibility $I = 0$ has a simple interpretation: there is no infection within the population. The second possibility is more interesting; it says that I' will be zero when

$$.00001\,S - \frac{1}{14} = 0 \qquad \text{or} \qquad S = \frac{100000}{14} \approx 7142.9.$$

If S is *greater* than 100000/14 and I is positive, then you can check that the formula

$$I' = I\left(.00001\,S - \frac{1}{14}\right) \text{ persons/day}$$

tells us I' is positive—so I is increasing. If, on the other hand, S is *less than* 100000/14, then I' is negative and I is decreasing. So $S = 100000/14$ represents a **threshold**. If S falls below the threshold, I decreases. If S exceeds the threshold, I increases. Finally, I reaches its peak when S equals the threshold.

The presence of a threshold value for S is purely a mathematical result. However, it has an interesting interpretation for the epidemic.

The threshold determines whether there will be an epidemic

As long as there are at least 7143 susceptibles, the infection will spread, in the sense that there will be more people falling ill than recovering each day. As new people fall ill, the number of susceptibles declines. Finally, when there are fewer than 7143 susceptibles, the pattern reverses: each day more people will recover than will fall ill.

If there were fewer than 7143 susceptibles in the population *at the outset*, then the number of infected would only decline with each passing day. The infection would simply never catch hold. The clear implication is that the noticeable surge in the number of cases that we associate with an "epidemic" disease is due to the presence of a large susceptible population. If the susceptible population lies below a certain threshold value, a surge just isn't possible. This is a valuable insight—and we got it with little effort. We didn't need to make lengthy calculations or call on the resources of a computer; a bit of algebra was enough.

1.1.7 Exercises

Reading a Graph. The graphs on pages and have no scales marked along their axes, so they provide mainly *qualitative* information. The graphs below do have scales, so you can now answer *quantitative* questions about them. For example, on day 20 there are about 18,000 susceptible people. Read the graphs to answer the following questions. (Note: $S + I + R$ is *not* constant in this example, so these graphs cannot be solutions to our model..)

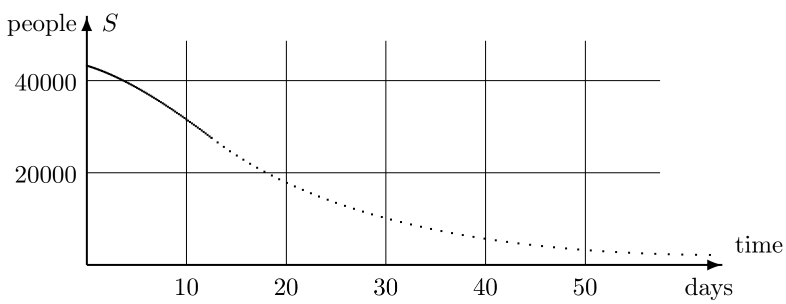


Figure 1.1.14

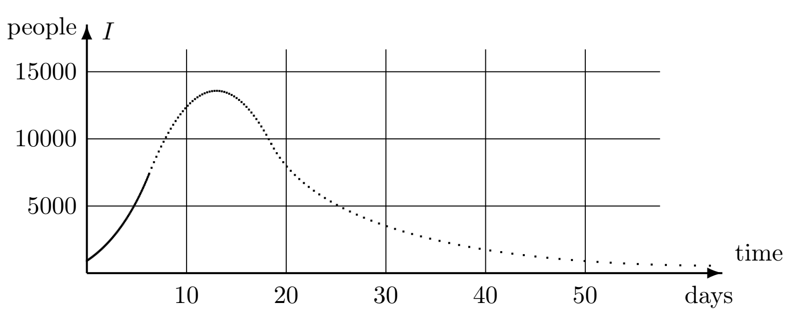


Figure 1.1.15

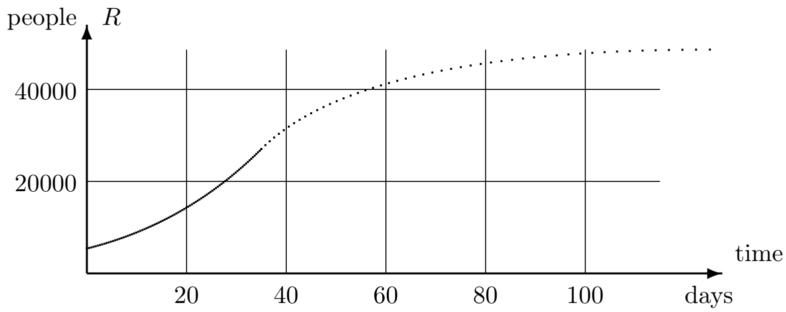


Figure 1.1.16

1. When does the infection hit its peak? How many people are infected at that time?
2. Initially, how many people are susceptible? How many days does it take for the susceptible population to be cut in half?
3. How many days does it take for the recovered population to reach 25,000? How many people *eventually* recover? Where did you look to get this information?
4. On what day is the size of the infected population increasing most rapidly? When is it decreasing most rapidly? How do you know?
5. How many people caught the illness at some time during the first 20 days? (Note that this is not the same as the number of people who are infected on day 20.) Explain where you found this information.
6. Copy the graph of R as accurately as you can, and then superimpose a sketch of S on it. Notice the time scales on the *original* graphs of S and R are different. Describe what happened to the graph of S when you superimposed it on the graph of R. Did it get compressed or stretched? Was this change in the horizontal direction or the vertical?

A Simple Model. These questions concern the rate equation $S' = -470$ persons per day that we used to model a susceptible population on pages –.

7. Suppose the initial susceptible population was 20,000 on Wednesday. Use the model to answer the following questions.

   (a) How many susceptibles will be left ten days later?

   (b) How many days will it take for the susceptible population to vanish entirely?

   (c) How many susceptibles were there on the previous Sunday?

   (d) How many days before Wednesday were there 30,000 susceptibles?

Mark Twain's Mississippi. The Lower Mississippi River meanders over its flat valley, forming broad loops called ox-bows. In a flood, the river can jump its banks and cut off one of these loops, getting shorter in the process. In his book *Life on the Mississippi* (1884), Mark Twain suggests, with tongue in cheek, that some day the river might even vanish! Here is a passage that shows us some of the pitfalls in using rates to predict the future and the past.

> In the space of one hundred and seventy six years the Lower Mississippi has shortened itself two hundred and forty-two miles. That is an average of a trifle over a mile and a third per year. Therefore, any calm person, who is not blind or idiotic, can see that in the Old Oölitic Silurian Period, just a million years ago next November, the Lower Mississippi was upwards of one million three hundred thousand miles long, and stuck out over the Gulf of Mexico like a fishing-pole. And by the same token any person can see that seven hundred and forty-two years from now the Lower Mississippi will be only a mile and three-quarters long, and Cairo [Illinois] and New Orleans will have joined their streets together and be plodding comfortably along under a single mayor and a mutual board of aldermen. There is something fascinating about science. One gets such wholesome returns of conjecture out of such a trifling investment of fact.

Let L be the length of the Lower Mississippi River. Then L is a variable quantity we shall analyze.

8. According to Twain's data, what is the exact *rate* at which L is changing, in miles per year? What approximation does he use for this rate? Is this a reasonable approximation? Is this rate *positive* or *negative*? Explain. In what follows, use Twain's approximation.

9. Twain wrote his book in 1884. Suppose the Mississippi that Twain wrote about had been 1100 miles long; how long would it have become in 1990?

10. Twain does not tell us how long the Lower Mississippi was in 1884 when he wrote the book, but he does say that 742 years later it will be only $1\frac{3}{4}$ miles long. How long must the river have been when he wrote the book?

11. Suppose t is the number of years since 1884. Write a formula that describes how much L has changed in t years. Your formula should complete the equation

$$\text{the change in } L \text{ in } t \text{ years} = \dots\ .$$

12. From your answer to question 10, you know how long the river was in 1884. From question 11, you know how much the length has changed t years after 1884. Now write a formula that describes how long the river is t years later.

13. Use your formula to find what L was a million years ago. Does your answer confirm Twain's assertion that the river was "upwards of 1,300,000 miles long" then?

14. Was the river ever 1,300,000 miles long; will it ever be $1\frac{3}{4}$ miles long? (This is called a **reality check**.) What, if anything, is wrong with the "trifling investment of fact" which led to such "wholesale returns of conjecture" that Twain has given us?

The Measles Epidemic. We consider once again the specific rate equations

$$\begin{aligned} S' &= -.00001\,SI, \\ I' &= .00001\,SI - I/14, \\ R' &= I/14, \end{aligned}$$

discussed in the text on pages –. We saw that at time $t = 1$,

$$S = 44446.6, \qquad I = 2903.4, \qquad R = 2650.0.$$

15. Calculate the current rates of change S', I', and R' when $t = 1$, and then use these values to determine S, I, and R one day later.

16. In the previous question you found S, I, and R when $t = 2$. Using these values, calculate the rates S', I', and R' and then determine the new values of S, I, and R when $t = 3$. See the table on page .

17. Double the time step. Go back to the starting time $t = 0$ and to the initial values

$$S = 45400, \qquad I = 2100, \qquad R = 2500.$$

Recalculate the values of S, I, and R at time $t = 2$ by using a time step of $\Delta t = 2$. You should perform only a single round of calculations, and use the rates S', I', and R' that are current at time $t = 0$.

18. There and back again. In the text we went one day into the future and then back again to the present. Here you'll go forward two days from $t = 0$ and then back again. There are two ways to do this: with a time step of $\Delta t = \pm 2$ (as in the previous question), and with a pair of time steps of $\Delta t = \pm 1$.

(a) ($\Delta t = \pm 2$). Using the values of S, I, and R at time $t = 2$ that you just got in the previous question, calculate the rates S', I', and R'. Then using a time step of $\Delta t = -2$, estimate new values of S, I, and R at time $t = 0$. How much do these new values differ from the original values 45,400, 2100, 2500?

(b) ($\Delta t = \pm 1$). Now make a new start, using the values

$$\begin{aligned} S &= 43156.1, & I &= 3986.5, & R &= 2857.4, \\ S' &= -1720.4, & I' &= 1435.7, & R' &= 284.7. \end{aligned}$$

that occur when $t = 2$ if we make estimates with a time step $\Delta t = 1$. (These values come from the table on page) Using two rounds of calculations with a time step of $\Delta t = -1$, estimate another set of new values for S, I, and R at time $t = 0$. How much do these new values differ from the original values 45,400, 2100, 2500?

(c) Which process leads to a *smaller* set of differences: a single round of calculations with $\Delta t = \pm 2$, or two rounds of calculations with $\Delta t = \pm 1$? Consequently, which process produces better estimates—in the sense in which we used to measure estimates on page ?

19. Quarantine. One of the ways to treat an epidemic is to keep the infected away from the susceptible; this is called quarantine. The intention is to reduce the chance that the illness will be transmitted to a susceptible person. Thus, quarantine alters the *transmission coefficient*.

(a) Suppose a quarantine is put into effect that cuts in half the chance that a susceptible will fall ill. What is the new transmission coefficient?

(b) On page it was determined that whenever there were fewer than

7143 susceptibles, the number of infected would decline instead of grow. We called 7143 a *threshold* level for S. Changing the transmission coefficient, as in part (a), changes the threshold level for S. What is the new threshold?

(c) Suppose we start with $S = 45{,}400$. Does quarantine eliminate the epidemic, in the sense that the number of infected immediately goes down from 2100, without ever showing an increase in the number of cases?

(d) Since the new transmission coefficient is not small enough to guarantee that I never goes up, can you find a smaller value that *does* guarantee I never goes up? Continue to assume we start with $S = 45400$.

(e) Suppose the initial susceptible population is 45,400. What is the *largest* value that the transmission coefficient can have and still guarantee that I never goes up? What level of quarantine does this represent? That is, do you have to reduce the chance that a susceptible will fall ill to one-third of what it was with no quarantine at all, to one-fourth, or what?

Other Diseases.

20. Suppose the spread of an illness similar to measles is modelled by the following rate equations:

$$\begin{aligned} S' &= -.00002\, SI, \\ I' &= .00002\, SI - .08\, I, \\ R' &= .08\, I. \end{aligned}$$

Note: the initial values $S = 45400$, etc. that we used in the text do not apply here.

(a) Roughly how long does someone who catches this illness remain infected? Explain your reasoning.

(b) How large does the susceptible population have to be in order for the illness to take hold—that is, for the number of cases to increase? Explain your reasoning.

(c) Suppose 100 people in the population are currently ill. According to the model, how many (of the 100 infected) will recover during the next 24 hours?

(d) Suppose 30 *new* cases appear during the same 24 hours. What does that tell us about S'?

(e) Using the information in parts (c) and (d), can you determine how large the current susceptible population is?

21. Construct the appropriate S-I-R model for a measles-like illness that lasts for 4 days. It is also known that a typical susceptible person meets only about 0.3% of infected population each day, and the infection is transmitted in only one contact out of six.

(a) How small does the susceptible population have to be for this illness to fade away without becoming an epidemic?

22. Consider the general S-I-R model for a measles-like illness:

$$\begin{aligned} S' &= -aSI, \\ I' &= aSI \qquad\qquad - bI, \\ R' &= \qquad\qquad\qquad\quad\; bI. \end{aligned}$$

(a) The threshold level for S—below which the number of infected will only decline—can be expressed in terms of the transmission coefficient a and the recovery coefficient b. What is that expression?

(b) Consider two illnesses with the same transmission coefficient a; assume they differ only in the length of time someone stays ill. Which one has the lower threshold level for S? Explain your reasoning.

What Goes Around Comes Around. Some relatively mild illnesses, like the common cold, return to infect you again and again. For a while, right after you recover from a cold, you are immune. But that doesn't last; after some weeks or months, depending on the illness, you become susceptible again. This means there is now a flow from the recovered population to the susceptible. These exercises ask you to modify the basic S-I-R model to describe an illness where immunity is temporary.

23. Draw a compartment diagram for such an illness. Besides having all the ingredients of the diagram on page , it should depict a flow from R to S. Call this **immunity loss**, and use c to denote the coefficient of immunity loss.

24. Suppose immunity is lost after about six weeks. Show that you can set $c = 1/42$ per day, and explain your reasoning carefully. A suggestion: adapt the discussion of recovery in the text.

25. Suppose this illness lasts 5 days and it has a transmission coefficient of .00004 in the population we are considering. Suppose furthermore that the total population is fixed in size (as was the case in the text). Write down rate equations for S, I, and R.

26. We saw in the text that the model for an illness that confers permanent immunity has a threshold value for S in the sense that when S is above the threshold, I increases, but when it is below, I decreases. Does *this* model have the same feature? If so, what is the threshold value?

27. For a mild illness that confers permanent immunity, the size of the recovered population can only grow. This question explores what happens when immunity is only temporary.

(a) Will R increase or decrease if

$$S = 45400, \qquad I = 2100, \qquad R = 2500?$$

(b) Suppose we shift 20000 susceptibles to the recovered population (so that $S = 25400$ and $R = 22500$), leaving I unchanged. Now, will R increase or will it decrease?

(c) Using a total population of 50,000, give two other sets of values for S, I, and R that lead to a decreasing R.

(d) In fact, the relative sizes of I and R determine whether R will increase or decrease. Show that

$$\begin{aligned} textif I > \tfrac{5}{42}R, &\quad \text{then } R \text{ will increase;} \\ \text{if } I < \tfrac{5}{42}R, &\quad \text{then } R \text{ will decrease.} \end{aligned}$$

Explain your argument clearly. A suggestion: consider the rate equation for R'.

28. The steady state. Any illness that confers only temporary immunity can appear to linger in a population forever. You may not always have a cold, but someone does, and eventually you catch another one. ("What goes around comes around.") Individuals gradually move from one compartment to the next. When they return to where they started, they begin another cycle.

Each compartment (in the diagram you drew in exercise 23) has an *inflow* and an *outflow*. It is conceivable that the two exactly balance, so that the *size* of the compartment doesn't change (even though its individual occupants do). When this happens for all three compartments simultaneously, the illness is said to be in a **steady state**. In this question you explore the steady state of the model we are considering. Recall that the total population is 50,000.

(a) What must be true if the inflow and outflow to the I compartment are to balance?

(b) What must be true if the inflow and outflow to the R compartment are to balance?

(c) If neither I nor R is changing, then the model must be at the steady state. Why?

(d) What is the value of S at the steady state?

(e) What is the value of R at the steady state? A suggestion: you know $R + I = 50000 -$ (the steady state value of S). You also have a connection between I and R at the steady state.

1.2 The Mathematical Ideas

A number of important mathematical ideas have already emerged in our study of an epidemic. In this section we pause to consider them, because they have a "universal" character. Our aim is to get a fuller understanding of what we have done so we can use the ideas in other contexts.

We often draw out of a few particular experiences a lesson that can be put to good use in new settings. This process is the essence of mathematics, and it has been given a name—abstraction—which means literally "drawing from." Of course abstraction is not unique to mathematics; it is a basic part of the human psyche.

1.2.1 Functions

A **function** describes how one quantity depends on another. In our study of a measles epidemic, Functions and their notation

the relation between the number of susceptibles S and the time t is a function. We write $S(t)$ to denote that S is a function of t. We can also write $I(t)$ and $R(t)$ because I and R are functions of t, too. We can even write $S'(t)$ to indicate that the rate S' at which S changes over time is a function of t. In speaking, we express $S(t)$ as "S of t" and $S'(t)$ as "S prime of t."

You can find functions everywhere. The amount of postage you pay for a letter is a function of the weight of the letter. The time of sunrise is a function of what day of the year it is. The crop yield from an acre of land is a function of the amount of fertilizer used. The position of a car's gasoline gauge (measured in centimeters from the left edge of the gauge) is a function of the amount of gasoline in the fuel tank. On a polygraph ("lie detector") there is a pen that records breathing; its position is a function of the amount of expansion of the lungs. The volume of a cubical box is a function of the length of a side. The last is a rather special kind of function because it can be described by an algebraic formula: if V is the volume of the box and s is the length of a side, then $V(s) = s^3$.

Most functions are *not* described by algebraic formulas, however. For instance, the postage function is given by a set of verbal instructions and the time of sunrise is given by a table in an almanac. The relation between a gas gauge and the amount of fuel in the tank is determined simply by making measurements. There is no algebraic formula that tells us how the number of susceptibles, S, depends upon t, either. Instead, we find $S(t)$ by carrying out the steps in the flow chart on page until we reach t days into the future.

In the function $S(t)$ the variable t is called the **input**

A function has input and output

and the variable S is called the **output**. In the sunrise function, the day of the year is the input and the time of sunrise is the output. In the function $S(t)$ we think of S as *depending on* t, so t is also called the independent variable and S the dependent variable. The set of values that the input takes is called the **domain** of the function. The set of values that the output takes is called the **range**.

The idea of a function is one of the central notions of mathematics. It is worth highlighting:

Observation 1.2.1 A **function** is a rule that specifies how the value of one variable, the input, determines the value of a second variable, the output.

Notice that we say *rule* here, and not *formula*. This is deliberate. We want the study of functions to be as broad as possible, to include all the ways one quantity is likely to be related to another in a scientific question.

1.2.1.1 Some technical details.

It is important not to confuse an expression like $S(t)$ with a product; $S(t)$ does *not* mean $S \times t$. On the contrary, the expression $S(1.4)$, for example, stands for the output of the function S when 1.4 is the input. In the epidemic model we then interpret it as the number of susceptibles that remain 1.4 days after today.

We have followed the standard practice in science by letting the single letter S designate both the *function*—that is, the *rule*—and the *output* of that function. Sometimes, though, we will want to make the distinction. In that case we will use two different symbols. For instance, we might write $S = f(t)$.

Then we are still using S to denote the output, but the new symbol f stands for the function rule.

The symbols we use to denote the input and the output of a function are just names; if we change them, we don't change the function. For example, here are three ways to describe the same function g:

$$\begin{aligned} g &: \text{multiply the input by 5, then subtract 3} \\ g(x) &= 5x - 3 \\ g(u) &= 5u - 3\,. \end{aligned}$$

It is important to realize that the *formulas* we just wrote in the last two lines are merely shorthand for the instructions stated in the first line. If you keep this in mind, then absurd-looking combinations like $g(g(2))$ can be decoded easily by remembering g of *anything* is just 5 times that thing, minus 3. We could thus evaluate $g(g(2))$ from the inside out (which is usually easier) as

$$g(g(2)) = g(5 \cdot 2 - 3) = g(7) = 5 \cdot 7 - 3 = 32,$$

or we could evaluate it from the outside in as

$$g(g(2)) = 5g(2) - 3 = 5(5 \cdot 2 - 3) - 3 = 5 \cdot 7 - 3 = 32,$$

as before.

Suppose f is some other rule, say $f(t) = t^2 + 4t - 1$. Remember that this is just shorthand for "Take the input (whatever it is), square it, add four times the input, and subtract 1." We could then evaluate

$$f(g(3)) = f(5 \cdot 3 - 3) = f(12) = 12^2 + 4 \cdot 12 - 1 = 144 + 48 - 1 = 191,$$

while

$$g(f(3)) = g(20) = 97.$$

This process of **chaining** functions together by using the output of one function as the input for another turns out to be very

Part of math is simply learning the language

important later in this course, and will be taken up again in chapter 3. For now, though, you should treat it simply as part of the formal language of mathematics, requiring a knowledge of the rules but no cleverness. It is analogous to learning how to conjugate verbs in French class—it's not very exciting for its own sake, but it allows us to read the interesting stuff later on!

A particularly important class of functions is composed of the **constant functions** which give the same output for every input. If h is the constant function that always gives back 17, then in formula form we would express this as $h(x) = 17$. Constant functions are so simple you might feel you are missing the point, but that's all there is to it!

1.2.2 Graphs

A graph describes a function in a visual form. Sometimes—as with a seismograph or a lie detector,

A graph is a function rule given visually

for instance—this is the *only* description we have of a particular function. The usual arrangement is to put the input variable on the horizontal axis and the output on the vertical—but it is a good idea when you are looking at a particular graph to take a moment to check; sometimes, the opposite convention is used! This is often the case in geology and economics, for instance.

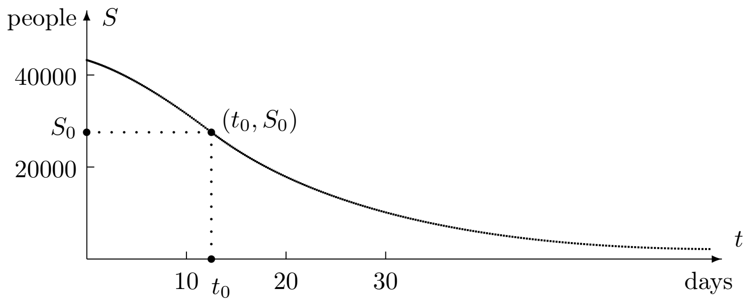


Figure 1.2.2

Sketched above is the graph of a function $S(t)$ that tells how many susceptibles there are after t days. Given any t_0, we "read" the graph to find $S(t_0)$, as follows: from the point t_0 on the t-axis, go vertically until you reach the graph; then go horizontally until you reach the S-axis. The value S_0 at that point is the output $S(t_0)$. Here t_0 is about 13 and S_0 is about 27,000; thus, the graph says that $S(13) \approx 27000$, or about 27,000 susceptibles are left after 13 days.

1.2.3 Linear Functions

1.2.3.1 Changes in input and output

If y depends on x, then Δy depends on Δx

Suppose y is a function of x. Then there is some rule that answers the question: What is the value of y for any given x? Often, however, we start by knowing the value of y for a particular x, and the question we really want to ask is: How does y respond to *changes* in x? We are still dealing with the same function—just looking at it from a different point of view. This point of view is important; we use it to analyze functions (like $S(t)$, $I(t)$, and $R(t)$) that are defined by rate equations.

The way Δy depends on Δx can be simple or it can be complex, depending on the function involved. The simplest possibility is that Δy and Δx are *proportional*:

$$\Delta y = m \cdot \Delta x, \quad \text{for some constant } m.$$

Thus, if Δx is doubled, so is Δy; if Δx is tripled, so is Δy.

The defining property of a linear function

A function whose input and output are related in this simple way is called a **linear function**, because the graph is a straight line. Let's take a moment to see why this is so.

1.2.3.2 The graph of a linear function

The graph consists of certain points (x, y) in the x, y-plane. Our job is to see how those points are arranged. Fix one of them, and call it (x_0, y_0). Let (x, y)

be any other point on the graph. Draw the line that connects this point to (x_0, y_0), as we have done in the figure at the right. Now set

$$\Delta x = x - x_0, \qquad \Delta y = y - y_0.$$

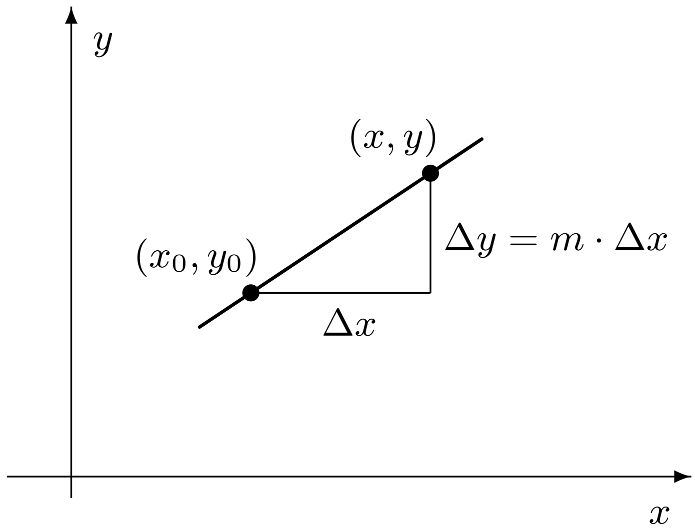


Figure 1.2.3

By definition of a linear function, $\Delta y = m \cdot \Delta x$, as the figure shows, so the slope of this line is $\Delta y/\Delta x = m$. Recall that m is a constant; thus, if we pick a new point (x, y), the slope of the connecting line won't change.

Since (x, y) is an arbitrary point on the graph, what we have shown is that

Observation 1.2.4 $m(x_0, y_0)$

. But there is only one such line—and all the points lie on it! That line must be the graph of the linear function.

Definition 1.2.5 A linear function is one that satisfies $\Delta y = m \cdot \Delta x$. ◆

Observation 1.2.6 Its graph is a straight line whose slope is m.

1.2.3.3 Rates, slopes, and multipliers

The interpretation of m as a slope is just one possibility; there are two other interpretations that are equally important. To illustrate them we'll use Mark Twain's vivid description of the shortening of the Lower Mississippi River (see page). This will also give us the chance to see how a linear function emerges in context.

Twain says "the Lower Mississippi has shortened itself … an average of a trifle over a mile and a third per year." Suppose we let L denote the length of the river, in miles, and t the time, in years. Then L depends on t, and Twain's statement implies that L is a *linear* function of t—in the sense in which we have just defined a linear function. Here is why. According to our definition, there must be some number m which makes $\Delta L = m \cdot \Delta t$. But notice that Twain's statement has exactly this form if we translate it into mathematical language. Convince yourself that it

Stop and do the translation

says

$$\Delta L \text{ miles} = -1\tfrac{1}{3}\ \frac{\text{miles}}{\text{year}} \times \Delta t \text{ years.}$$

Thus we should take m to be $-1\frac{1}{3}$ miles per year.

The role of m here is to convert one quantity (Δt years) into another (ΔL miles) by multiplication. All linear functions work this way. In the defining equation $\Delta y = m \cdot \Delta x$, multiplication by m converts Δx into Δy. Any change in x produces a change in y that is m times as large. For this reason we give m its second interpretation as a **multiplier**.

It is easier to understand why the usual symbol for *slope* is m—instead of s—when you see that a slope can be interpreted as a multiplier.

It is important to note that, in our example, m is not simply $-1\frac{1}{3}$; it is $-1\frac{1}{3}$ *miles per year*. In other words, m is the **rate** at which the river is getting shorter. All linear functions work this way, too. We can rewrite the equation $\Delta y = m \cdot \Delta x$ as a ratio

$$m = \frac{\Delta y}{\Delta x} = \text{the rate of change of } y \text{ with respect to } x.$$

For these reasons we give m its third interpretation as a **rate of change**.

Observation 1.2.7 For a linear function satisfying $\Delta y = m \cdot \Delta x$, the coefficient m is the rate of change (slope) and serves as a multiplier.

We already use y' to denote the rate of change of y, so we can now write $m = y'$ when y is a linear function of x. In that case we can also write

$$\Delta y = y' \cdot \Delta x.$$

This expression should recall a pattern very familiar to you. (If not, change y to S and x to t!) It is the fundamental formula we have been using to calculate future values of S, I, and R. We can approach the relation between y and x the same way. That is, if y_0 is an "initial value" of y, when $x = x_0$, then *any* value of y can be calculated from

$$y = y_0 + y' \cdot \Delta x \quad \text{or} \quad y = y_0 + m \cdot \Delta x.$$

1.2.3.4 Units

Suppose x and y are quantities that are

If x and y have units, so does m

measured in specific units. If y is a linear function of x, with $\Delta y = m \cdot \Delta x$, then m must have units too. Since m is the multiplier that "converts" x into y, the units for m must be chosen so they will convert x's units into y's units. In other words,

$$\text{units for } y = \text{units for } m \times \text{units for } x.$$

This implies

$$\text{units for } m = \frac{\text{units for } y}{\text{units for } x}.$$

For example, the multiplier in the Mississippi River problem converts years to miles, so it must have units of miles per year. The rate equation $R' = bI$ in the S-I-R model is a more subtle example. It says that R' is a linear function of I. Since R' is measured in persons per day and I is measured in persons, we must have

$$\text{units for } b = \frac{\text{units for } R'}{\text{units for } I} = \frac{\frac{\text{persons}}{\text{days}}}{\text{persons}}$$

1.2.3.5 Formulas for linear functions

The expression $\Delta y = m \cdot \Delta x$ declares that y is a linear function of x, but it doesn't quite tell us what y itself *looks like* directly in terms of x. In fact, there are several equivalent ways to write the relation $y = f(x)$ in a formula, depending on what information we are given about the function.

- **The initial-value form**

  Here is a very common situation: we know the value of y at an "initial" point—let's say $y_0 = f(x_0)$—and we know the rate of change—let's say it is m. Then the graph is the straight line of slope m that passes through the point (x_0, y_0). The formula for f is

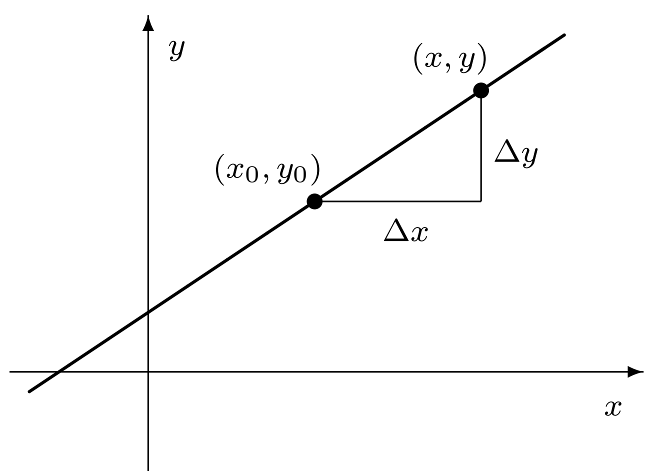
  

  Figure 1.2.8

  $$y = y_0 + \Delta y = y_0 + m \cdot \Delta x = y_0 + m(x - x_0) = f(x).$$

  What you should note particularly about this formula is that it expresses y in terms of the initial data x_0, y_0, and m—as well as x. Since that data consists of a point (x_0, y_0) and a slope m, the initial-value formula is also referred to as the **point-slope form** of the equation of a line. It may be more familiar to you with that name.

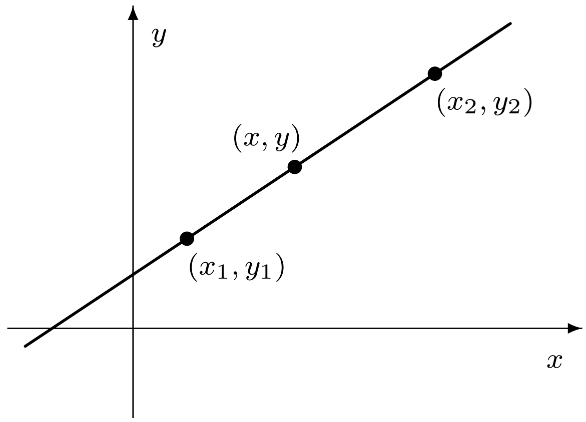


Figure 1.2.9

- **The interpolation form**

  This time we are given the value of y at *two* points—let's say $y_1 = f(x_1)$ and $y_2 = f(x_2)$. The graph is the line that passes through (x_1, y_1) and (x_2, y_2), and its slope is therefore

  $$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

  Now that we know the slope of the graph we can use the point-slope form (taking (x_1, y_1) as the "point", for example) to get the equation. We have

  $$y = y_1 + m(x - x_1) = y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) = f(x).$$

  Notice how, once again, y is expressed in terms of the initial data—which consists of the two points (x_1, y_1) and (x_2, y_2).

  The process of finding values of a quantity between two given values is called **interpolation**. Since our new expression does precisely that, it is called the interpolation formula. (Of course, it also finds values outside the given interval.) Since the initial data is a pair of points, the interpolation formula is also called the **two-point formula** for the equation of a line.

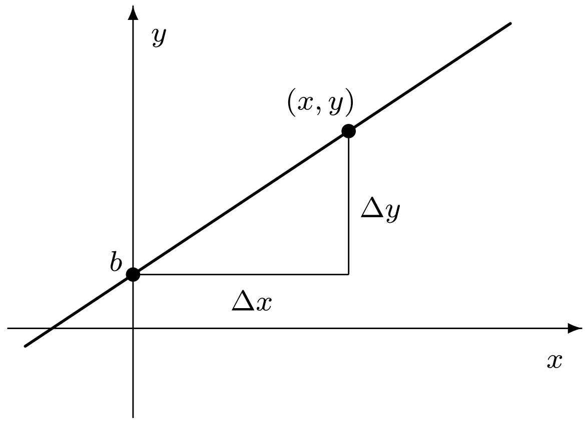


Figure 1.2.10

- **The slope-intercept form**

  This is a special case of the initial-value form that occurs when the initial $x_0 = 0$. Then the point (x_0, y_0) lies on the y-axis, and it is frequently written in the alternate form $(0, b)$. The number b is called the **y-intercept**. The equation is

  $$y = mx + b = f(x).$$

  In the past you may have thought of this as *the* formula for a linear function, but for us it is only one of several. You will find that we will use the other forms more often.

1.2.4 Functions of Several Variables

1.2.4.1 Language and notation.

Many functions depend on more than one variable. For example, sunrise depends on the day of the year but it also depends on the latitude

A function can have several input variables

(position north or south of the equator) of the observer. Likewise, the crop yield from an acre of land depends on the amount of fertilizer used, but it also depends on the amount of rainfall, on the composition of the soil, on the amount of weeding done—to mention just a few of the other variables that a farmer has to contend with.

The rate equations in the S-I-R model also provide examples of functions with more than one input variable. The equation

$$I' = .00001\,SI - I/14$$

says that we need to specify both S and I to find I'. We can say that

$$F(S, I) = .00001\,SI - I/14$$

is a function whose input is the **ordered pair** of variables (S, I). In this case F is given by an algebraic formula. While many other functions of several variables also have formulas—and they are extremely useful—not all functions do. The sunrise function, for example, is given by a two-way table (see page) that shows the time of sunrise for different days of the year and different latitudes.

As a technical matter it is important to note that the input variables S and I of the function $F(S, I)$ above appear in a particular *order*, and that order is part of the definition of the function. For example, $F(1, 0) = 0$, but $F(0, 1) = -1/14$. (Do you see why? Work out the calculations yourself.)

1.2.4.2 Parameters

Suppose we rewrite the rate equation for I', replacing .00001 and 1/14 with the general values a and b:

$$I' = aSI - bI.$$

This makes it clear that I' depends on a and b, too. But note that a and b are not variables in quite the same way that S and I are. For example, a and b will vary if we switch from one disease to another or from one population to another. However, they will stay fixed while we consider a particular disease in a particular population. By contrast, S and I will *always* be treated as variables. We call a quantity like a or b a **parameter**.

To emphasize that I' depends on the parameters as well as S and I, we can write I' as the output of a new function

$$I' = I'(S, I, a, b) = aSI - bI$$

whose input is the set of *four* variables

Some functions depend on parameters

(S, I, a, b), in that order. The variables S, I, and R must also depend on the parameters, too, and not just on t. Thus, we should write $S(t, a, b)$, for example, instead of simply $S(t)$. We implicitly used the fact that S, I, and R depend on a and b when we discovered there was a threshold for an epidemic (page). In exercise 22 of section 1 (page), you made the relation explicit. In that problem you show I will simply decrease over time (i.e., there will be no "burst" of infection) if

$$S < \frac{b}{a}.$$

There are even more parameters lurking in the S-I-R problem. To uncover them, recall that we needed *two* pieces of information to estimate S, I, and R over time:

1. the rate equations;
2. the initial values S_0, I_0, and R_0.

We used $S_0 = 45400$, $I_0 = 2100$, and $R_0 = 2500$ in the text, but if we had started with other values then S, I, and R would have ended up being different functions of t. Thus, we should really write

$$S = S(t, a, b, S_0, I_0, R_0)$$

to tell a more complete story about the inputs that determine the output S. Most of the time, though, we do *not* want to draw attention to the parameters; we usually write just $S(t)$.

1.2.4.3 Further possibilities.

Steps I, II, and III on page are also functions, because they have well-defined input and output. They are unlike the other examples we have discussed up to this point because they have more than one output variable. You should see, though, that there is nothing more difficult going on here.

In our study of the S-I-R model it was natural not to separate functions that have one input variable from those that have several. This is the pattern we shall follow in the rest of the course. In particular, we will want to deal with parameters, and we will want to understand how the quantities we are studying depend on those parameters.

1.2.5 The Beginnings of Calculus

While functions, graphs, and computers are part of the general fabric of mathematics, we can also abstract from the S-I-R model some important aspects of the calculus itself. The first of these is the idea of a **rate of change**. In this chapter we just assumed the idea was intuitively clear. However, there are some important questions not yet answered; for example, how do you deal with a quantity whose rate of change is itself always changing? These questions, which lead to the fundamental idea of a **derivative**, are taken up in chapter 3.

Rate equations—more commonly called **differential equations**—lie at the very heart of calculus. We will have much more to say about them, because many processes in the physical, biological, and social realms can be modelled by rate equations. In our analysis of the S-I-R model, we used rate equations to estimate future values by assuming that rates stay fixed

How the next three chapters are connected

for a whole day at a time. The discussion called "there and back again" on page points up the shortcomings of this assumption. In chapter 2 we will develop a procedure, called Euler's method, to address this problem. In chapter 4 we will return to differential equations in a general way, equipped with Euler's method and the concept of the derivative.

1.2.6 Exercises

Functions and Graphs.

1. Sketch the graph of each of the following functions. Label each axis, and mark a scale of units on it. For each line that you draw, indicate
i) its slope;
ii) its y-intercept;
iii) its x-intercept (where it crosses the x-axis).

(a) $y = -\frac{1}{2}x + 3$ c) $5x + 3y = 12$

(b) $y = (2x - 7)/3$

2. Graph the following functions. Put labels and scales on the axes.

(a) $V = .3Z - 1$; b) $W = 600 - P^2$.

3. Sketch the graph of each of the following functions. Put labels and scales on the axes. For each graph that you draw, indicate
i) its y-intercept;
ii) its x-intercept(s).

For part (d) you will need the **quadratic formula**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

for the roots of the **quadratic equation** $ax^2 + bx + c = 0$.

a) $y = x^2$
b) $y = x^2 + 1$
c) $y = (x+1)^2$
d) $y = 3x^2 + x - 1$

The next four questions refer to these functions:

$c(x, y) = 17$	a constant function
$j(z) = z$	the identity function
$r(u) = 1/u$	the reciprocal function
$D(p, q) = p - q$	the difference function
$s(y) = y^2$	the squaring function
$Q(v) = \dfrac{2v+1}{3v-6}$	a rational function

$$H(x) = \begin{cases} 5 & \text{if } x < 0 \\ x^2 + 2 & \text{if } 0 \le x < 6 \\ 29 - x & \text{if } 6 \le x \end{cases}$$

$$T(x, y) = r(x) + Q(y)$$

4. Determine the following values:

Table 1.2.11

$c(5, -3)$	$j(17)$	$c(a, b)$	$j(u^2 + 1)$
$j(c(3, -5))$	$s(1.1)$	$r(1/17)$	$Q(0)$
$Q(2)$	$Q(3/7)$	$D(5, -3)$	$D(-3, 5)$
$H(1)$	$H(7)$	$H(4)$	$H(H(H(-3)))$
$r(s(-4))$	$r(Q(3))$	$Q(r(3))$	$T(3, 7)$

5. True or false. Give reasons for your answers: if you say true, explain why; if you say false, give an example that shows why it is false.

(a) For every non-zero number x, $r(r(x)) = j(x)$.

(b) If $a > 1$, then $s(a) > 1$.

(c) If $a > b$, then $s(a) > s(b)$.

(d) For all real numbers a and b, $s(a + b) = s(a) + s(b)$.

(e) For all real numbers a, b, and c, $D(D(a, b), c)) = D(a, D(b, c))$.

6. Find all numbers x for which $Q(x) = r(Q(x))$.

7. The **natural domain** of a function f is the largest possible set of real numbers x for which $f(x)$ is defined. For example, the natural domain of $r(x) = 1/x$ is the set of all non-zero real numbers.

(a) Find the natural domains of Q and H.

(b) Find the natural domains of $P(z) = Q(r(z))$; $R(v) = r(Q(v))$.

(c) What is the natural domain of the function $W(t) = \sqrt{\dfrac{1-t^2}{t^2-4}}$?

Computer Graphing. The purpose of these exercises is to give you some experience using a "graphing package" on a computer. This is a program that will draw the graph of a function $y = f(x)$ whose formula you know. You must type in the formula, using the following symbols to represent the basic arithmetic operations:

Table 1.2.12

to indicate	**type**
addition	`+`
subtraction	`-`
multiplication	
division	`/`
an exponent	`^`

The caret "`^`" appears above the "`6`" on a keyboard (Shift-6).

Here is an example:

Table 1.2.13

to enter:	type:
$\dfrac{7x^5 - 9x^2}{x^3+1}$	`(7*x^5 - 9*x^2)/(x^3 + 1)`

The parentheses you see here are important. If you do not include them, the computer will interpret your entry as

$$7x^5 - \frac{9x^2}{x^3} + 1 = 7x^5 - \frac{9}{x} + 1 \neq \frac{7x^5 - 9x^2}{x^3+1}.$$

In some graphing packages, you do not need to use to indicate a multiplication. If this is true for the package you use, then you can enter the fractional expression above in the somewhat simpler form

`(7x^5 - 9x^2)/(x^3 + 1).`

To do the following exercises, follow the specific instructions for the graphing package you are using.

8. Graph the function $f(x) = .6x + 2$ on the interval $-4 \le x \le 4$.

(a) What is the y-intercept of this graph? What is the x-intercept?

(b) Read from the graph the value of $f(x)$ when $x = -1$ and when $x = 2$. What is the difference between these y values? What is the difference between the x values? According to these differences, what is the slope of the graph? According to the *formula*, what is the slope?

9. Graph the function $f(x) = 1 - 2x^2$ on the interval $-1 \le x \le 1$.

(a) What is the y-intercept of this graph? The graph has two x-intercepts; use algebra to find them.

You can also find an x-intercept using the computer. The idea is to

magnify the graph near the intercept until you can determine as many decimal places in the x coordinate as you want. For a start, graph the function on the interval $0 \le x \le 1$. You should be able to see that the graph on your computer monitor crosses the x-axis somewhere around .7. Regraph $f(x)$ on the interval $.6 \leq x \leq .8$. You should then be able to determine that the x-intercept lies between .70 and .71. This means $x = .7...$; that is, you know the location of the x-intercept to one decimal place of accuracy.

(b) Regraph $f(x)$ on the interval $.70 \leq x \leq .71$ to get two decimal places of accuracy in the location of the x-intercept. Continue this process until you have at least 7 places of accuracy. What is the x-intercept?

The circular functions. Graphing packages "know" the familiar functions of trigonometry. Trigonometric functions are qualitatively different from the functions in the preceding problems. Those functions are defined by algebraic formulas, so they are called **algebraic functions**. The trigonometric functions are defined by explicit "recipes," but *not* by algebraic formulas; they are called

transcendental functions. For calculus, we usually use the definition of the trigonometric functions as **circular functions**. This definition begins with a unit circle centered at the origin. Given the input number t, locate a point P on the circle by tracing an arc of length t along the circle from the point $(1, 0)$. If t is positive, trace the arc counterclockwise; if t is negative, trace it clockwise. Because the circle has radius 1, the arc of length t subtends a central angle of **radian** measure t.

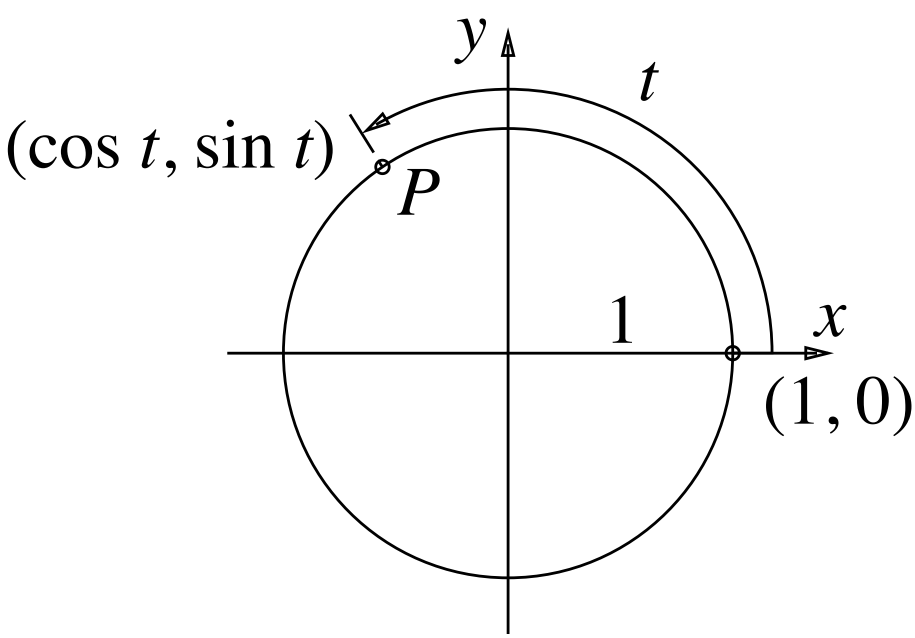


Figure 1.2.14 image

The circular (or trigonometric) functions $\cos t$ and $\sin t$ are defined as the coordinates of the point P,

$$P = (\cos t, \sin t).$$

The other trigonometric functions are defined in terms of the sine and cosine:

$$\begin{aligned} &tant = \sin t/\cos t, && \sec t = 1/\cos t, \\ &\cot t = \cos t/\sin t, && \csc t = 1/\sin t. \end{aligned}$$

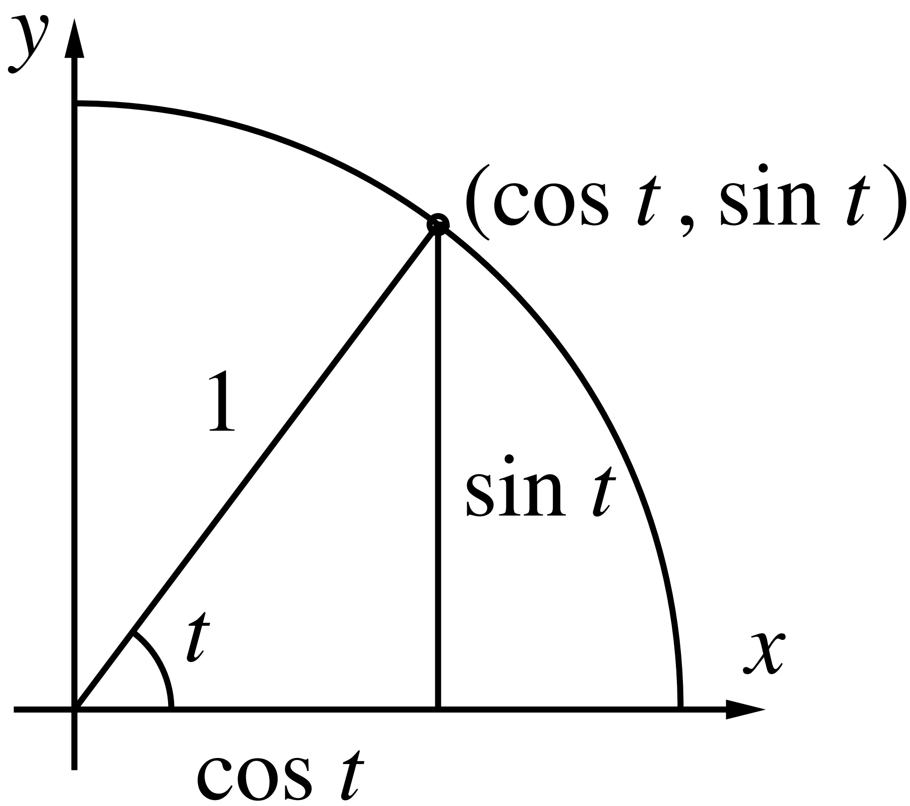


Figure 1.2.15 image

Notice that when t is a positive acute angle, the circle definition agrees with the right triangle definitions of the sine and cosine:

$$\sin t = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \text{and} \qquad \cos t = \frac{\text{adjacent}}{\text{hypotenuse}}.$$

However, the circle definitions of the sine and cosine have the important advantage that they produce functions whose domains are the set of *all* real numbers. (What are the domains of the tangent, secant, cotangent and cosecant functions?)

In calculus, angles are also always measured in radians. To convert between radians and degrees, notice that the circumference of a unit circle is 2π, so the radian measure of a semi-circular arc is half of this, and thus we have

$$\pi \text{ radians } = 180 \text{ degrees.}$$

As the course progresses, you

Simplicity determines the choice of radians for measuring angles

will see why radians are used rather than degrees (or mils or any other unit for measuring angles)—it turns out that the formulas important to calculus take their simplest form when angles are expressed in radians.

Graphing packages "know" the trigonometric functions in exactly this form: circular functions with the input variable given in radians. You might wonder, though, how a computer or calculator "knows" that $\sin(1) = .017452406\ldots$. It certainly isn't drawing a very accurate circle somewhere and measuring the y coordinate of some point. While the circular function approach is

How do we get values for the circular functions?

a useful way to think about the trigonometric functions conceptually, it isn't very helpful if we actually want values of the functions. One of the achievements of calculus, as you will see later in this course, is that it provides effective methods for computing values of functions like the circular functions that aren't given by algebraic formulas.

The following exercises let you review the trigonometric functions and explore some of the possibilities using computer graphing.

10. Graph the function $f(x) = \sin(x)$ on the interval $-2 \le x \le 10$.

(a) What are the x-intercepts of $\sin(x)$ on the interval $-2 \le x \le 10$? Determine them to two decimal places accuracy.

(b) What is the largest value of $f(x)$ on the interval $-2 \le x \le 10$? Which value of x makes $f(x)$ largest? Determine x to two decimal places accuracy.

(c) Regraph $f(x)$ on the very small interval $-.001 \le x \le .001$. Describe what you see. Can you determine the slope of this graph?

11. Graph the function $f(x) = \cos(x)$ on the interval $0 \le x \le 14$. On the same screen graph the *second* function $g(x) = \cos(2x)$.

(a) How far apart are the x-intercepts of $f(x)$? How far apart are the x-intercepts of $g(x)$?

(b) The graph of $g(x)$ has a pattern that repeats. How wide is this pattern? The graph of $f(x)$ also has a repeating pattern; how wide is *it*?

(c) Compare the graphs of $f(x)$ and $g(x)$ to one another. In particular, can you say that one of them is a stretched or compressed version of the other? Is the compression (or stretching) in the vertical or the horizontal direction?

(d) Construct a *new* function $f(x)$ whose graph is the same shape as the graph of $g(x) = \cos(2x)$, but make the graph of $f(x)$ twice as tall as the graph of $g(x)$. [A suggestion: either deduce what $f(x)$ should be, or make a guess. Then test your choice on the computer. If your choice doesn't work, think how you might modify it, and then test your modifications the same way.]

12. The aim here is to find a solution to the equation $\sin x = \cos(3x)$. There is no purely *algebraic* procedure to solve this equation. Because the sine and cosine are not defined by *algebraic* formulas, this should not be particularly surprising. (Even for algebraic equations, there are only a few very special cases for which there are formulas like the quadratic formula. In chapter 5 we will look at a method for solving equations when formulas can't help us.)

(a) Graph the two functions $f(x) = \sin(x)$ and $g(x) = \cos(3x)$ on the interval $0 \le x \le 1$.

(b) Find a solution of the equation $\sin(x) = \cos(3x)$ that is accurate to six decimal places.

(c) Find *another* solution of the equation $\sin(x) = \cos(3x)$, accurate to four decimal places. Explain how you found it.

13. Use a graphing program to make a sketch of the graph of each of the following functions. In each case, make clear the domain and the range of the function, where the graph crosses the axes, and where the function has a maximum or a minimum.

a) $F(w) = (w-1)(w-2)(w-3)$ b) $Q(a) = \dfrac{1}{a^2+5}$

[3pt]c) $E(x) = x + \dfrac{1}{x}$ d) $e(x) = x - \dfrac{1}{x}$

[3pt]e) $g(u) = \sqrt{\dfrac{u-1}{u+1}}$ f) $M(u) = \dfrac{u^2-2}{u^2+2}$

14. Graph on the same screen the following three functions:

$$f(x) = 2^x, \qquad g(x) = 3^x, \qquad h(x) = 10^x.$$

Use the interval $-2 \leq x \leq 1.2$.

(a) Which function has the largest value when $x = -2$?

(b) Which is climbing most rapidly when $x = 0$?

(c) Magnify the picture at $x = 0$ by resetting the size of the interval to $-.0001 \leq x \leq .0001$. Describe what you see. Estimate the slopes of the three graphs.

Proportions, Linear Functions, and Models.

15. Go back to the three functions given in problem 1. For each function, choose an initial value x_0 for x, find the corresponding value y_0 for y, and express the function in the form $y - y_0 = m \cdot (x - x_0)$.

16. You should be able to answer all parts of this problem without ever finding the equations of the functions involved.

(a) Suppose $y = f(x)$ is a linear function with multiplier $m = 3$. If $f(2) = -5$, what is $f(2.1)$? $f(2.0013)$? $f(1.87)$? $f(922)$?

(b) Suppose $y = G(x)$ is a linear function with multiplier $m = -2$. If $G(-1) = 6$, for what value of x is $G(x) = 8$? $G(x) = 0$? $G(x) = 5$? $G(x) = 491$?

(c) Suppose $y = h(x)$ is a linear function with $h(2) = 7$ and $h(6) = 9$. What is $h(2.046)$? $h(2 + a)$?

17. In Massachusetts there is a sales tax of 5%. The tax T, in dollars, is proportional to the price P of an object, also in dollars. The constant of proportionality is $k = 5\% = .05$. Write a formula that expresses the sales tax as a linear function of the price, and use your formula to compute the tax on a television set that costs \$289.00 and a toaster that costs \$37.50.

18. Suppose $W = 213 - 17\,Z$. How does W change when Z changes from 3 to 7; from 3 to 3.4; from 3 to 3.02? Let ΔZ denote a change in Z and ΔW the change thereby produced in W. Is $\Delta W = m\,\Delta Z$ for some constant m? If so, what is m?

19. In the following table, q is a linear function of p. Fill in the blanks in the table.

Table 1.2.16

(a) Find a formula to express Δq as a function of Δp, and another to express q as a function of p.

20. Thermometers. There are two scales in common use to measure the temperature, called **Fahrenheit degrees** and the **Celsius degrees**. Let F and C, respectively, be the temperature on each of these scales. Each of these quantities is a linear function of the other; the relation between them in determined by the following table:

Table 1.2.17

physical measurement	C	F
freezing point of water	0	32
boiling point of water	100	212

(a) Which represents a larger change in temperature, a Celsius degree or a Fahrenheit degree?

(b) How many Fahrenheit degrees does it take to make the temperature go up one Celsius degree? How many Celsius degrees does it take to make it go up one Fahrenheit degree?

(c) What is the multiplier m in the equation $\Delta F = m \cdot \Delta C$? What is the multiplier μ in the equation $\Delta C = \mu \cdot \Delta F$? (The symbol μ is the Greek letter *mu*.) What is the relation between μ and m?

(d) Express F as a linear function of C. Graph this function. Put scales and labels on the axes. Indicate clearly the slope of the graph and its vertical intercept.

(e) Express C as a linear function of F and graph this function. How are the graphs in parts (d) and (e) related? Give a clear and detailed explanation.

(f) Is there any temperature that has the same reading on the two temperature scales? What is it? Does the temperature of the air ever reach this value? Where?

21. The Greenhouse Effect. The concentration of carbon dioxide (CO_2) in the atmosphere is increasing. The concentration is measured in parts per million (PPM). Records kept at the South Pole show an increase of .8 PPM per year during the 1960s.

(a) At that rate, how many years does it take for the concentration to increase by 5 PPM; by 15 PPM?

(b) At the beginning of 1960 the concentration was about 316 PPM. What would it be at the beginning of 1970; at the beginning of 1980?

(c) Draw a graph that shows CO_2 concentration as a function of the time since 1960. Put scales on the axes and label everything clearly.

(d) The *actual* CO_2 concentration at the South Pole was 324 PPM at the beginning of 1970 and 338 PPM at the beginning of 1980.

Plot these values on your graph, and compare them to your calculated values.

(e) Using the actual concentrations in 1970 and 1980, calculate a new rate of increase in concentration. Using that rate, estimate what the increase in CO_2 concentration was between 1970 and 1990. Estimate the CO_2 concentration at the beginning of 1990.

(f) Using the rate of .8 PPM per year that held during the 1960s, determine how many years before 1960 there would have been *no* carbon dioxide at all in the atmosphere.

22. Thermal Expansion. Measurements show that the length of a metal bar increases in proportion to the increase in temperature. An aluminum bar that is exactly 100 inches long when the temperature is 40°F becomes 100.0052 inches long when the temperature increases to 80°F.

(a) How long is the bar when the temperature is 60°F? 100°F?

(b) What is the multiplier that connects an increase in length ΔL to an increase in temperature ΔT?

(c) Express ΔL as a linear function of ΔT.

(d) How long will the bar be when $T = 0$°F?

(e) Express L as a linear function of T.

(f) What temperature change would make $L = 100.01$ inches?

(g) For a *steel* bar that is also 100 inches long when the temperature is 40° F, the relation between ΔL and ΔT is $\Delta L = .00067\,\Delta T$. Which expands more when the temperature is increased; aluminum or steel?

(h) How long will this steel bar be when $T = 80$°F?

23. Falling Bodies. In the simplest model of the motion of a falling body, the velocity increases in proportion to the increase in the time that the body has been falling. If the velocity is given in feet per second, measurements show the constant of proportionality is approximately 32.

(a) A ball is falling at a velocity of 40 feet/sec after 1 second. How fast is it falling after 3 seconds?

(b) Express the change in the ball's velocity Δv as a linear function of the change in time Δt.

(c) Express v as a linear function of t.

The model can be expanded to keep track of the *distance* that the body has fallen. If the distance d is measured in feet, the units of d' are feet per second; in fact, $d' = v$. So the model describing the motion of the body is given by the rate equations

$$\begin{aligned} d' &= v \quad \text{feet per second;} \\ v' &= 32 \quad \text{feet per second per second.} \end{aligned}$$

(d) At what rate is the distance increasing after 1 second? After 2 seconds? After 3 seconds?

(e) Is d a linear function of t? Explain your answer.

In many cases, the rate of change of a variable quantity is proportional to the quantity itself. Consider a human population as an example. If a city of 100,000 is increasing at the rate of 1500 persons per year, we would expect a similar city of 200,000 to be increasing at the rate of 3000 persons per year. That is, if P is the population at time t, then the **net growth rate** P' is proportional to P:

$$P' = k\,P.$$

In the case of the two cities, we have

$$P' = 1500 = k\,P = k \times 100000 \quad \text{so} \quad k = \frac{1500}{100000} = .015.$$

24. In the equation $P' = k\,P$, above, explain why the units for k are

$$\frac{\text{persons per year}}{\text{person}}.$$

The number k is called the **per capita growth rate**. ("Per capita" means "per person"—"per *head*", literally.)

25. **Poland and Afghanistan**. In 1985 the per capita growth rate in Poland was 9 persons per year per thousand persons. (That is, $k = 9/1000 = .009$.) In Afghanistan it was 21.6 persons per year per thousand.

(a) Let P denote the population of Poland and A the population of Afghanistan. Write the equations that govern the growth rates of these populations.

(b) In 1985 the population of Poland was estimated to be 37.5 million persons, that of Afghanistan 15 million. What are the net growth rates P' and A' (as distinct from the *per capita* growth rates)? Comment on the following assertion: When comparing two countries, the one with the larger per capita growth rate will have the larger net growth rate.

(c) On the average, how long did it take the population to increase by one person in Poland in 1985? What was the corresponding time interval in Afghanistan?

26. **Bacterial Growth**. A colony of bacteria on a culture medium grows at a rate proportional to the present size of the colony. When the colony weighed 32 grams it was growing at the rate of 0.79 grams per hour. Write an equation that links the growth rate to the size of the population.

(a) What is ΔP if $\Delta t = 1$ minute? Estimate how long it would take to make $\Delta P = .5$ grams.

27. **Radioactivity**. In radioactive decay, radium slowly changes into lead. If one sample of radium is twice the size of a second lump, then the larger sample will produce twice as much lead as the second in any given time. In other words, the rate of decay is proportional to the amount of radium present. Measurements show that 1 gram of radium decays into lead at the rate of 1/2337 grams per year. Write an equation that links the decay rate to the size of the radium sample. How does your equation indicate that the process involves *decay* rather than *growth*?

28. **Cooling**. Suppose a cup of hot coffee is brought into a room at 70°F. It will cool off, and it will cool off *faster* when the temperature difference between the coffee and the room is greater. The simplest assumption we can make is that the rate of cooling is proportional to this temperature difference (this is called Newton's law of cooling). Let C denote the temperature of the coffee, in °F, and C' the rate at which it is cooling, in °F per minute. The new element here is that C' is proportional, not to C, but to the *difference* between C and the room temperature of 70°F.

(a) Write an equation that relates C' and C. It will contain a proportionality constant k. How did you indicate that the coffee is *cooling* and not *heating up*?

(b) When the coffee is at 180°F it is cooling at the rate of 9°F per minute. What is k?

(c) At what rate is the coffee cooling when its temperature is 120°F?

(d) Estimate how long it takes the temperature to fall from 180°F to 120°F. Then make a better estimate, and explain why it is better.

1.3 Using a Program

1.3.1 Computers

A computer changes the way we can use calculus as a tool, and it vastly enlarges the range of questions that we can tackle. No longer need we back away from a problem that involves a lot of computations. There are two aspects to the power of a computer. First, it is fast. It can do a million additions in the time it takes us to do one. Second, it can be programmed. By arranging computations into a loop—as we did on page —we can construct a program with only a few instructions that will carry out millions of repetitive calculations.

The purpose of this section is to give you practice using a computer program that estimates values of S, I, and R in the epidemic model. As you will see, it carries out the three rounds of calculations you have already done by hand. It also contains a loop that will allow you to do a hundred, or a million, rounds of calculations with no extra effort.

1.3.1.1 The Program SIR

The program below calculates values of S, I, and R. It is a set of instructions—sometimes called **code**—that is designed to be read by you and by a computer. These instructions mirror the operations we performed by hand to generate the table on page .

Read a program line by line from the top

```
t = 0
S = 45400
I = 2100
R = 2500
deltat = 1
```

```
PRINT t, S, I, R
FOR k = 1 TO 3
        Sprime = -.00001 * S * I
        Iprime = .00001 * S * I - I / 14
        Rprime = I / 14
        deltaS = Sprime * deltat
        deltaI = Iprime * deltat
        deltaR = Rprime * deltat
        t = t + deltat
        S = S + deltaS
        I = I + deltaI
        R = R + deltaR
        PRINT t, S, I, R
NEXT k
```

The code here is similar to what it would be in most programming languages. The line numbers, however, are not part of the program; they are there to help us refer to the lines. A computer reads the code one line at a time, starting at the top. Each line is a complete instruction which causes the computer to do something. The purpose of nearly every instruction in this program is to assign a numerical value to a symbol. Watch for this as we go down the lines of code.

The first line, `t = 0`,

Lines 1–5

is the instruction "Give t the value 0." The next four lines are similar. Notice, in the fifth line, how Δt is typed out as `deltat`. It is a common practice for the name of a variable to be several letters long. A few lines later S' is typed out as `Sprime`, for instance. The instruction on the sixth line

Line 6

is the first that does not assign a value to a symbol. Instead, it causes the computer to print the following on the computer monitor screen:

`0`

Skip over the line that says `FOR k = 1 TO 3`.

Line 7

It will be easier to understand after we've read the rest of the program.

Look at the first three indented lines.

Lines 8–10

You should recognize them as coded versions of the rate equations

$$\begin{aligned} S' &= -.00001\, SI \\ I' &= .00001\, SI - I/14 \\ R' &= I/14 \end{aligned}$$

for the measles epidemic. (The program uses to denote multiplication.) They are instructions to assign numerical values to the symbols S', I', and R'. For instance, `Sprime = -.00001 * S * I` (line 8) says

Give S' the value $-.00001\, SI$;use the current values of S and I to get $-.00001\, SI$.

Now the computer knows that the current values of S and I are 45400 and 2100, respectively. (Can you see why?) So it calculates the product $-.00001 \times 45400 \times 2100 = -953.4$ and then gives S' the value -953.4. There is an extra step to calculate the product.

Notice that the first three indented lines are bracketed together and labelled "Step I," because they carry out Step I in the flow chart.

Lines 11–13

The next three indented lines carry out Step II in the flow chart. They assign values to three more symbols—namely ΔS, ΔI, and ΔR—using the current values of S', I', R' and Δt.

The next four indented lines present a puzzle.

Lines 14–17

They don't make sense if we read them as ordinary mathematics. For example, in an expression like `t = t + deltat`, we would cancel the `t`'s and conclude `deltat = 0`. The lines *do* make sense when we read them as computer instructions, however. As a computer instruction, `t = t + deltat` says

Make the new value of t equal to the current value of $t + \Delta t$.

(To make this clear, some computer languages express this instruction in the form `let t = t + deltat`.) Once again we have an instruction that assigns a numerical value to a symbol, but this time the symbol (t, in this case) already has a value before the instruction is carried out. The instruction gives it a *new* value. (Here the value of t is changed from 0 to 1.) Likewise, the instruction `S = S + deltaS` gives S a new value. What was the old value, and what is the new?

Compare the three lines of code

How a program computes new values

that produce new values of S, I, and R with the original equations that we used to define Step III back on page :

`S = S + deltaS`	new $S =$ current $S + \Delta S$,
`I = I + deltaI`	new $I =$ current $I + \Delta I$,
`R = R + deltaR`	new $R =$ current $R + \Delta R$.

The words "new" and "current" aren't needed in the computer code because they are automatically understood to be there. Why? First of all, a symbol (like `S`) always has a *current* value, but an instruction can give it a *new* value. Second, a computer instruction of the form `A = B` is always understood to mean "new `A` = current `B`."

Notice that the instructions `A = B` and `B = A` mean different things. The second says "new `B` = current `A`." Thus, in `A = B`, `A` is altered to equal `B`, while in `B = A`, `B` is altered to equal `A`. To emphasize that the symbol on the left is always the one affected, some programming languages use a modified equal sign, as in `A := B`. We sometimes read this as "`A` gets `B`".

The next line is another `PRINT` statement,

Line 18

exactly like the one on line 6. It causes the current values of t, S, I, and R to be printed on the computer monitor screen. But this time what appears is

```
1
```

The values were changed by the previous four instructions. It is important to remember that the computer carries out instructions in the order they are written. Had the second `PRINT` statement appeared right after line 13, say, the old values of t, S, I, and R would have appeared on the monitor screen a second time.

We will take the last line and line 7 together.

Lines 7 and 19: the loop

They are the instructions for the loop. Consider the situation when we reach the last line. The variables t, S, I, and R now have their "day 1" values.

To continue, we need an instruction that will get us back to line 8, because the instructions on lines 8–17 will convert the current (day 1) values of t, S, I, and R into their "day 2" values. That's what lines 7 and 19 do.

Here is the meaning of the instruction `FOR k = 1 TO 3` on line 7:

Give k the value 1,

```
FOR k = 1 TO 3
```

and be prepared later togive it the value 2 and then the value 3.

The variable k plays the role of a **counter**, telling us how many times we have gone around the loop. Notice that k did not appear in our hand calculations. However, when we said we had done three rounds of calculations, for example, we were really saying $k = 3$.

After the computer reads and executes line 7, it carries out all the instructions from lines 8 to 18, arriving finally at the last line. The computer then interprets the instruction `NEXT k` on the last line as follows:

Give k the next value

```
NEXT k
```

that the `FOR command allows, and move back to the line immediately after the FOR command.`

After the computer carries out this instruction, k has the value 2 and the computer is set to carry out the instruction on line 8. It then executes that instruction, and continues down the program, line by line, until it reaches line 19 once again. This sets the value of k to 3 and moves the computer back to line 8. Once again it continues down the program to line 19.

How the program stops

This time there is no allowable value that k can be given, so the program stops.

The `NEXT k` command is different from all the others in the program. It is the only one that directs the computer to go to a different line. That action causes the program to **loop**. Because the loop involves all the indented instructions between the `FOR` statement and the `NEXT` statement, it is called a **FOR–NEXT loop**. This is just one kind of loop. Computer programs can contain other sorts that carry out different tasks. In the next chapter we will see how a **DO–WHILE** loop is used.

1.3.2 Exercises

The program SIR. The object of these exercises is to verify that the program SIR works the way the text says it does. Follow the instructions for running a program on the computer you are using.

1. Run the program to confirm that it reproduces what you have already calculated by hand (table, page).

2. On a copy of the program, mark the instructions that carry out the following tasks:

(a) give the input values of S, I, and R;

(b) say that the calculations take us 1 day into the future;

(c) carry out step II (see page);

(d) carry out step III;

(e) give us the output values of S, I, and R;

(f) take us once around the whole loop;

(g) say how many times we go around the loop.

3. Delete all the lines of the program from line 7 onward (or else type in the first 6 lines). Will this program run? What will it do? Run it and report what you see. Is this what you expected?

4. Starting with the original SIR program on page , delete lines 7 and 19. These are the ones that declare the FOR–NEXT loop. Will this program run? What will it do? Run it and report what you see. Is this what you expected?

5. Using the 17-line program you constructed in the previous question, remove the PRINT statement from the last line and insert it between what appear as lines 13 and 14 on page . Will this program run? What will it do? Run it and report what you see. Is this what you expected?

6. Starting with the original SIR program on page , change line 7 so it reads `FOR k = 26 TO 28`. Thus, the counter k takes the values 26, 27, and 28. Will this program run? What will it do? Run it and report what you see. Is this what you expected?

Programs to practice on. In this section there are a number of short programs for you to analyze and run.

7. When Program 1 runs it will print the values of A and B that are current when the program stops. What values will it print? Type in this program and run it to verify your answers.

8. What will Program 2 do when it runs? Type in this program and run it to verify your answers.

9. After each line in Program 3 write the values that A and B have *after* that line has been carried out. What values of A and B will it print? Type in this program and run it to verify your answers.

The next three programs have an element not found in the program SIR. In each of them, there is a FOR–NEXT loop, and the counter k actually appears in the statements within the loop.

10. What output does Program 4 produce? Type in the code and run the program to confirm your answer.

11. What is the difference between the code in Program 5 and the code in Program 4? What is the output of Program 5? Does it differ from the output of Program 4? If so, why?

12. What output does Program 6 produce? Type in the code and run the program to confirm your answer.

13. Program 7 prints five lines of output. What are they? Type in the program and run it to confirm your answers.

14. What is the output of Program 8? Type in the program and run it to confirm your answers.

15. Describe exactly how the *codes* for Programs 8 and 9 differ. How do the *outputs* differ?

Analyzing the measles epidemic.

16. Alter the program SIR to have it calculate estimates for S, I, and R over the first *six* days. Construct a table that shows those values.

17. Alter the program to have it estimate the values of S, I, and R for the first *thirty* days.

(a) What are the values of S, I, and R when $t = 30$?

(b) According to these figures, on what day does the infection peak? What values do you get for S, I, and R?

(c) How can you reconcile the value you just got for S with the value we obtained algebraically on page ?

18. By adding an appropriate PRINT statement after line 10 you can also get the program to print values for S', I', and R'. Do this, and check that you get the values shown in the table on page .

19. According to these estimates, on what day do the largest number new infections occur? How many are there? Explain where you got your information.

20. On what day do you estimate that the largest number of recoveries occurs? Do you see a connection between this question and 17 (b)?

21. On what day do you estimate the infected population grows most rapidly? Declines most rapidly? What value does I' have on those days?

22. Alter the original SIR program so that it will go *backward* in time, with time steps of 1 day. Specify the changes you made in the program. Use this altered program to obtain estimates for the values of S, I, and R yesterday. Compare your estimates with those in the text (page).

(a) Estimate the values of S, I, and R *three* days before today.

23. According to the S-I-R model, when did the infection begin? That is, how many days before today was the estimated value of I approximately 1?

24. **There and back again**. Use the SIR program, modified as necessary, to carry out the calculations described in exercise 18 on page . Do your computer results agree with those you obtained earlier?

25. In exercises 22–26 at the end of section 2 (pages –) you set up rate equations to model some other systems. Choose a couple of these and think of some interesting questions that could be answered using a suitably modified SIR program. Make the modifications and report your results.

1.4 Chapter Summary

1.4.1 The Main Ideas

- Natural processes like the spread of disease can often be described by **mathematical models**. Initially, this involves identifying **numerical quantities** and relations between them.

- A relation between quantities often takes the form of a **function**. A function can be described in many different ways; **graphs**, **tables**, and **formulas** are among the most common.

- **Linear functions** make up a special but important class. If y is a linear function of x, then $\Delta y = m \cdot \Delta x$, for some constant m. The constant m is a **multiplier**, **slope**, and **rate of change**.

- If $y = f(x)$, then we can consider the **rate of change** y' of y with respect to x. A mathematical model whose variables are connected by **rate equations** can be analyzed to predict how those variables will change.

- Predicted changes are **estimates** of the form $\Delta y = y' \cdot \Delta x$.

- The computations that produce estimates from rate equations can be put into a **loop**, and they are readily carried out on a **computer**.

- A computer increases the **scope** and **complexity** of the problems we can consider.

1.4.2 Expectations

- You should be able to work with functions given in various forms, to find the output for any given input.

- You should be able to read a graph. You should also be able to construct the graph of a linear function directly, and the graph of a more complicated function using a computer graphing package.

- You should be able to determine the natural domain of a function given by a formula.

- You should be able to express proportional quantities by a linear function, and interpret the constant of proportionality as a multiplier.

- Given any two of these quantities for a linear function—multiplier, change in input, change in output—you should be able to determine the third.

- You should be able to model a situation in which one variable is proportional to its rate of change.

- Given the value of a quantity that depends on time and given its rate of change, you should be able to estimate values of the quantity at other times.

- For a set of quantities determined by rate equations and initial conditions, you should be able to estimate how the quantities change.

- Given a set of rate equations, you should be able to determine what happens when one of the quantities reaches a maximum or minimum, or remains unchanged over time.

- You should be able to understand how a computer program with a FOR–NEXT loop works.

1.4.3 Chapter Exercises

Exercises

A Model of an Orchard. If an apple orchard occupies one acre of land, how many trees should it contain so as to produce the largest apple crop? This is an example of an **optimization** problem. The word *optimum* means "best possible"—especially, the best under a given set of conditions. These exercises seek an optimum by analyzing a simple mathematical model of the orchard. The model is the function that describes how the total yield depends on the number of trees.

An immediate impulse is just to plant a lot of trees, on the principle: more trees, more apples. But there is a catch: if there are too many trees in a single acre, they crowd together. Each tree then gets less sunlight and nutrients, so it produces fewer apples. For example, the relation between the *yield per tree*, Y, and the *number of trees*, N, may be like that shown in the graph drawn on the left, below.

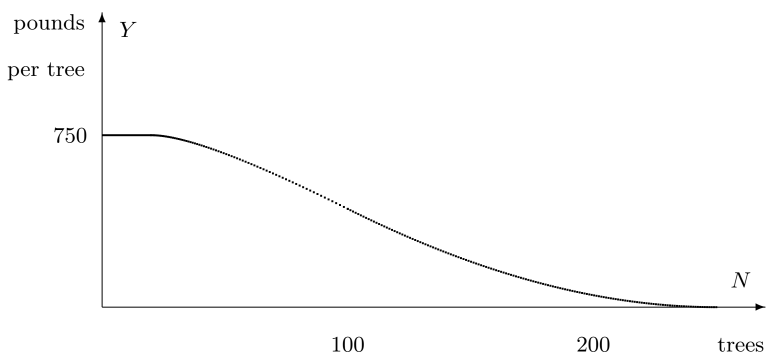


Figure 1.4.1

When there are only a few trees, they don't get in each other's way, and they produce at the maximum level—say, 750 pounds per tree. Hence the graph starts off level. At some point, the trees become too crowded to produce *anything*! In between, the yield per tree drops off as shown by the curved middle part of the graph.

We want to choose N so that the *total yield*, T, will be as large as possible. We have $T(N) = Y(N) \cdot N$, but since we don't know $Y(N)$ very precisely, it is difficult to analyze $T(N)$. To help, let's replace $Y(N)$ by the approximation shown in the graph on the right. Now carry out an analysis using this graph to represent $Y(N)$.

1. Find a formula for the straight segment of the new graph of $Y(N)$ on the interval $40 \leq N \leq 180$. What is the formula for $T(N)$ on the same interval?
2. What are the formulas for $T(N)$ when $0 \leq N \leq 40$ and when $180 \leq N$? Graph T as a function of N. Describe the graph in words.
3. What is the maximum possible total yield T? For which N is this maximum attained?
4. Suppose the endpoints of the sloping segment were P and Q, instead of 40 and 180, respectively. Now what is the formula for $T(N)$? (Note

that P and Q are *parameters* here. Different values of P and Q will give different models for the behavior of the total output.) How many trees would then produce the maximum total output? Expect the maximum to depend on the parameters P and Q.

Rate Equations. Do the following exercises by hand. You may wish to check your answers by using suitable modifications of the program SIR.

5. Radioactivity. From exercise 25 of section 2 we know a sample of R grams of radium decays into lead at the rate

$$R' = \frac{-1}{2337} R \quad \text{grams per year.}$$

Using a step size of 10 years, estimate how much radium remains in a 0.072 gram sample after 40 years.

6. Poland and Afghanistan. If P and A denote the populations of Poland and Afghanistan, respectively, then their net per capita growth rates imply the following equations:

$$P' = .009\,P \quad \text{persons per year;}$$
$$A' = .0216\,A \quad \text{persons per year.}$$

(See exercise 23 of section 2.) In 1985, $P = 37.5$ million, $A = 15$ million. Using a step size of 1 year, estimate P and A in 1990.

7. Falling bodies. If d and v denote the distance fallen (in feet) and the velocity (in feet per second) of a falling body, then the motion can be described by the following equations:

$$d' = v \quad \text{feet per second;}$$
$$v' = 32 \quad \text{feet per second per second.}$$

(See exercise 21 of section 2.) Assume that when $t = 0$, $d = 0$ feet and $v = 10$ feet/sec. Using a step size of 1 second, estimate d and v after 3 seconds have passed.

Chapter 2

Successive Approximations

In this chapter we continue exploring the mathematical implications of the S-I-R model. In the last chapter we calculated future values of S, I, and R by assuming that the rates S', I', and R' stayed fixed for a whole day. Since the rates are *not* fixed—they change with S, I, and R—the values of S, I, and R we obtained have to be considered as estimates only. In this chapter we will see how to build a succession of better and better estimates that get us as close as we wish to the true values implied by the model.

This method of **successive approximation** is a basic tool of calculus. It is the one fundamentally new process you will encounter, the ingredient that sets calculus apart from the mathematics you have already studied. With it you will be able to solve a vast array of problems that other methods can't handle.

2.1 Making Approximations

In chapter 1 we looked at the specific S-I-R model:

$$\begin{aligned} S' &= -.00001\,SI, \\ I' &= .00001\,SI - I/14, \\ R' &= I/14, \end{aligned}$$

with initial values at time $t = 0$:

$$S = 45400, \quad I = 2100, \quad R = 2500.$$

We originally developed this model as a *description* of the relations among the different components of an epidemic. Almost immediately, though, we began using the rate equations in the model as a *recipe* for predicting what happens over the course of the epidemic: If we know at some time t the values of $S(t), I(t),$ and $R(t)),$

Rate equations tell us where to go next

then the equations tell us how to estimate values of the functions at other times. We used this approach in the last chapter to move backwards and forwards in time, calculating the values of S, I, and R as we went.

While we got numbers, there were some questions about how accurate these numbers were—that is, how exactly they represented the values implied by the model. In the process we called "there and back again" we used current values of S, I, and R to find the rates, used these rates to go forward one day, recalculate the rates, and come back to the present—and we got different values from the ones we started with! Resolving this discrepancy will be an important feature of the technique developed in this section.

2.1.1 The Longest March Begins with a Single Step

So far, in generating numbers from the S-I-R rate equations, we have assumed that the rates remained constant over an entire day, or longer. Since the rates aren't constant—they depend on the values of S, I, and R, which are always changing—the values we calculated for the variables at times other than the given initial time are, at best, estimates. These estimates, while incorrect, are not useless. Let's see how they behave in the "there and back again" process of chapter 1 as we recalculate the rates more and more frequently, producing a sequence of approximations to the values we are looking for.

There and Back Again Again. On page in chapter 1 we used the rate equations to go forward a day and come back again. We started with the initial values

$$S(0) = 45400, \quad I(0) = 2100, \quad R(0) = 2500,$$

calculated the rates, went forward a day to $t = 1$, recalculated the rates, and came back a day to $t = 0$. We ended up with the estimates

$$S(0) = 45737.1, \quad I(0) = 1820.3, \quad R(0) = 2442.6$$

—which are rather far from the values of $S(0)$, $I(0)$, and $R(0)$ we started with.

A clue to the resolution of this discrepancy appeared in problem 18, page . There you were asked to go forward two days and come back again in two different ways, using $\Delta t = 2$ (a total of 2 steps) in the first case and $\Delta t = 1$ (a total of 4 steps) in the second. Here are the resulting values calculated for $S(0)$ in each case and the discrepancy between this value and the original value $S(0) = 45400$:

Table 2.1.1

step size	**new $S(0)$**	**discrepancy**
$\Delta t = 2$	46717.6	1317.6
$\Delta t = 1$	46021.3	621.3

While the discrepancy is fairly large in either case, $\Delta t = 1$ clearly does better than $\Delta t = 2$. But if smaller

Smaller steps generate a smaller discrepancy

is better, why stop at $\Delta t = 1$? What happens if we take even smaller time steps, get the corresponding new values of S, I, and R, and use these values to recalculate the rates each time?

Recall that the rate S' (or I' or R') is simply the multiplier which gives ΔS—the (estimated) change in S—for a given change Δt in t

$$\Delta S = S' \cdot \Delta t\,.$$

This relation holds for any value of Δt, integer or not. Once we have this value for ΔS, we can then calculate

$$\text{new (estimated) } S = \text{current (estimated) } S + \Delta S$$

in the usual way. Note that we have written "(estimated)" throughout to emphasize the fact that if S' is not constant over the entire time Δt, then the value we get for ΔS will typically be only an approximation to the real change in S.

Let's try going forward one day and coming back, using different values for Δt. As we reduce Δt the number of calculations will increase. The program SIR we used in the last chapter can still be used to do the tedious calculations. Thus if we decide to use 10 steps of size $\Delta t = .1$, we would just change two lines in that program:If we now run SIR with these modifications we can verify the following sequence of values (The values have been rounded off, and the `PRINT` statement has been modified to show the new values of the rates at each step as well):

Estimated values of $S, I,$ and R for step sizes $\Delta t = .1$

Table 2.1.2

t	$S(t)$	$I(t)$	$R(t)$	$S'(t)$	$I'(t)$	$R'(t)$
0.0	45 400.0	2100.0	2500.0	–953.4	803.4	150.0
0.1	45 304.7	2180.3	2515.0	–987.8	832.1	155.7
0.2	45 205.9	2263.6	2530.6	–1023.3	861.6	161.7
0.3	45 103.6	2349.7	2546.7	–1059.8	892.0	167.8
1.0	44 278.7	3042.9	2678.4	–1347.7	1130.0	217.4

Having arrived at $t = 1$, we can now use SIR to turn around and go back to $t = 0$. Here's how:

- Change the initial line of the program to `t = 1` to reflect our new starting time.
- Change the next three lines to use the values we just calculated for $S(1)$, $I(1)$, and $R(1)$ as our *starting* values in

  The *sign* of `deltat` determines whether we move forward or backward in time

  SIR.
- Change the value of `deltat` to be `-.1` (so each time the program executes the command `t = t + deltat` it reduces the value of `t` by .1).

With these changes SIR will yield the desired estimates for $S(0)$, $I(0)$, and $R(0)$, and we get

$$S(0) = 45433.5, \quad I(0) = 2072.3, \quad R(0) = 2494.3.$$

This is clearly a considerable improvement over the values obtained with $\Delta t =$ 1.

With this promising result, the obvious thing to do is to try even smaller values of Δt, perhaps $\Delta t = .01$. We could continue using SIR, making the needed modifications each time. Instead, though, let's rewrite SIR slightly to make it better suited to our current needs. Look at the program SIRVALUE below and compare it with SIR.

Program: SIRVALUE

```
tinitial = 0
tfinal = 1
t = tinitial
S = 45400
```

```
I = 2100
R = 2500
numberofsteps = 10
deltat = (tfinal - tinitial)/numberofsteps
FOR k = 1 TO numberofsteps
      Sprime = -.00001 * S * I
      Iprime = .00001 * S * I - I / 14
      Rprime = I / 14
      deltaS = Sprime * deltat
      deltaI = Iprime * deltat
      deltaR = Rprime * deltat
      t = t + deltat
      S = S + deltaS
      I = I + deltaI
      R = R + deltaR
NEXT k
PRINT t, S, I, R
```

Program: SIR

```
t = 0
S = 45400
I = 2100
R = 2500
deltat = .1
FOR k = 1 TO 10
      Sprime = -.00001 * S * I
      Iprime = .00001 * S * I - I / 14
      Rprime = I / 14
      deltaS = Sprime * deltat
      deltaI = Iprime * deltat
      deltaR = Rprime * deltat
      t = t + deltat
      S = S + deltaS
      I = I + deltaI
      R = R + deltaR
      PRINT t, S, I, R
NEXT k
```

You will see that the major change is to place the `PRINT` statement outside the loop, so only the final values of S, I, and R get printed. This speeds up the work, since otherwise, with $\Delta t = .001$, for instance, we would be asking the computer to print out 1000 lines—about 30 screens of text! Another change is that the value of `deltat` no longer needs to be specified—it is automatically determined by the values of `tinitial`, `tfinal`, and `numberofsteps`.

As written above, SIR and SIRVALUE both run for 10 steps of size 0.1 . By changing the value of `numberofsteps` in the program we can quickly get estimates

With a computer we can generate lots of data and look for patterns

for $S(1)$, $I(1)$, and $R(1)$ for a wide range of values for Δt . Moreover, once we have these estimates we can use SIRVALUE again to go backwards in time to $t = 0$, by making changes similar to those we made in SIR earlier. First, we need to change the value of `tinitial` to 1 and the value of `tfinal` to 0. Notice that this automatically will make `deltat` a negative quantity, so that each time we run through the loop we step back in time. Second, we need to set the starting values of `S, I,` and `R` to the values we just obtained for $S(1)$, $I(1)$, and $R(1)$. With these changes, SIRVALUE will give us the corresponding estimated values for $S(0)$, $I(0)$, and $R(0)$.

If we use SIRVALUE with Δt ranging from 1 to .00001 (which means letting `numberofsteps` range from 1 to 100,000) we get the table below. This table lists the computed values of $S(1)$, $I(1)$, and $R(1)$ for each Δt, followed by the estimated value of $S(0)$ obtained by running SIRVALUE backward in time from these new values, and, finally, the discrepancy between this estimated value of $S(0)$ and the original value $S(0) = 45400$.

Estimated values of S, I, and R when $t = 1$,for step sizes $\Delta t = 10^{-N}$, $N = 0, \ldots 5$,together with the corresponding backwards estimate for $S(0)$.

Table 2.1.3

Δt	$S(1)$	$I(1)$	$R(1)$	**new** $S(0)$	**discrepancy**
1.0	44 446.6	2903.4	2650.0	45 737.0626	337.0626
0.1	44 278.6648	3042.9241	2678.4111	45 433.4741	33.4741
0.01	44 257.8301	3060.1948	2681.9751	45 403.3615	3.3615
0.001	44 255.6960	3061.9633	2682.3406	45 400.3363	.3363
0.000 1	44 255.4821	3062.1406	2682.3773	45 400.0336	.0336
0.000 01	44 255.4607	3062.1584	2682.3809	45 400.0034	.0034

There are several striking features of this table. The first is that if we go forward one day and come back again, we can get back as close as

Smaller steps generate a discrepancy which can be made as small as we like

we want to our initial value of $S(0)$ *provided we recalculate the rates frequently enough.* After 200,000 rounds of calculations ($\Delta t = .00001$) we ended up only .0034 away from our starting value. In fact, there is a clear pattern to the values of the errors as we decrease the step size. In the exercises it is left for you to explore this pattern and show that similar results hold for I and for R.

A second feature is that as we read down the column under $S(1)$, we find each digit **stabilizes**—that is, after changing for a while, it eventually becomes fixed at a particular value. The initial digits 44 are the first to stabilize, and that happens by the time $\Delta t = 0.1$. Then the third digit 2 stabilizes, when $\Delta t = 0.01$. Roughly speaking, one more digit stabilizes at each successive level. The table is revealing to us, digit by digit, the true value of $S(1)$. By the fifth stage we learn that the integer part of $S(1)$ is 44255. By the sixth stage we can say that the true value of $S(1)$ is $44255.4\ldots$.

When we write $S(1) = 44255.4\ldots$

Approximations lead to exact values

we are expressing $S(1)$ to **one decimal place accuracy**. This says, first, that the decimal expansion of $S(1)$ begins with exactly the six digits shown and, second, that there are further digits after the 4 (represented by the three dots "…"). In this case, we can identify further digits simply by continuing the table. Since our step sizes have the form $\Delta t = 10^{-N}$, we just need to increase N. For example, to express $S(1)$ accurately to six decimal places, we need to stabilize the first eleven digits in our estimates of $S(1)$. The table suggests that Δt should probably be about 10^{-10}—i.e., $N = 10$.

The true value of $S(1)$ emerges through a process that generates a sequence of successive approximations. We say $S(1) = 44255.4$…is the **limit** of this sequence as Δt is made smaller and smaller or, equivalently, as N is made larger and larger. We also say that the sequence of successive approximations **converges** to the limit $S(1)$. Here is a mathematical notation that expresses these statements more compactly:

$$\begin{aligned} S(1) &= \lim_{\Delta t \to 0} \{\text{the estimate of } S(1)\} \qquad \text{or, equivalently,} \\ &= \lim_{N \to \infty} \{\text{the estimate of } S(1) \text{ with } \Delta t = 10^{-N}\}. \end{aligned}$$

The symbol ∞ stands for "infinity," and the expression $N \to \infty$ is often pronounced "as N goes to infinity." However, it is often more instructive to say "as N gets larger and larger, without bound."

You should check that similar patterns are occurring in the $I(1)$ and $R(1)$ columns as well.

The limit concept lies at the heart of calculus. Later on we'll give a precise definition, but you should first see limits at work in a number of contexts and begin to develop some intuitions about what they are. This approach mirrors the historical development of calculus—mathematicians freely used limits for well over a century before a careful, rigorous definition was developed.

2.1.2 One Picture Is Worth a Hundred Tables

As we noted, the program SIRVALUE prints out only the final values of S, I, and R because it would typically take too much space to print out the intermediate values. However, if instead of printing these values we plot them graphically, we can convey all this intermediate information in a compact and comprehensible form.

Suppose, for instance, that we wanted to record the calculations leading up to $S(3)$ by plotting all the points. The graphs below plot all the pairs of values (t, S) that are calculated along the way for the cases $\Delta t = 1$ (4 points), $\Delta t = .1$ (31 points), and $\Delta t = .01$ (301 points).

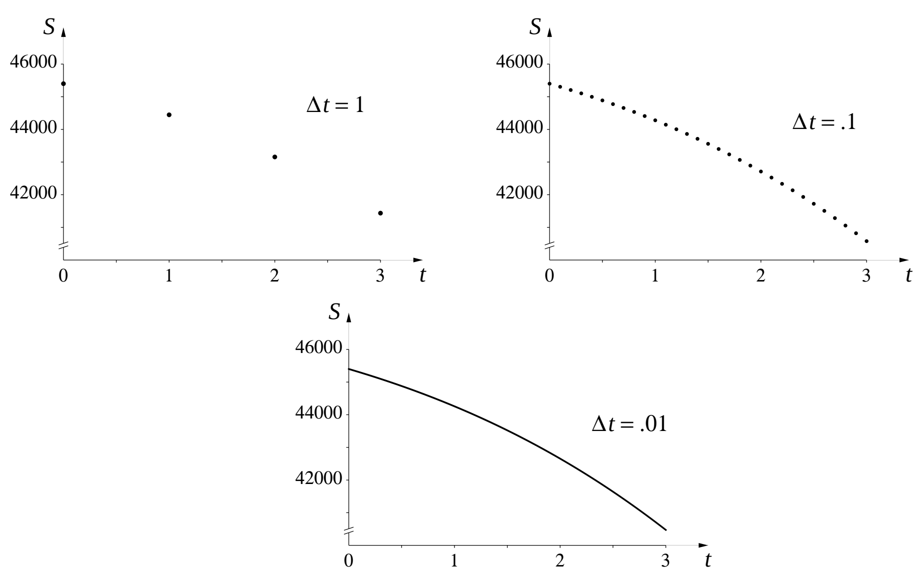


Figure 2.1.4 image

By the time we get to steps of size .01, the resulting graph begins to look like a continuous curve. This suggests that instead of simply plotting the points we might want to draw lines connecting the points as they're calculated.

We can easily modify SIRVALUE to do this—the only changes will be to replace the `PRINT` command with a command to draw a line and to move this command inside the loop (so that it is executed every time new values are computed). We will also need to add a line or two at the beginning to tell the computer to set up the screen to plot points. This usually involves opening a —i.e., specifying the horizontal and vertical ranges the screen should depict. Since programming languages vary slightly in the way this is done, we use italicized text "*Set up GRAPHICS*" to make clear that this statement is **not** part of the program—you will have to express this in the form your programming language specifies. Similarly, the command

Plot the line from `(t, S)` *to* `(t + deltat, S + deltaS)`

will have to be stated in the correct format for your language. The computational core of SIRVALUE is unchanged. Here is what the new program looks like if we want to use $\Delta t = .1$ and connect the points with straight lines:

Program: SIRPLOT

Set up GRAPHICS

```
tinitial = 0
tfinal = 3
t = tinitial
S = 45400
I = 2100
R = 2500
numberofsteps = 30
deltat = (tfinal - tinitial)/numberofsteps
FOR k = 1 TO numberofsteps
      Sprime = -.00001 * S * I
      Iprime = .00001 * S * I - I / 14
      Rprime = I / 14
      deltaS = Sprime * deltat
      deltaI = Iprime * deltat
      deltaR = Rprime * deltat
```

Plot the line from `(t, S)` *to* `(t + deltat, S + deltaS)`

```
      t = t + deltat
      S = S + deltaS
      I = I + deltaI
      R = R + deltaR
NEXT k
```

If we had wanted just to plot the points, we could have used a command of the form *Plot the point* `(t, S)` in place of the command to plot the line, moving this command down two lines so it came after we had computed the new values of `t` and `S`. We would also need to place that command before the loop so that the initial point corresponding to $t = 0$ gets plotted.

When we "connect the dots" like this we emphasize graphically the underlying assumption we have been making in all our estimates: that the function $S(t)$ is linear (i.e., it is changing at a constant rate) over each interval Δt. Let's see what the graphs look like when we do this for the three values of Δt we used above. To compare the results more readily we'll plot the graphs on the same set of axes. (We will look at a program for doing this in the next section.) We get the following picture:

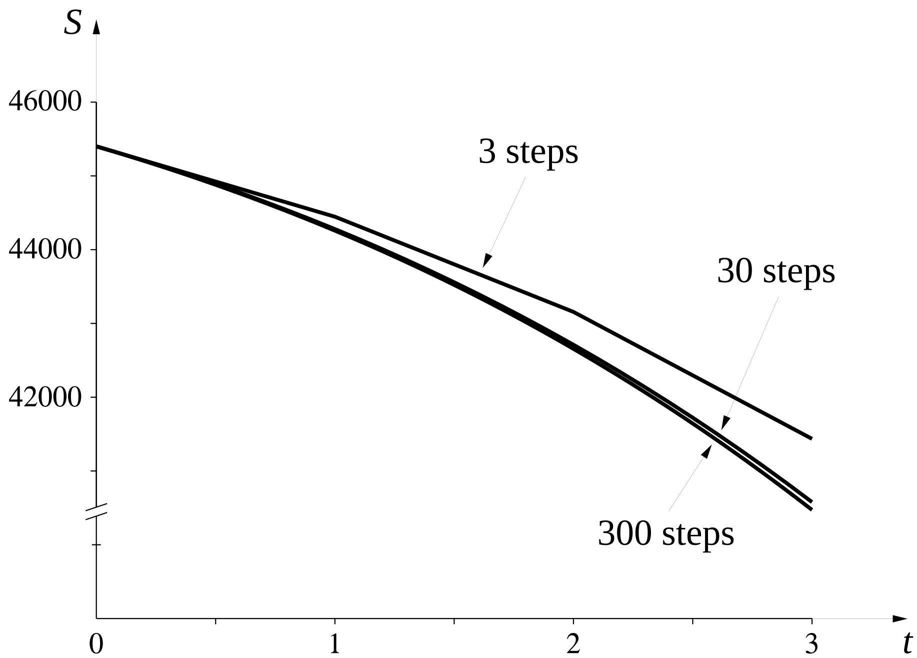


Figure 2.1.5 image

The graphs become indistinguishable from each other and increasingly look like smooth

Graphs made up of line segments look like smooth curves if the segments are short enough

curves as the number of segments increases. If we plotted the 3000-step graph as well, it would be indistinguishable from the 300-step graph at this scale. If we now shift our focus from the end value $S(3)$ and look at all the intermediate values as well, we find that each graph gives an approximate value for $S(t)$ for *every* value of t between 0 and 3. We are seeing the entire function $S(t)$ over this interval.

Just as we wrote

$$S(3) = \lim_{N\to\infty} \{\text{the estimate of } S(3) \text{ with } \Delta t = 10^{-N}\}.$$

We can also write

$$\text{graph of } S(t) = \lim_{N\to\infty} \{\text{line-segment approximations with } \Delta t = 10^{-N}\}.$$

The way we see the graph of $S(t)$ emerging from successive approximations is our first example of a fundamental result. It has wide-ranging implications which will occupy much of our attention for the rest of the course.

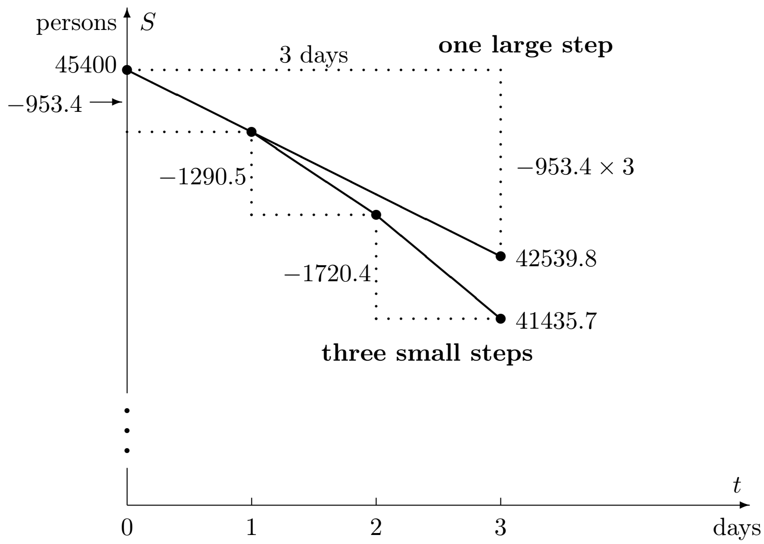
2.1.3 Piecewise Linear Functions

Let's examine the implications of this approach more closely by considering the "one-step" ($\Delta t = 3$) approximation to $S(t)$ and the "three-step" ($\Delta t = 1$) approximation over the time interval $0 \le t \le 3$. In the first case we are making the simplifying assumption that S decreases at the rate $S' = -953.4$ persons per day for the entire three days. In the second case we use three shorter steps of length $\Delta t = 1$, with the slopes of the corresponding segments given by the table on page in Chapter 1, summarized below (note that since $\Delta t = 1$ day we have that the magnitude of $\Delta S = S' \cdot \Delta t$ is the same as the magnitude of S'):

Table 2.1.6

t	S	S'
0	45 400.0	−953.4
1	44 446.6	−1 290.5
2	43 156.1	−1 720.4
3	41 435.7	

Here are the corresponding graphs we get:



Two approximations to S during the first three days

Figure 2.1.7

The "one-step" estimate. Assuming that S decreases at the rate $S' = -953.4$ persons per day for the entire three days is equivalent to assuming that S follows the upper graph—a straight line with slope –953.4 persons/day. In other words, the one-step approach approximates S by a **linear function** of t. If we use the notation $S_1(t)$ to denote this (one-step) linear approximation we have

$$\text{One-step estimate: } S(t) \approx S_1(t) = 45400 - 953.4\,t\,.$$

Because the one-step estimate is actually a function we can find the value of $S_1(t)$ for *all* t in the interval $0 \leq t \leq 3$, not just $t = 3$, and thereby get corresponding estimates for $S(t)$ as well. For example,

$$S_1(2) = 45400 - 953.4 \times 2 = 43493.2$$
$$S_1(1.7) = 45400 - 953.4 \times 1.7 = 43779.22\,.$$

The "three-step" estimate. With three smaller steps of size $\Delta t = 1$, we get a function whose graph is composed of three line segments, each starting at the (t, S) point at the

A single function may not be specified by a single equation beginning of each day and with a slope equal to the corresponding rate of new infections at the beginning of the day. Let's call this function $S_3(t)$. The three-step estimate $S_3(t)$ is hence not a linear function, strictly speaking. However, since its graph is made up of several straight pieces, it is called a **piecewise linear function**. Recalling that the equation of a line through the point (x_0, y_0) with slope m can be written in the form $y = m(x - x_0) + y_0$, we can use the values for S (which correspond to the y-values) and S' (which give us the slopes of the segments) at times $t = 0$, $t = 1$, and $t = 2$ calculated above to get an explicit formula for $S_3(t)$:

$$S_3(t) = \begin{cases} y = -953.4(t-0) + 45400 & \text{if } 0 \le t \le 1 \\ y = -1290.5(t-1) + 44446.6 & \text{if } 1 \le t \le 2 \\ y = -1720.4(t-2) + 43156.1 & \text{if } 2 \le t \le 3 \end{cases}$$

Note that we have used $\le$ in the defining formulas since at the values $t = 1$ and $t = 2$ it doesn't matter which equation we use. This is equivalent to saying that the straight line segments are connected to each other. Since the slopes of the three segments of $S_3(t)$ are progressively more negative, the piecewise linear graph gets progressively steeper as t increases. This explains why the value $S_3(3)$ is *lower* than the value of $S_1(3)$. While it is rare that we would actually need to write down an explicit formula like this for the piecewise-linear approximation—it is easier, and usually more informative, just to define $S_3(t)$ by its graph—it is nevertheless important to realize that there really is an approximating function defined for all values of t in the interval $[0, 3]$, not just at the finite set of t values where we make the recalculations.

By the time we are dealing with the 300-step function $S_{300}(t)$ we can't even tell by its graph that it is piecewise linear unless we zoom in very close. In principle, though, we could still write down a simple linear formula for each of its segments (see the exercises).

An appraisal. The graph of $S_1(t)$ gives us a rough idea of what is happening to the true function $S(t)$ during the first three days. It starts off at the same rate as S', but subsequently the rates move apart. The value of S_1' never changes, while S' changes with the (ever-changing) values of S and I.

The graph of $S_3(t)$ is a distinct improvement because it changes its direction twice, modifying its slope at the beginning of each day to come back into agreement with the rate equation. But since the three-step graph is still piecewise linear, it continues to suffer from the same shortcoming as the one-step: once we restrict our attention to a single straight segment (for example, where $1 \le t \le 2$), then the three-step graph *also* has a constant slope, while S' is always changing. Nevertheless, $S_3(t)$ does satisfy the rate equation in our original model three times—at the beginning of each segment—and isn't too far off at other times. When we get to $S_{300}(t)$ we have a function which satisfies the rate equation at 300 times and is very close in between.

Each of these graphs gives us an idea of the behavior of the true function $S(t)$ during the time interval $0 \le t \le 3$. None is strictly correct, but none is hopelessly wrong, either. All are **approximations** to the truth. Moreover, $S_3(t)$ is a **better** approximation than $S_1(t)$—because it reflects at least some of the variability in S'—and $S_{300}(t)$ is better still. Thus, even before we have a clear picture of the shape of the true function $S(t)$, we would expect it to be closer to $S_{300}(t)$ than to $S_3(t)$. As we saw above, when we take piecewise linear approximations with smaller and smaller step sizes, it is reasonable to think that they will approach the true function S in the limit. Expressing this in the notation we have used before,

$$\text{the function } S(t) = \lim_{N\to\infty} \{\text{the chain of linear functions with } \Delta t = 10^{-N}\}.$$

2.1.4 Approximate versus Exact

You may find it unsettling that our efforts give us only a sequence of approximations to $S(3)$, and not the exact value, or only a sequence of piecewise-linear approximations to $S(t)$, not the "real" function itself. In what sense can we say we "know" the number $S(3)$ or the function $S(t)$? The answer is: in

What does it mean to "know" a number like π?

the same sense that we "know" a number like $\sqrt{2}$ or π. There are two distinct aspects to the way we know a number. On the one hand, we can **characterize** a number precisely and completely:

Table 2.1.8

π:	the ratio of the circumference of a circle to its diameter;
$\sqrt{2}$:	the positive number whose square is 2;

On the other hand, when we try to **construct** the decimal expansion of a number, we usually get only approximate and incomplete results. For example, when we do calculations by hand we might use the rough estimates $\sqrt{2} \approx 1.414$ and $\pi \approx 3.1416$. With a desk-top computer we might have $\sqrt{2} \approx 1.414\,213\,562\,373\,095$ and $\pi \approx 3.141\,592\,653\,589\,793$, but these are still approximations, and we are really saying

$$\begin{aligned} \sqrt{2} &= 1.414\,213\,562\,373\,09\ldots \\ \pi &= 3.141\,592\,653\,589\,79\ldots\ . \end{aligned}$$

The *complete* decimal expansions for $\sqrt{2}$ and π are unknown! The exact values exist as limits of approximations that involve successively longer strings of digits, but we never see the limits—only approximations. In the final section of this chapter we will see ways of generating these approximations for $\sqrt{2}$ and for π.

What we say about π and $\sqrt{2}$ is true for $S(3)$ in exactly the same way. We can *characterize* it quite precisely, and we can *construct* approximations to its numerical value to any desired degree of accuracy. Here, for example, is a characterization of $S(3)$:

> The S-I-R problem for which $a = .00001$ and $b = 1/14$ and for which $S = 45400$, $I = 2100$, $R = 2500$ when $t = 0$ determines three functions $S(t)$, $I(t)$, and $R(t)$. The number $S(3)$ is the value that the function $S(t)$ has when $t = 3$.

You should try to extend this argument to describe the sense in which we "know" the function $S(t)$ by knowing its piecewise-linear approximations. Try to convince yourself that this is operationally no different from the way we "know" functions like $f(x) = \sqrt{x}$. In each instance we can **characterize** the function completely, but we can only **construct** an approximation to most values of the function or to its graph.

All this discussion of approximations may strike you as an unfortunate departure from the accuracy and precision you may have been led to expect

Being able to approximate a number to 12 decimal places is usually as good as knowing its value precisely

in mathematics up until now. In fact, it is precisely this ability to make quick and accurate approximations to problems that is one of the most powerful features of mathematics. This is what goes on every time you use your calculator to evaluate $\log 3$ or $\sin 37$. Your calculator doesn't really know what these numbers are—but it does know how to approximate them quickly to 12 decimal places. Similar kinds of approximations are also at the heart of how bridges are built and spaceships are sent to the moon.

A Caution:. The fact that computers and calculators are really only dealing with approximations

However …

when we think they are being exact occasionally leads to problems, the most common of which involves **roundoff errors**. You can probably generate a relatively harmless manifestation of this on your computer with the SIRVALUE program. Modify the `PRINT` line so it prints out the final value of `t` to 10 or 12 digits, and try running it with a high value for `numberofsteps`, say 1 million or 10 million. You would expect the final value of `t` to be exactly 1 in every case, since you are adding `deltat = 1/numberofsteps` to itself `numberofsteps` times. The catch is that the computer doesn't store the exact value `1/numberofsteps` unless `numberofsteps` is a power of 2. In all other cases it will only be using an approximation, and if you add up enough quantities that are slightly off, their cumulative error will begin to show. We will encounter a somewhat less benign manifestation of roundoff error in the next chapter.

2.1.5 Exercises

There and back again.

1. Look at the table on page . What is your best guess of the *exact* value of $I(1)$? (Use the "…" notation introduced on page .)

  (a) What is the exact value of $R(1)$?

2. We noted that the discrepancy (the difference between the new estimate for $S(0)$ and the original value) seemed to decrease as Δt decreased.

  (a) What is your best estimate (using only the information in the table) for the value of Δt needed to produce a discrepancy of .001 ?

  (b) More generally, express as precisely as you can the apparent relation between the size of the discrepancy and the size of Δt.

3. Suppose you wanted to try going three days forward and then coming back, using $\Delta t = .01$. What changes would you have to make in SIRVALUE to do this?

  (a) Make a table similar to the one on page for going three days forward and coming back for $\Delta t = 1$, .1, .01, and.001.

  (b) In this new table how does the size of the discrepancy for a given value of Δt compare with the value in the original table?

  (c) What value of Δt do you think you would need to determine the integer parts of $S(3)$ and $R(3)$ exactly?

Piecewise linear functions.

4. Using this three-step approximation, what is $S_3(1.7)$? What is $S_3(2.5)$?

5. How would you modify SIRVALUE to get $S_{3000}(3)$? Do it; what do you get?

6. What additional changes would you make to get the values of t, S, and S' at the beginning of the 193rd segment of $S_{300}(t)$? [HINT: You only need to alter the `FOR k = 1 TO numberofsteps` line (since you don't want to go all the way to the end) and the `PRINT` line. (Note that after running the loop for, say, 20 times, the values of `t` and `S` are the values for the beginning of the 21st segment, while the value of `Sprime` will still be the slope of the 20th segment.)]

7. Suppose we wanted to determine the value of $S_{300}(2.84135)$.

- **(a)** In which of the 300 segments of the graph of $S_{300}(t)$ would we look to find this information?
- **(b)** What are the values of t, S, and S' at the beginning of this segment?
- **(c)** What is the equation of this segment?
- **(d)** What is $S_{300}(2.84135)$?

8. How would you modify SIRVALUE to calculate estimates for S, I, and R when $t = -6$, using $\Delta t = .05$? Do it; what do you get?

9. We want to use SIRPLOT to look at the graph of $S(t)$ over the first 20 days, using $\Delta t = .01$.

- **(a)** What changes would we have to make in the program?
- **(b)** Sketch the graph you get when you make these changes.
- **(c)** If you wanted to plot the graph of $I(t)$ over this same time interval, what additional modifications to SIRPLOT would be needed? Make them, and sketch the result. When does the infection appear to hit its peak?
- **(d)** Modify SIRPLOT to sketch on the same graph all three functions over the first 70 days. Sketch the result.

The DO–WHILE loop.

A difficulty in giving a precise answer to the last question was that we had to get all the values for 20 days, then go back to estimate by eye when the peak occurred. It would be helpful if we could write a program that ran until it reached the point we were looking for, and then stopped. To do this, we need a different kind of loop—a *conditional* loop that keeps looping only while some specified condition is true. A DO–WHILE loop is one useful way to do this. Here's how the modified SIRPLOT program would look:

Set up GRAPHICS

```
tinitial = 0
t = tinitial
S = 45400
I = 2100
R = 2500
Iprime = .00001 * S * I - I / 14
deltat = .01
DO WHILE Iprime > 0
      Sprime = -.00001 * S * I
```

```
    Iprime = .00001 * S * I - I / 14
    Rprime = I / 14
    deltaS = Sprime * deltat
    deltaI = Iprime * deltat
    deltaR = Rprime * deltat
```

Plot the line from `(t, S)` *to* `(t + deltat, S + deltaS)`

```
    t = t + deltat
    S = S + deltaS
    I = I + deltaI
    R = R + deltaR
LOOP
PRINT t - deltat
```

The changes we have made are:

- Since we don't know what the final time will be, eliminate the `tfinal = 20` and the `numberofsteps = 2000` commands.
- Since the condition in our loop is keyed to the value of I', we have to calculate the initial value of I' before the loop starts.
- Instead of `deltat = (tfinal - tinitial)/numberofsteps` use the statement `deltat = .01`.
- The key change is to replace the `FOR k = 1 TO numberofsteps` line by the line `DO WHILE Iprime > 0`.
- To denote the end of the loop we replace the `NEXT k` command with the command `LOOP`.
- After the `LOOP` command add the line `PRINT t - deltat`. (The reason we had to back up one step at the end is because due to the way the program was written, the computer takes one final step with a negative value for `Iprime` before it stops.)

The net effect of all this is that the program will continue working as before, calculating values and plotting points `(t, S)`, but *only as long as the condition in the* `DO WHILE` *statement is true.* The condition we used here was that I' had to be positive—this is the condition that ensures that values of I are still getting bigger. While this condition is true, we can always get a larger value for I by going forward another increment Δt. As soon as the condition is false—i.e., as soon as I' is negative—the values for I will be decreasing, which means we have passed the peak and so want to stop.

10. Make these modifications; what value for t do you get?

11. You could modify the `PRINT t` command to also print out other quantities.

  (a) What is the value of I at its peak?

  (b) What is the value of S when I is at its peak? Does this agree with the threshold value we predicted in chapter 1?

12. Suppose you change the initial value of S to be 5400 and run the previous program. Now what happens? Why?

13. Suppose we wanted to know how long the epidemic lasts. We could use DO–WHILE and keep stepping forward using, say $\Delta t = .1$, so long as $I \geq 1$. As soon as I was less than 1 we would want to stop and see what the value of t was.

(a) What modifications would you make in SIRVALUE to get this information?

(b) Run your modified program. What value do you get for t?

(c) Run the program using $\Delta t = .01$. Now what is your estimate for the duration of the epidemic? What can you say about the actual time required for I to drop below 1?

14. If we think the epidemic started with a single individual, we can go backwards in time until I is no longer greater than 1, and see what the corresponding time is. Do this for $\Delta t = -1, -.1,$ and $-.01$. What is your best estimate for the time that the infection arrived?

(a) How many Recovereds were there at the start of the epidemic? This would be the people who had presumably been infected in a previous epidemic and now had immunity.

2.2 The Mathematical Implications—Euler's Method

2.2.1 Approximate Solutions

In the last section we approximated the function $S(t)$ by piecewise-linear approximations using steps of size $\Delta t = 1$, .1, and .01. This process can clearly be extended to produce approximations with an *arbitrary* number of steps. For any given step size Δt, the result is a piecewise linear graph whose segments are Δt days wide. This graph then provides us with an estimate for $S(t)$ for every value of t in the interval $0 \leq t \leq 3$. We call this process of obtaining a function by constructing a sequence of increasingly better approximations **Euler's method**, after the Swiss mathematician Leonhard Euler (1707–1783). Euler was interested in the general problem of finding the functions determined by a set of rate equations, and in 1768 he proposed this method to approximate them. The method is conceptually simple and can indeed be used to get solutions for an enormous range of rate equations. For this reason we will make it a basic tool.

To begin to get a sense of the general utility of Euler's method, let's use it in a new setting. Here is a simple problem that involves just a single variable y that depends on t.

$$\begin{aligned} mboxrateequation: \quad & y' = .1\,y\left(1 - \frac{y}{1000}\right), \\ \text{initial condition:} \quad & y = 100 \\ & mboxwhen\ t = 0. \end{aligned}$$

To solve the problem we must find the function $y(t)$ determined by this rate equation and initial condition. We'll work without a context, in order to emphasize the purely mathematical nature of Euler's method. However, this rate equation is one member of a family called **logistic equations** frequently used in population models. We will explore this context in the exercises.

Let's now construct the function that approximates $y(t)$ on the interval $0 \leq t \leq 75$, using $\Delta t = .5$. We can make the suitable modifications in SIRPLOT or SIRVALUE to get this approximation. Suppose we want to view the approximation graphically. Here's what the modified SIRPLOT would look like:

Program: modified SIRPLOT

Set up GRAPHICS

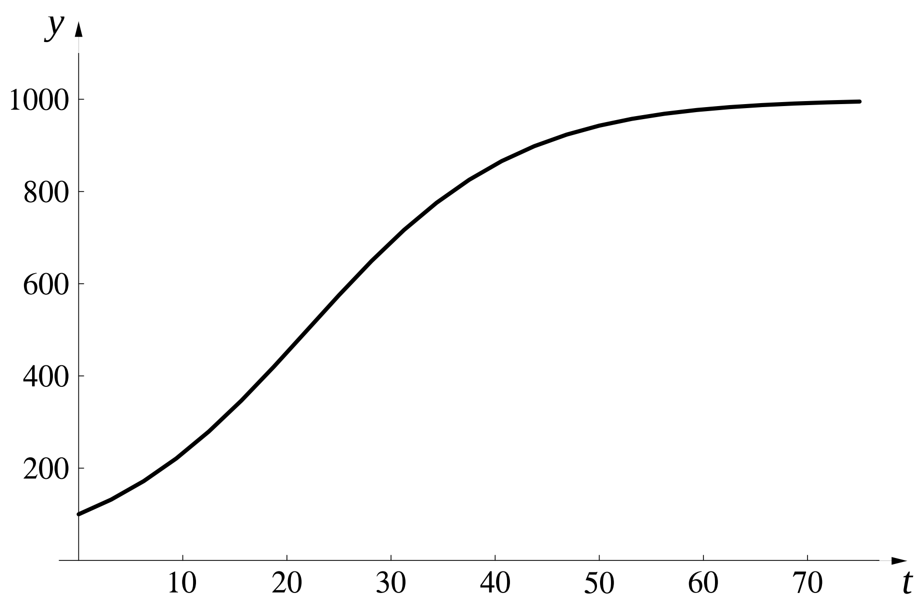
```
tinitial = 0
tfinal = 75
t = tinitial
y = 100
numberofsteps = 150
deltat = (tfinal - tinitial)/numberofsteps
FOR k = 1 TO numberofsteps
      yprime = .1 * y * (1 - y / 1000)
      deltay = yprime * deltat
```

Plot the line from `(t, y)` *to* `(t + deltat, y + deltay)`

```
      t = t + deltat
      y = y + deltay
NEXT k
```

As before, words in italics, like "*Plot the line from...to*" need to be translated into the specific formulation required by the computer language you are using.

When we run this program, we get the following graph (axes and scales have been added):



As before, what we see here is only an approximation to the true solution $y(t)$. How can we get some idea of how good this approximation is?

2.2.2 Exact Solutions

For any chosen step size, we can produce an *approximate* solution to a rate equation problem. We will call such an approximation an **Euler approximation**. In the last section we saw that we can improve the accuracy of the approximation by making the steps smaller and using more of them.

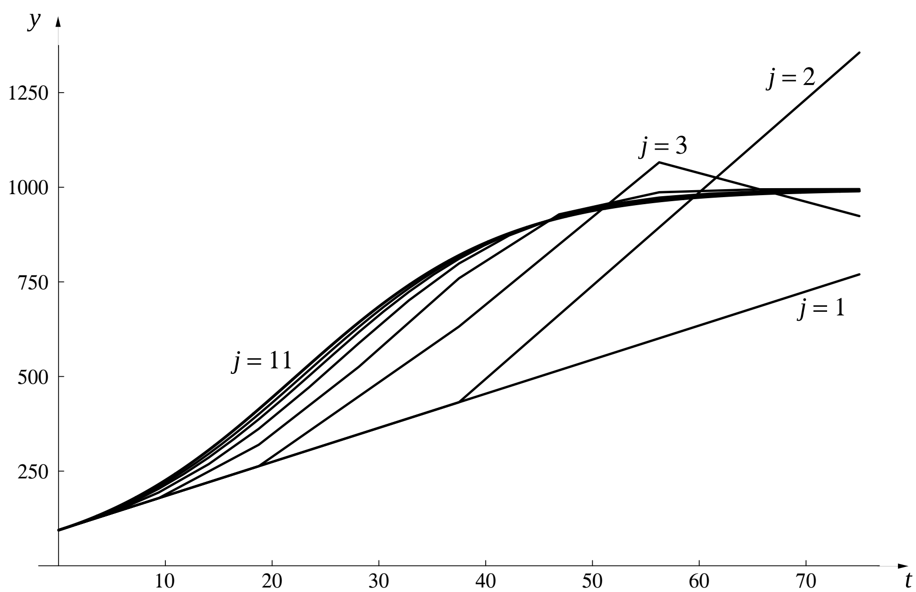
For example, we have already found an approximate solution to the problem

$$y' = .1y\left(1 - \frac{y}{1000}\right); \qquad y(0) = 100$$

on the interval $0 \le t \le 75$, using 150 steps of size $\Delta t = 1/2$. Consider a **sequence** of Euler approximations to this problem that are obtained by increasing the number of steps from one stage to the next. To be systematic, let the first approximation have 1 step, the next 2, the next 4, and so on. (The important feature is that the number of steps increases from one approximation to the next, not necessarily that they double—going up by powers of 10 would be just as good. A slight advantage in using powers of 2 is to maximize computer accuracy.) The number of steps thus has the form 2^{j-1}, where $j = 1$, 2, 3, ... If we use $y_j(t)$ to denote the approximating function with 2^{j-1} steps, then we have an unending list:

$$\begin{array}{rl} y_1(t): & \text{Euler's approximation with 1 step} \\ y_2(t): & \text{Euler's approximation with 2 steps} \\ y_3(t): & \text{Euler's approximation with 4 steps} \\ \vdots & \qquad\vdots \\ y_j(t): & \text{Euler's approximation with } 2^{j-1} \text{ steps} \\ \vdots & \qquad\vdots \end{array}$$

Here are the graphs of $y_j(t)$ for $j = 1, 2, \ldots, 7$ plus the graph of $y_{11}(t)$:



These functions form a sequence of **successive approximations** to the true solution $y(t)$, which is obtained by taking the **limit**, as we did in the last section:

$$y(t) = \lim_{j\to\infty} y_j(t).$$

Earlier we noticed how the digits in the estimates for $S(3)$ stabilized. If we

Functions and graphs can be limits, too

plot the approximations $y_j(t)$ together we'll find that they stabilize, too. Each graph in the sequence is different from the preceding one, but the differences diminish the larger j becomes. Eventually, when j is large enough, the graph of y_{j+1} does not differ noticeably from the graph of y_j. That is, the position of the graph **stabilizes** in the coordinate plane. In this example, at the scale in the graph above, this happens around $j = 11$. If we had drawn the graph of y_{15} or y_{20}, it would not have been distinguishable

Euler's method is the process of finding solutions through a sequence of successive approximations

from the graph of y_{11}. It is this entire process of calculating a sequence of successive approximations using increasingly many steps as far as is needed to get the desired level of stabilization that is meant when we talk about Euler's method.

The program SEQUENCE shown below plots 14 Euler approximations to $y(t)$, increasing the number of steps by a factor of 2 each time. It demonstrates how the graphs of $y_j(t)$ stabilize to define $y(t)$ as their limit.

Program: SEQUENCEA sequence of graphs for $y' = .1y(1-y/1000)$; $y(0) = 100$

Set up GRAPHICS

```
FOR j = 1 TO 14

tinitial = 0
tfinal = 75
t = tinitial
y = 100
numberofsteps = 2 ^ (j - 1)
deltat = (tfinal - tinitial) / numberofsteps
FOR k = 1 TO numberofsteps
      yprime = .1 * y * (1 - y / 1000)
      deltay = yprime * deltat
```

Plot the line from `(t, y)`
to `(t + deltat, y + deltay)`
Color the line with color `j`

```
      t = t + deltat
      y = y + deltay
NEXT k
```

Program: modified SIRPLOT

```
NEXT j
```

Notice that SEQUENCE contains the program SIRPLOT embedded in a loop that executes SIRPLOT 14 times. In this way SEQUENCE plots 14 different graphs. The only new element that has been added to SIRPLOT is "*Color* *the line with color* `j`". When you express this in your programming language it instructs the computer to draw the j-th graph using color number j in the computer's "palette." In the exercises you are asked to use the program SEQUENCE to explore the solutions to a number of rate equation problems.

Approximate solutions versus exact. By constructing successive approximations to the solution of a rate equation problem, using a sequence of step sizes `deltat` $= \Delta t$ that shrink to 0, we obtain the *exact* solution in the limit.

In practice, though, all we can ever get are particular approximations. However, we can control the level of precision in our approximations by adjusting the step size. If we are dealing with a model of some real process, then this is typically all we need. For example, when it comes to interpreting the S-I-R model, we might be satisfied to predict that there will be about 40500 susceptibles remaining in the population after three days. The table on page indicates we would get that level of precision using a step size of about $\Delta t = 10^{-2}$. Greater precision than this may be pointless, because the modelling process—which converts reality to mathematics—is itself only an approximation.

The question we asked in the last section—In what sense do we know a number?—applies equally to the functions we obtain using Euler's method. That is, even if we can characterize a function quite precisely as the solution of a particular rate equation, we may be able to evaluate it only approximately.

2.2.3 A Caution

We have now seen how to take a set of rate equations and find approximations to the solution of these equations to any degree of accuracy desired. It is important to remember that all these mathematical manipulations are only drawing inferences about the model. We are essentially saying that *if* the original equations capture the internal dynamics of the situation being modelled, *then* here is what we would expect to see. It is still essential at some point to go back to the reality being modeled and check these predictions to see whether our original assumptions were in fact reasonable, or need to be modified. As Alfred North Whitehead has said:

> There is no more common error than to assume that, because prolonged and accurate mathematical calculations have been made, the application of the result to some fact of nature is absolutely certain.

2.2.4 Exercises

Approximate solutions.

1. Modify SIRVALUE and SIRPLOT to analyze the population of Poland (see exercise 25 of chapter 1, page). We assume the population $P(t)$ satisfies the conditions

$$P' = .009\,P \quad \text{and} \quad P(0) = 37{,}500{,}000,$$

where t is years since 1985. We want to know P 100 years into the future; you can assume that P does not exceed 100,000,000.

(a) Estimate the population in 2085.

(b) Sketch the graph that describes this population growth.

The Logistic Equation. Suppose we were studying a population of rabbits. If we turn 100 rabbits loose in a field and let $y(t)$ be the number of rabbits at time t measured in months, we would like to know how this function behaves. The next several exercises are designed to explore the behavior of the rate equation

$$y' = .1y\left(1 - \frac{y}{1000}\right); \qquad y(0) = 100$$

and see why it might be a reasonable model for this system.

2. By modifying SIRVALUE in the way we modified SIRPLOT to get SEQUENCE, obtain a sequence of estimates for $y(37)$ that allows you to specify the exact value of $y(37)$ to two decimal places accuracy.

3. Referring to the graph of $y(t)$ obtained in the text on page , what can you say about the behavior of y as t gets large?

  (a) Suppose we had started with $y(0) = 1000$. How would the population have changed over time? Why?

  (b) Suppose we had started with $y(0) = 1500$. How would the population have changed over time? Why?

  (c) Suppose we had started with $y(0) = 0$. How would the population have changed over time? Why?

  (d) The number 1000 in the denominator of the rate equation is called the **carrying capacity** of the system. Can you give a physical interpretation for this number?

4. Obtain graphical solutions for the rate equation for different values of the carrying capacity. What seems to be happening as the carrying capacity is increased? (Don't restrict yourself to $t = 37$ here.) In this problem and the next you should sketch the different solutions on the same set of axes.

5. Keep everything in the original problem unchanged except for the constant .1 out front. Obtain graphical solutions with the value of this constant = .05, .2, .3, and .6. How does the behavior of the solution change as this constant changes?

6. Returning to the original logistic equation, modify SIRVALUE or DO–WHILE to find the value for t such that $y(t) = 900$.

7. Suppose we wanted to fit a logistic rate equation to a population, starting with $y(0) = 100$. Suppose further that we were comfortable with the 1000 in the denominator of the equation, but weren't sure about the .1 out front. If we knew that $y(20) = 900$, what should the value for this constant be?

Using SEQUENCE.

8. Each Euler approximation is made up of a certain number of straight line segments. What instruction in the program SEQUENCE determines the number of segments in a particular approximation? The first graph drawn has only a single segment. How many does the fifth have? How many does the fourteenth have?

9. What is the slope of the first graph? What are the slopes of the two parts of the second graph? [You should be able to answer these questions without resorting to a computer.]

10. Modify the line in the program SEQUENCE which determines the number of steps by having it read `numberofsteps = j`, and run the modified program. Again, we are getting a sequence of approximations, with the number of steps increasing each time, but the approximations don't seem to be getting all that close to anything. Explain why this modified program isn't as effective for our purposes as the original.

11. Modify SEQUENCE to produce a sequence of Euler approximations to the function $y(t)$ that satisfies the conditions

$$y' = .2\,y(5-y) \quad \text{and} \quad y(0) = 1.$$

on the interval $0 \le t \le 10$. [You need to change the final t value in the program, and you also need to ensure that the graphs will fit on your screen.]

(a) What is $y(10)$? [If you add the line `PRINT j, y` just before the line `NEXT j`, a sequence of 14 estimates for $y(10)$ will appear on the screen with the graphs.]

(b) Make a rough sketch of the graph that is the limit of these approximations. The right half of the limit graph has a distinctive feature; what is it?

(c) Without doing any calculations, can you estimate the value of $y(50)$? How did you arrive at this value?

(d) Change the **initial condition** from $y(0) = 1$ to $y(0) = 9$. Construct the sequence of Euler approximations beginning with `numberofsteps` being 1, and make a rough sketch of the limit graph. What is $y(10)$ now? Explain why the first several approximations look so strange.

12. Modify SEQUENCE to construct a sequence of Euler approximations for population of Poland (from exercise 1, above). Sketch the limit graph $P(t)$, and mark the values of $P(0)$ and $P(100)$ at the two ends.

13. Construct a sequence of Euler approximations to the function $y(t)$ that satisfies the conditions

$$y' = 2\,t \quad \text{and} \quad y(0) = 0$$

over the interval $0 \le t \le 2$. Note that this time the rate y' is given in terms of t, not y. Euler's method works equally well. Using your sequence of approximations, estimate $y(2)$. How accurate is your estimate?

14. Construct a sequence of Euler approximations to the function $y(t)$ that satisfies the conditions

$$y' = \frac{4}{1+t^2} \quad \text{and} \quad y(0) = 0$$

over the interval $0 \le t \le 1$. Estimate $y(1)$. How accurate is your estimate? [Note: the exact value of $y(1)$ is π, which your estimates may have led you to expect. By using special methods we shall develop much later we can prove that $y(1) = \pi$.]

2.3 Approximate Solutions

Our efforts to find the functions that were determined by the rate equations for the *S*-*I*-*R* model have brought to light several important issues:

- We often have to deal with a question that does not have a simple, straightforward answer; perhaps we are trying to determine a quantity (like the square root of 2, or $S(3)$ in the *S*-*I*-*R* model), to find some function (like $S(t)$), or to understand a process (like an epidemic, or buying and selling in a market). An **approximation** can get us started.
- In many instances, we can make repeated improvements in the approximation. If these **successive approximations** get arbitrarily close to the unknown, and they do it quickly enough, that may answer the question for all practical purposes. In many cases, there is no alternative.
- The information that successive approximations give us is conveyed in the form of a **limit**.
- The method of successive approximations can be used to evaluate many kinds of mathematical objects, including numbers, graphs, and functions.
- **Limit processes** give us a valuable tool to probe difficult questions. They lie at the heart of calculus.

Even the process of building a mathematical model for a physical system can be seen as an instance of successive approximations. We typically start with a simple model (such as the *S*-*I*-*R* model) and then add more and more features to it (e.g., in the case of the *S*-*I*-*R* model we might divide the population into different subgroups, have the parameters in the model depend on the season of the year, make immunity of limited duration, etc.). Is it always possible, at least in theory, to get a sequence of approximating mathematical models that approaches reality in the limit?

In the following chapters we will apply the process of successive approximation to many different kinds of problems. For example, in chapter 3 the problem will be to get a better understanding of the notion of a rate of change of one quantity with respect to another. Then, in chapter 4, we will return to the task of solving rate equations using Euler's method. Chapter 6 introduces the integral, defining it through a sequence of successive approximations. As you study each chapter, pause to identify the places where the method of successive approximations is being used. This can give you insight into the special role that calculus plays within the broader subject of mathematics.

To illustrate the general utility of the method, we end this chapter by returning to the problem raised in section 1 of constructing the values of $\sqrt{2}$ and π to an arbitrary number of decimal places.

2.3.1 Calculating π—The Length of a Curve

Humans were grappling with the problem of calculating π at least 3000 years ago. In his work *Measurement of the Circle*, Archimedes (287–212 b.c.) used the method of successive approximations to calculate $\pi = 3.14\ldots$. He did this by starting with a circle of diameter 1, constructing an inscribed and a circumscribed hexagon, and calculating the lengths of their perimeters. The perimeter of the circumscribed hexagon was clearly an overestimate for π, while the perimeter of the inscribed hexagon was an underestimate. He then improved these estimates by going from hexagons to inscribed and circumscribed

12-sided polygons and again calculating the perimeters. He repeated this process of doubling the number of sides until he had inscribed and circumscribed polygons with 96 sides. These left him with his final estimate

$$3.1409\ldots = 3\frac{284\frac{1}{4}}{2017\frac{1}{4}} < \pi < 3\frac{667\frac{1}{2}}{4673\frac{1}{2}} = 3.1428\ldots$$

In grade school we learned a nice simple formula for the length of a circle, but that was about it. We were never taught formulas for the lengths of other simple curves like elliptic or parabolic arcs, for a very good reason—there are no such formulas. There are various physical approaches we might take. For example, we could get a rough approximation by laying a piece of string along the curve, then picking up the string and measuring it with a ruler. Instead of a physical solution, we can use the essence of Archimedes' insight of approximating a circle by an inscribed "polygon"—what we have earlier called a piecewise linear graph—to determine the length of any curve. The basic idea is reminiscent of the way we made successive approximations to the functions $S(t)$, $I(t)$, and $R(t)$ in the first section of this chapter. Here is how we will approach the problem:

- approximate the curve by a chain of straight line segments;
- measure the lengths of the segments;
- use the sum of the lengths as an approximation to the true length of the curve.

Repeat this process over and over, each time using a chain that has shorter segments (and therefore more of them) than the last one. The length of the curve emerges as the limit of the sums of the lengths of the successive chains.

Distance Formula.

If we are given two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ in the plane, then the distance between them is $d = \sqrt{(\Delta x)^2 + (\Delta y)^2}$, equivalently $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

This follows directly from the Pythagorean theorem, as shown in the figure below.

We'll demonstrate how this process works on a parabola. Specifically, consider the graph of $y = x^2$ on the interval $0 \leq x \leq 1$. At the right we have sketched the graph and our initial approximation. It is a piecewise linear approximation with two segments whose end points have equally spaced x-coordinates.

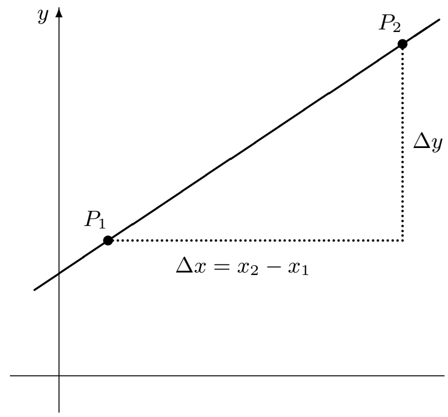


Figure 2.3.1

We can use the distance formula to find the lengths of the two segments.

$$\begin{aligned} mboxfirstsegment: & \sqrt{(.5-0)^2+(.25-0)^2} = .559016994 \\ \text{second segment}: & \sqrt{(1-.5)^2+(1-.25)^2} = .901387819 \end{aligned}$$

Their total length is the sum

$$.559016994 + .901387819 = 1.460404813.$$

The following program prints out the lengths of the two **segments** and their **total** length.

Program: LENGTHEstimating the length of $y = x^2$ over $0 \leq x \leq 1$

```
DEF fnf (x) = x ^ 2
xinitial = 0
xfinal = 1
numberofsteps = 2
deltax = (xfinal - xinitial) / numberofsteps
total = 0
FOR k = 1 TO numberofsteps
      xl = xinitial + (k - 1) * deltax
      xr = xinitial + k * deltax
      yl = fnf(xl)
      yr = fnf(xr)
      segment = SQR((xr - xl) ^ 2 + (yr - yl) ^ 2)
      total = total + segment
      PRINT k, segment
NEXT k
PRINT numberofsteps, total
```

2.3.2 Finding Roots with a Computer

When we casually turn to our calculator

Calculators and computers really work by making approximations

and ask it for the value of $\sqrt{2}$, what does it really do? Like us, the calculator can only add, subtract, multiply, and divide. Anything else we ask it to do must be reducible to these operations. In particular, the calculator doesn't really "know" the value of $\sqrt{2}$. What it does know is how to approximate $\sqrt{2}$ to, say, 12 significant figures using only elementary arithmetic. In this section we will look at two ways we might do this. Apart from the fact that both approaches use successive approximations, they are remarkably different in flavor. One works graphically, using a computer graphing package, and the other is a numerical algorithm that is about 4000 years old.

A geometric approach. Exercise 9 on page considered the problem of finding the roots of $f(x) = 1 - 2x^2$. A bit of algebra confirms that $\sqrt{2}/2$ is a root—i.e., $f(\sqrt{2}/2) = 0$. The question is: what is the *numerical value* of $\sqrt{2}/2$?

We'll answer this question by constructing a sequence of approximations that add digits, one at a time, to an estimate for $\sqrt{2}/2$. Since the root lies at the point where the graph of f crosses the x-axis, we just magnify the graph at this point over and over again, "trapping" the point between x values that can be made arbitrarily close together.

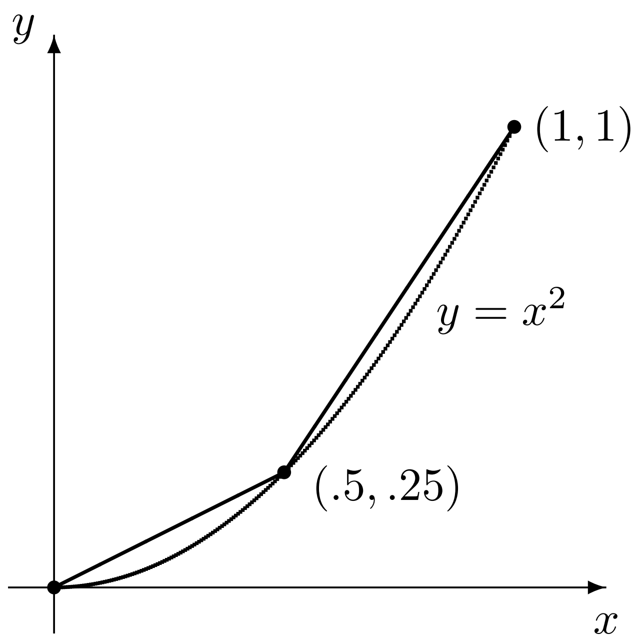


Figure 2.3.2

If we make each stage a ten-fold magnification over the previous one, then, as we zoom in on the next smaller interval that contains the root, one more digit in our estimate will be stabilized. The first six stages are described in the table below. They tell us $\sqrt{2}/2 = .707106\ldots$ to six decimal places accuracy.

Since this method of finding roots requires only that we be able to plot successive magnifications of the graph of f on a computer screen, the method can be applied to any function that can be entered into a computer.

The positive root of $1 - 2x^2$

Table 2.3.3

when the root lies between:

Table 2.3.4 **Table 2.3.5**

lower value	**upper value**	o
.70	.80	.7
.700	.710	.7
.7070	.7080	.7
.70710	.70720	.7
.707100	.707110	.7
.7071060	.7071070	.7

An algebraic approach – the Babylonian algorithm. About 4000 years ago Babylonian builders had a method for constructing the square root of a number from a sequence of successive approximations. To demonstrate the method, we'll construct $\sqrt{5}$. We want to find x so that

$$x^2 = 5 \qquad \text{or} \qquad x = \frac{5}{x}.$$

The second expression may seem a peculiar way to characterize x, but it is at least equivalent to the first. The advantage of the second expression is that it gives us *two* numbers to consider: x and $5/x$.

For example, suppose we guess that $x = 2$. Of course this is incorrect, because $2^2 = 4$, not 5. The two numbers we get from the second expression are 2 and $5/2 = 2.5$. One is smaller than $\sqrt{5}$, the other is larger (because $2.5^2 = 6.25$). Perhaps their *average* is a better estimate. The average is 2.25, and $2.25^2 = 5.0625$.

Although 2.25 is not $\sqrt{5}$, it is a better estimate than either 2.5 or 2. If we change x to 2.25, then

$$x = 2.25\,, \quad \frac{5}{x} = 2.222\,, \quad \text{and their average is } \ 2.236\,.$$

Is 2.236 a better estimate than 2.5 or 2.222? Indeed it is: $2.236^2 = 4.999696$. If we now change x to 2.236, a remarkable thing happens:

$$x = 2.236\,, \quad \frac{5}{x} = 2.236\,, \quad \text{and their average is } \ 2.236\,!$$

In other words, if we want accuracy to three decimal places, we have already found $\sqrt{5}$. All the digits have stabilized.

Suppose we want *greater* accuracy? Our routine readily obliges. If we set $x = 2.236000$, then

$$x = 2.236000\,, \quad \frac{5}{x} = 2.236136\,, \quad \text{and their average is } \ 2.236068\,.$$

In fact, $\sqrt{5} = 2.236068\ldots$ is accurate to six decimal places.

Here is a summary of the argument we have just developed, expressed in terms of $\sqrt{a}$ for an arbitrary positive number a.

If x is an estimate for $\sqrt{a}$, then the average of x and a/x is a better estimate.

Once we choose an *initial* estimate, this argument constructs a sequence of successive approximations to $\sqrt{a}$. The process of constructing the sequence is called the **Babylonian algorithm** for square roots.

A procedure that tells us how to carry out a sequence of steps, one at a time, to reach a specific goal is called an **algorithm**. Many algebraic processes are algorithms. The word is a Latinization of the name of the Persian astronomer Muhammad al-Khwārizmī (c.780 a.d.–c.850 a.d.), who lived and worked in Baghdad. The title of his seminal book *His ā b al-jabr wal-muq ā bala* (830 a.d.)—usually referred to simply as *al-Jabr*—has a Latin form that is even more familiar to mathematics students.

The Babylonian algorithm takes the current estimate x for $\sqrt{a}$ that we have at each stage and says "replace x by the average of x and a/x." This kind of instruction is ideally suited to a computer, because `A = B` in a computer program means "replace the current value of `A` by the current value of `B`." In the program BABYLON printed below, the algorithm is realized by a `FOR - NEXT` loop with a single line that reads "`x = (x + a / x) / 2`".

The three-step procedure that we used in chapter 1 to obtain values of S, I, and R in the epidemic problem is also an algorithm, and for that reason it was a straightforward matter to express it as the computer program SIRVALUE.

Program: BABYLON An algorithm to find $\sqrt{a}$

```
a = 5
x = 2
n = 6
FOR k = 1 TO n
      x = (x + a / x) / 2
      PRINT x
NEXT k
```

Output:

```
2.25
2.236111
2.236068
2.236068
2.236068
2.236068
```

2.3.3 Exercises

In many of the questions below you are asked to do things like divide an arc into 2,000,000 segments or find a certain number to 8 decimal places. This assumes you have fast computers

Use common sense in deciding how close an approximation to make

available. If you are using slower machines or programmable calculators, you should certainly feel free to scale back what is called for—perhaps to 200,000 segments or 6 decimal places. Use your own common sense; there's no value in sitting in front of a screen for an hour waiting for an answer to emerge. To approximate a length by 200,000 segments will take 10 times as long as to approximate it by 20,000 segments, which in turn will take 10 times as long as the 2,000 segment approximation. If you start with the cruder approximations, you should be able to get a good sense of what is reasonable to attempt with the facilities you have available, and modify what the problems call for accordingly.

1. By using a computer to graph $y = x^2 - 2^x$, find the solutions of the equation $x^2 = 2^x$ to four decimal place accuracy.

The program LENGTH.

2. Run the program LENGTH to verify that it gives the lengths of the individual segments and their total length.

3. What line in the program gives the instruction to work with the function $f(x) = x^2$? What line indicates the number of segments to be measured?

4. Each segment has a left and a right endpoint. What lines in the program designate the x- and y-coordinates of the left endpoint; the right endpoint?

5. Where in the program is the length of the k-th segment calculated? The segment is treated as the hypotenuse of a triangle whose length is measured by the Pythagorean theorem. How is the base of that triangle denoted in the program? How is the altitude of that triangle denoted?

6. Modify the program so that it uses 20 segments to estimate the length of the parabola. What is the estimated value?

7. Modify the program so that it estimates the length of the parabola using $200,\ 2,000,\ 20,000,\ 200,000,$ and $2,000,000$ segments. Compare your results with those tabulated below. [To speed the process up, you will certainly want to delete the `PRINT k, segment` statement that appears inside the loop. Do you see why?]

Table 2.3.6

Number of line segments	Sum of their lengths
2	1.460 404 813
20	1.478 756 512
200	1.478 940 994
2 000	1.478 942 839
20 000	1.478 942 857
200 000	1.478 942 857
2 000 000	1.478 942 857

8. What is the length of the parabola $y = x^2$ over the interval $0 \le x \le 1$, correct to 8 decimal places? What is the length, correct to 12 decimal places?

9. Starting at the origin, and moving along the parabola $y = x^2$, where are you when you've gone a total distance of 10?

10. Modify the program to find the length of the curve $y = x^3$ over the interval $0 \le x \le 1$. Find a value that is correct to 8 decimal places.

11. **Back to the circle.** Consider the unit circle centered at the origin. Pythagoras' Theorem shows that a point (x, y) is on the circle if and only if $x^2 + y^2 = 1$. If we solve this for y in terms of x, we get $y = \pm\sqrt{1 - x^2}$, where the plus sign gives us the upper half of the circle and the minus sign gives the lower half. This suggests that we look at the function $g(x) = \sqrt{1 - x^2}$. The arclength of $g(x)$ over the interval $-1 \le x \le 1$ should then be exactly π.

(a) Divide the interval into 100 pieces—what is the corresponding length?

(b) How many pieces do you have to divide the interval into to get an accuracy equal to that of Archimedes?

(c) Find the length of the curve $y = g(x)$ over the interval $-1 \le$

$x \leq 1$, correct to eight decimal places accuracy.

12. This question concerns the function $h(x) = \dfrac{4}{1+x^2}$.

(a) Sketch the graph of $y = h(x)$ over the interval $-2 \leq x \leq 2$.

(b) Find the length of the curve $y = h(x)$ over the interval $-2 \leq x \leq 2$.

13. Find the length of the curve $y = \sin x$ over the interval $0 \leq x \leq \pi$.

The program BABYLON.

14. Run the program BABYLON on a computer to verify the tabulated estimates for $\sqrt{5}$.

15. In whatever computer language you are using, it should be possible to tell the computer to print out more decimals. Your teacher can tell you how this is handled on your computers. Modify BABYLON to run with at least 14-digit precision in this and the following problems. Also modify the program so it prints out the square of the estimate each time as well. What is the estimated value of $\sqrt{5}$ in this circumstance? How many steps were needed to get this value? Use the square of this estimate as a measure of its accuracy. What is the square?

16. Use the Babylonian algorithm to find $\sqrt{80}$.

(a) First use 2 as your initial estimate. How many steps are needed for the calculations to stabilize—that is, to reach a value that doesn't change from one step to the next?

(b) Since $9^2 = 81$, a good first estimate for $\sqrt{80}$ is 9. How many steps are needed this time for the calculations to stabilize? Are the final values in (a) and (b) the same?

17. Use the Babylonian algorithm to find $\sqrt{250}$ and $\sqrt{1990}$. If you use 2 as the initial estimate in each case, how many steps are needed for the calculations to stabilize? If you use the integer nearest to the final answer as your initial estimate, then how many steps are needed? Square your answers to measure their accuracy.

18. The Babylonian algorithm is considered to be **very fast**, in the sense that each stage roughly doubles the number of digits that stabilize. Does your work on the preceding exercises confirm this observation? By comparison, is the routine that got the estimates for $S(3)$ (computed with the program SIRVALUE) faster or slower than the Babylonian algorithm?

2.4 Chapter Summary

2.4.1 The Main Ideas

- The exact numerical value of a quantity may not be known; the value is often given by an **approximation**.
- A numerical quantity is often given as the **limit** of a sequence of **successive approximations**.

- When a particular digit in a sequence of successive approximations **stabilizes**, that digit is assumed to appear in the limit.
- **Euler's method** is a procedure to construct a sequence of increasingly better **approximations** of a function defined by a set of rate equations and initial conditions. Each approximation is a piecewise linear function.
- The exact function defined by a set of rate equations and initial conditions can be expressed as the **limit** of a sequence of **successive Euler approximations** with smaller and smaller step sizes.

2.4.2 Expectations

- You should be able to use a program to **construct a sequence** of estimates for S, I, and R, given a specific S-I-R model with initial conditions.
- You should be able to **modify** the SIRVALUE and SIRPLOT programs to construct a sequence of estimates for the values of functions defined by other rate equations and initial conditions.
- You should be able to use programs that construct a sequence of **Euler approximations** for the function defined by a rate equation with an initial condition. The programs should provide both tabular and graphical output.
- You should be able to estimate the values of the roots of an equation $f(x) = 0$ using a **computer graphing package**.
- You should be able to find a square root using the **Babylonian algorithm**.
- You should be able to find the length of any piece of any curve.

2.4.3 Chapter Exercises

1. We have considered the logistic equation

$$y' = .1y\left(1 - \frac{y}{1000}\right)$$

On page we looked at the resulting graph of y vs. t, using the starting value $y(0) = 100$. There is another graphical interpretation, though, that is also instructive. Note that the logistic equation specifies y' as a function of y. Sketch the graph of this function. That is, plot y values on the horizontal axis and y' values on the vertical axis; points on the graph will thus be of the form (y, y'), where the y'-coordinate is the value given by the logistic equation for the given y value. What shape does the graph have?

   (a) For what value of y does this graph take on its largest value? Where does this y value appear in the graph of $y(t)$ versus t?

   (b) For what values of y does the graph of y' versus y cross the y-axis? Where do these y values appear in the graph of $y(t)$ versus t?

Grids on Graphs. When writing a graphing program it is often useful to have the computer draw a grid on the screen. This makes it easier to estimate numerical values, for instance. We can use a simple FOR–NEXT loop inside a program to do this. For instance, suppose we had written a program (SIRPLOT or SEQUENCE, for instance) with the graphics already in place. Suppose the screen window covered values from 0 to 100 horizontally, and 0 to 50,000 vertically. If we wanted to draw 21 vertical lines (including both ends) spaced 5 units apart and 11 horizontal lines spaced 5,000 units apart, the following two loops inserted in the program would work:

`FOR k = 0 TO 20`

Plot the line from `(5 * k, 0)` *to* `(5 * k, 50000)`

`NEXT k`
`FOR k = 0 TO 10`

Plot the line from `(0, 5000 * k)` *to* `(100, 5000 * k)`

`NEXT k`

You should make sure you see how these loops work and that you can modify them as needed.

1. If the screen window runs from -20 to 120 horizontally, and 250 to 750 vertically, how would you modify the loops above to create a vertical grid spaced 10 units apart and a horizontal grid spaced 25 units apart?

2. Go back to our basic S-I-R model. Modify SIRPLOT to calculate and plot on the same graph the values of $S(t)$, $I(t)$, and $R(t)$ for t going from 0 to 120, using a stepsize of $\Delta t = .1$. Include a grid with a horizontal spacing of 5 days, and a vertical spacing of 2000 people. You should get something that looks like this:

Chapter 3

The Derivative

In developing the S-I-R model in chapter 1 we took the idea of the rate of change of a population as intuitively clear. The rate at which one quantity changes with respect to another is a central concept of calculus and leads to a broad range of insights. The chief purpose of this chapter is to develop a fuller understanding—both analytic and geometric—of the connection between a function and its rate of change. To do this we will introduce the concept of the **derivative** of a function.

3.1 Rates of Change

3.1.1 The Changing Time of Sunrise

The sun rises at different times,

The time of sunrise is a function

depending on the date and location. At 40° N latitude (New York, Beijing, and Madrid are about at this latitude) in the year 1990, for instance, the sun rose at

Table 3.1.1

7:16	on January 23,
5:58	on March 24,
4:52	on July 25.

Clearly the time of sunrise is a function of the date. If we represent the time of sunrise by T (in hours and minutes) and the date by d (the day of the year), we can express this functional relation in the form $T = T(d)$. For example, from the table above we find $T(23) = 7{:}16$. It is not obvious from the table, but it is also true that the the rate at which the time of sunrise is changing is different at different times of the year—T' varies as d varies. We can see how the rate varies by looking at some further data for sunrise at the same latitude, taken from *The Nautical Almanac for the Year 1990*:

Table 3.1.2

date

Table 3.1.3 **Table 3.1.4**

January 20
23
26

Let's use this table to estimate the rate at which the time of sunrise is changing on January 23.

Calculate the rate using earlier and later dates

We'll use the times three days earlier and three days later, and compare them. On January 26 the sun rose 4 minutes earlier than on January 20. This is a change of -4 minutes in 6 days, so the rate of change is

$$-4\ \frac{\text{minutes}}{6\text{ days}} \approx -.67 \text{ minutes per day.}$$

We say this is the **rate** at which sunrise is changing on January 23, and we write

$$T'(23) \approx -.67\ \frac{\text{minutes}}{\text{day}}.$$

The rate is negative because the time of sunrise is decreasing—the sun is rising earlier each day.

Similarly, we find that around March 24 the time of sunrise is changing approximately $-9/6 = -1.5$ minutes per day, and around July 25 the rate is $5/6 \approx +.8$ minutes per day. The last value is positive, since the time of sunrise is increasing—the sun is rising later each day in July. Since March 24 is the 83rd day of the year and July 25 is the 206th, using our notation for rate of change we can write

$$T'(83) \approx -1.5\ \frac{\text{minutes}}{\text{day}}; \qquad T'(206) \approx .8\ \frac{\text{minutes}}{\text{day}}.$$

Notice that, in each case, we have calculated the rate on a given day by using times *shortly before* and *shortly after* that day. We will continue this pattern wherever possible. In particular, you should follow it when you do the exercises about a falling object, at the end of the section.

Once we have the rates, we can estimate the time of sunrise for dates not given in the table. For instance, January 28 is five days after January 23, so the total change in the time of sunrise from January 23 to January 28 should be approximately

$$\Delta T \approx -.67\ \frac{\text{minutes}}{\text{day}} \times 5 \text{ days} = -3.35 \text{ minutes.}$$

In whole numbers, then, the sun rose 3 minutes earlier on January 28 than on January 23. Since sunrise was at 7:16 on the 23rd, it was at 7:13 on the 28th.

By letting the change in the number of days be negative, we can use this same reasoning to tell us the time of sunrise on days shortly *before* the given dates. For example, March 18 is -6 days away from March 24, so the change in the time of sunrise should be

$$\Delta T \approx -1.5\ \frac{\text{minutes}}{\text{day}} \times -6 \text{ days} = +9 \text{ minutes.}$$

Therefore, we can estimate that sunrise occurred at 5:58 + 0:09 = 6:07 on March 18.

3.1.2 Changing Rates

Suppose instead of using the tabulated values for March we tried to use our January data to *predict* the time of sunrise in March. Now March 24 is 60 days after January 23, so the change in the time of sunrise should be approximately

$$\Delta T \approx -.67\ \frac{\text{minutes}}{\text{day}} \times 60 \text{ days} = -40.2 \text{ minutes},$$

and we would conclude that sunrise on March 24 should be at about 7:16 − 0:40 = 6:37, which is more than half an hour later than the actual time! This is a problem we met often in estimating future values in the S-I-R model.

Predictions over long time spans are less reliable

implicitly assume that the time of sunrise changes at the fixed rate of −.67 minutes per day over the entire 60-day time-span. But this turns out not to be true: the rate actually varies, and the variation is too great for us to get a useful estimate. Only with a much smaller time-span does the rate not vary too much.

Here is the same lesson in another context. Suppose you are travelling in a car along a busy road at rush hour and notice that you are going 50 miles per hour. You would be fairly confident that in the next 30 seconds (1/120 of an hour) you will travel about

$$\Delta\,\text{position} \approx 50\ \frac{\text{miles}}{\text{hour}} \times \frac{1}{120} \text{ hour} = \frac{5}{12} \text{ mile} = 2200 \text{ feet}.$$

The actual value ought to be within 50 feet of this, making the estimate accurate to within about 2% or 3%. On the other hand, if you wanted to estimate how far you would go in the next 30 minutes, your speed would probably fluctuate too much for the calculation

$$\Delta\,\text{position} \approx 50\ \frac{\text{miles}}{\text{hour}} \times \frac{1}{2} \text{ hours} = 25 \text{ miles}$$

to have the same level of reliability.

3.1.3 Other Rates, Other Units

In the S-I-R model the rates we analyzed were **population growth rates**. They told us how the three populations changed over time, in units of persons per day. If we were studying the growth of a colony of mold, measuring its size by its weight (in grams), we could describe *its* population growth rate in units of grams per hour. In discussing the motion of an automobile, the rate we consider is the **velocity** (in miles per hour), which tells us how the distance from some starting point changes over time. We also pay attention to the rate at which *velocity* changes over time. This is called **acceleration**, and can be measured in miles per hour per hour.

While many rates do involve changes with respect to time, other rates do not. Two examples are the survival rate for a disease (survivors per thousand infected persons) and the dose rate for a medicine (milligrams per pound of body weight).

Examples of rates

Other common rates are the annual birth rate and the annual death rate, which might have values like 19.3 live births per 1,000 population and 12.4 deaths per 1,000 population. Any quantity expressed as a percentage, such as an interest rate or an unemployment rate, is a rate of a similar sort. An unemployment rate of 5%, for instance, means 5 unemployed workers per 100 workers. There are many other examples of rates in the economic world that make use of a variety of units—exchange rates (e.g., francs per dollar), marginal return (e.g., dollars of profit per dollar of change in price).

Sometimes we even want to know the rate of change of one rate with respect to another rate.

The rate of change of a rate

For example, automobile fuel economy (in miles per gallon—the first rate) changes with speed (in miles per hour—the second rate), and we can measure the rate of change of fuel economy with speed. Take a car that goes 22 miles per gallon of fuel at 50 miles per hour, but only 19 miles per gallon at 60 miles per hour. Then its fuel economy is changing approximately at the rate

$$dfrac\Delta\,\text{fuel economy}\Delta\,\text{speed} = \frac{19-22\text{ miles per gallon}}{60-50\text{ miles per hour}}$$

$$[5pt] = -.3\text{ miles per gallon } \textit{per} \text{ mile per hour.}$$

3.1.4 Exercises

A falling object. These questions deal with an object that is held motionless 10,000 feet above the surface of the ocean and then dropped. Start a clock ticking the moment it is dropped, and let D be the number of feet it has fallen after the clock has run t seconds. The following table shows some of the values of t and D.

Table 3.1.9

time (seconds)	distance (feet)
0	0.00
1	15.07
2	56.90
3	121.03
4	203.76
5	302.00
6	413.16
7	535.10

1. What units do you use to measure velocity—that is, the rate of change of distance with respect to time—in this problem?

2. Make a careful graph that shows these eight data points. Put *time* on the horizontal axis. Label the axes and indicate the units you are using on each.

  (a) The slope of any line drawn on this time–distance graph has the units of a *velocity*. Explain why.

3. Make three estimates of the velocity of the falling object at the 2 second mark using the distances fallen between these times:
i) from 1 second to 2 seconds;
ii) from 2 seconds to 3 seconds;
iii) from 1 second to 3 seconds.

4. Each of the estimates in the previous question corresponds to the slope of a particular line you can draw in your graph. Draw those lines and label each with the corresponding velocity.

(a) Which of the three estimates in the previous question do you think is best? Explain your choice.

5. Using your best method, estimate the velocity of the falling object after 4 seconds have passed.

6. Is the object speeding up or slowing down as it falls? How can you tell?

7. Approximate the velocity of the falling object after 7 seconds have passed. Use your answer to estimate the number of feet the object has fallen after 8 seconds have passed. Do you think your estimate is too high or too low? Why?

8. For those of you attending a school at which you pay tuition, find out what the tuition has been for each of the last four years at your school.

(a) At what rate has the tuition changed in each of the last three years? What are the units? In which year was the rate the greatest?

(b) A more informative rate is often the **inflation rate** which doesn't look at the dollar change per year, but at the percentage change per year—the dollar change in tuition in a year's time expressed as a percentage of the tuition at the beginning of the year. What is the tuition inflation rate at your school for each of the last three years? How do these rates compare with the rates you found in part (a)?

(c) If you were interested in seeing how the inflation rate was changing over time, you would be looking at the rate of change of the inflation rate. What would the units of this rate be? What is the rate of change of the inflation rate at your school for the last two years?

(d) Using all this information, what would be your estimate for next year's tuition?

9. Your library should have several reference books giving annual statistics of various sorts. A good one is the *Statistical Yearbook* put out by the United Nations with detailed data from all over the world on manufacturing, transportation, energy, agriculture, tourism, and culture. Another is *Historical Statistics of the United States, Colonial Times–1970*. Select an interesting quantity and compare its growth rate at different times or for different countries. Calculate this growth rate over a stretch of four or five years and report whether there are any apparent patterns. Calculate the rate of change of this growth rate and interpret its values.

10. Oceanographers are very interested in the **temperature profile** of the part of the ocean they are studying. That is, how does the temperature T (in degrees Celsius) vary with the depth d (measured in meters). A typical temperature profile might look something like the following:

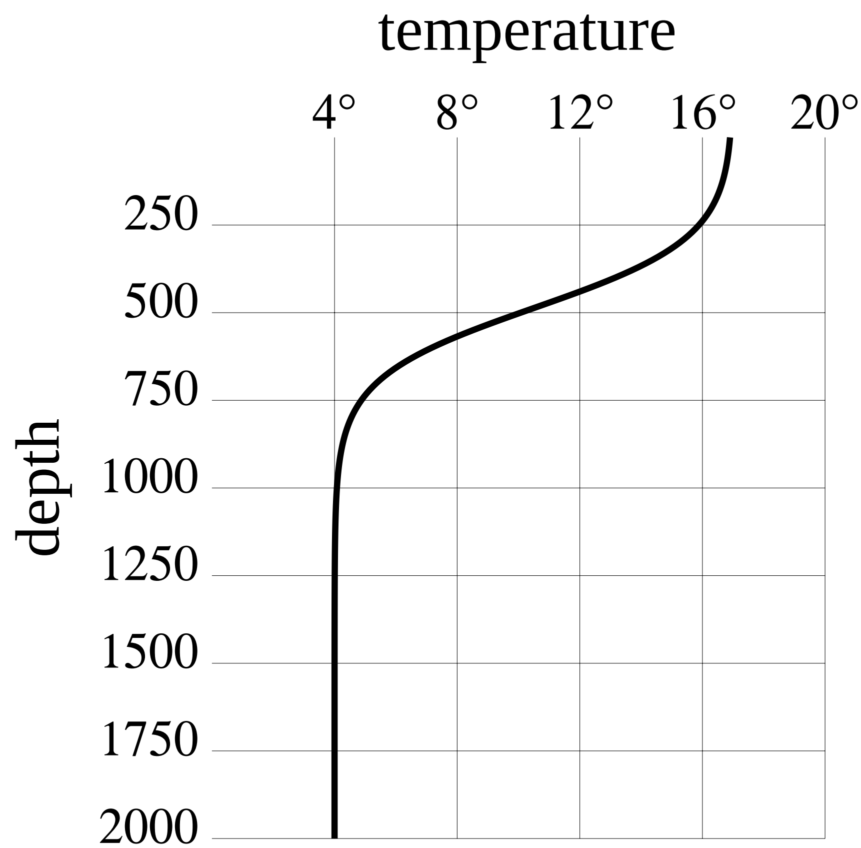


Figure 3.1.10 image

(Note that since water is densest at 4° C, water at that temperature settles to the bottom.)

(a) What are the units for the rate of change of temperature with depth?

(b) In this graph will the rate be positive or negative? Justify your answer.

(c) At what rate is the temperature changing at a depth of 0 meters? 500 meters? 1000 meters?

(d) Sketch a possible temperature profile for this location if the surface is iced over.

(e) This graph is not oriented the way our graphs have been up til now. Why do you suppose oceanographers (and geologists and atmospheric scientists) often draw graphs with axes positioned like this?

3.2 Microscopes and Local Linearity

3.2.1 The Graph of Data

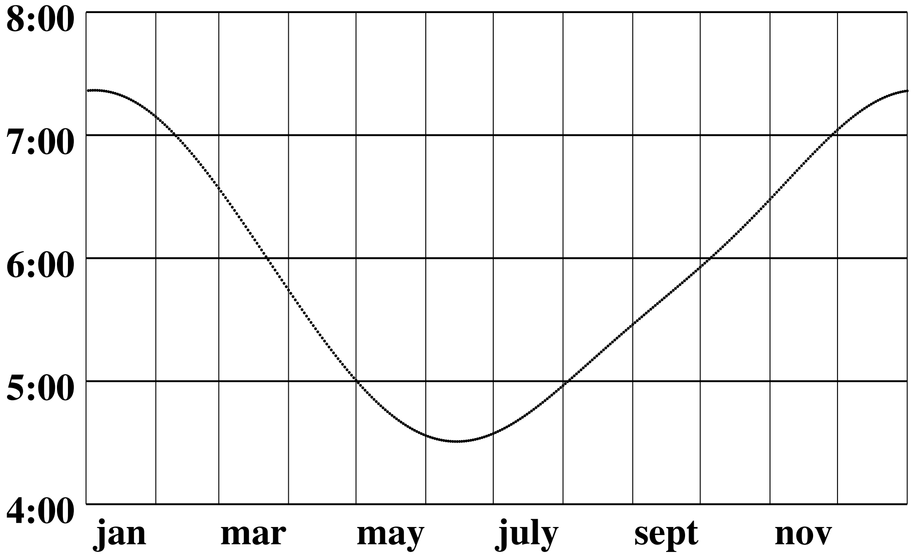


Figure 3.2.1 image

This section is about seeing rates geometrically. We know from chapter 1 that we can visualize the rate of change of a *linear* function as the slope of its graph. Can we say the same thing about the sunrise function? The graph of this function appears at the right; it plots the time of sunrise (over the

course of a year at 40° N latitude), as a function of the date. The graph is curved, so the sunrise function is not linear. There is no immediately obvious connection between rate and slope. In fact, it isn't even clear what we might mean by the *slope* of this graph! We can make it clear by using a **microscope**.

Imagine we have a microscope

Zoom in on the graph with a microscope

that allows us to "zoom in" on the graph near each of the the three dates we considered in section 1. If we put each magnified image in a window, then we get the following:

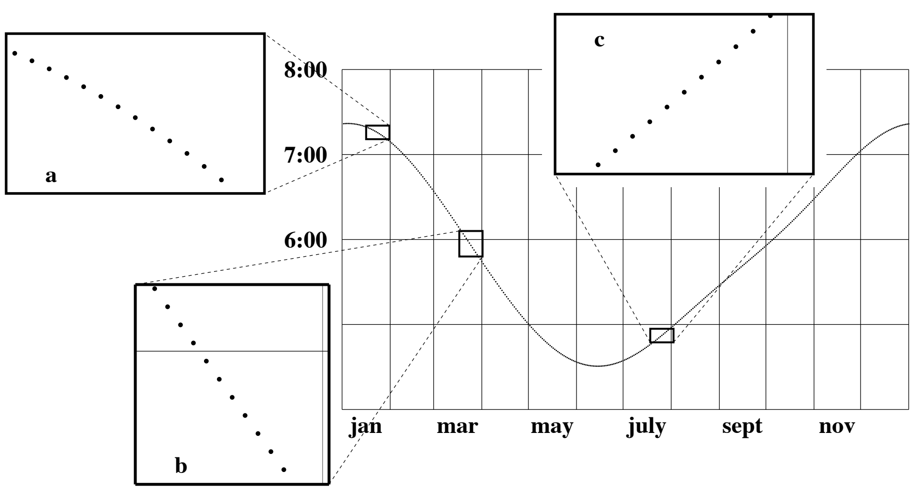


Figure 3.2.2 image

Notice how different the graph looks under the microscope. First of all, it now shows up clearly as a collection of separate points—one for each day of the year.

The graph looks straight under a microscope

Second, the points in a particular window lie on a line that is essentially straight. The straight lines in the three windows have very different slopes, but that is only to be expected.

What is the connection between these slopes and the rates of change we calculated in the last section? To decide, we should calculate the slope in each window. This involves choosing a pair of points (d_1, T_1) and (d_2, T_2) on the graph and calculating the ratio

$$\frac{\Delta T}{\Delta d} = \frac{T_2 - T_1}{d_2 - d_1}.$$

In window **a** we'll take the two points that lie three days on either side of the central date, January 23. These points have coordinates $(20, 7{:}18)$ and $(26, 7{:}14)$ (table, page). The slope is thus

$$\frac{\Delta T}{\Delta d} = \frac{7{:}14 - 7{:}18}{26 - 20} = \frac{-4 \text{ minutes}}{6 \text{ days}} = -.67 \, \frac{\text{minutes}}{\text{day}}.$$

If we use the same approach in the other two windows we find that the line in window **b** has slope -1.5 min/day,

Slope and rate calculations are the same

while the line in window **c** has slope $+.8$ min/day. These are exactly the same calculations we did in section 1 to determine the rate of change of the time of sunrise around January 23, March 24, and July 25, and they produce the same values we obtained there:

$$T'(23) \approx -.67 \, \frac{\text{min}}{\text{day}}, \qquad T'(83) \approx -1.5 \, \frac{\text{min}}{\text{day}}, \qquad T'(206) \approx .8 \, \frac{\text{min}}{\text{day}}.$$

This is a crucial observation which we use repeatedly in other contexts; let's pause and state it in general terms:

The **rate of change of a function** at a point is equal tothe **slope of its graph** at that point, if the graphlooks straight when we view it under a microscope.

3.2.2 The Graph of a Formula

Rates and slopes are really the same thing—that's what we learn by using a microscope to view the graph of the sunrise function. But the sunrise graph consists of a finite number of disconnected points—a very common situation when we deal with data. In such cases it doesn't make sense to magnify the graph too much. For instance, we would get no useful information from a window that was narrower than the space between the data points.

High-power magnification is possible with a formula

There is no such limitation if we use a microscope to look at the graph of a function given by a formula, though. We can zoom in as close as we wish and still see a continuous curve or line. By using a high-power microscope, we can learn even more about rates and slopes.

Consider this rather complicated-looking function:

$$f(x) = \frac{2 + x^3 \cos x + 1.5^x}{2 + x^2}.$$

Let's find $f'(27)$, the rate of change of f when $x = 27$. We need to zoom in on the graph of f at the point $(27, f(27)) = (27, 69.859043)$. We do this in stages, producing a succession of windows that run clockwise from the upper left. Notice how the graph gets straighter with each magnification.

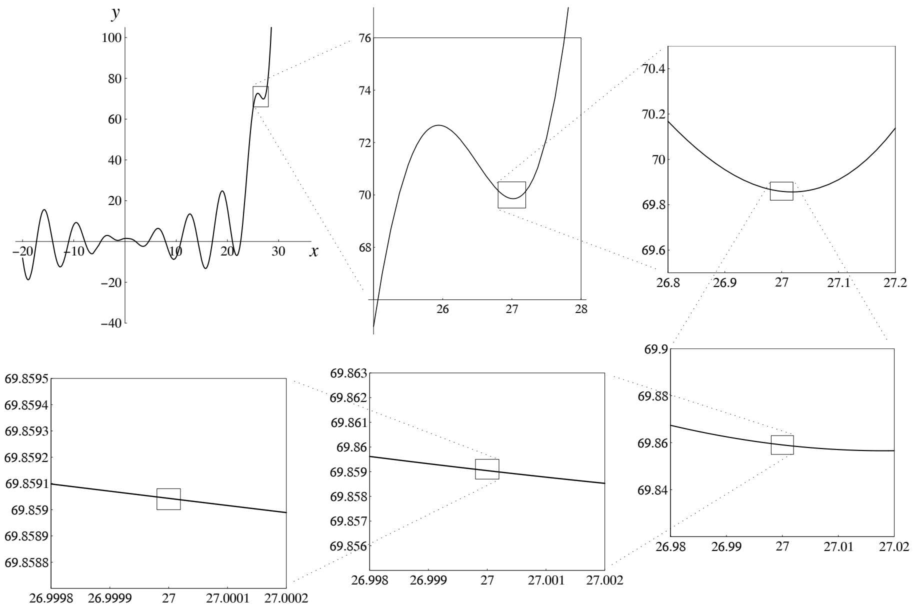


Figure 3.2.3 image

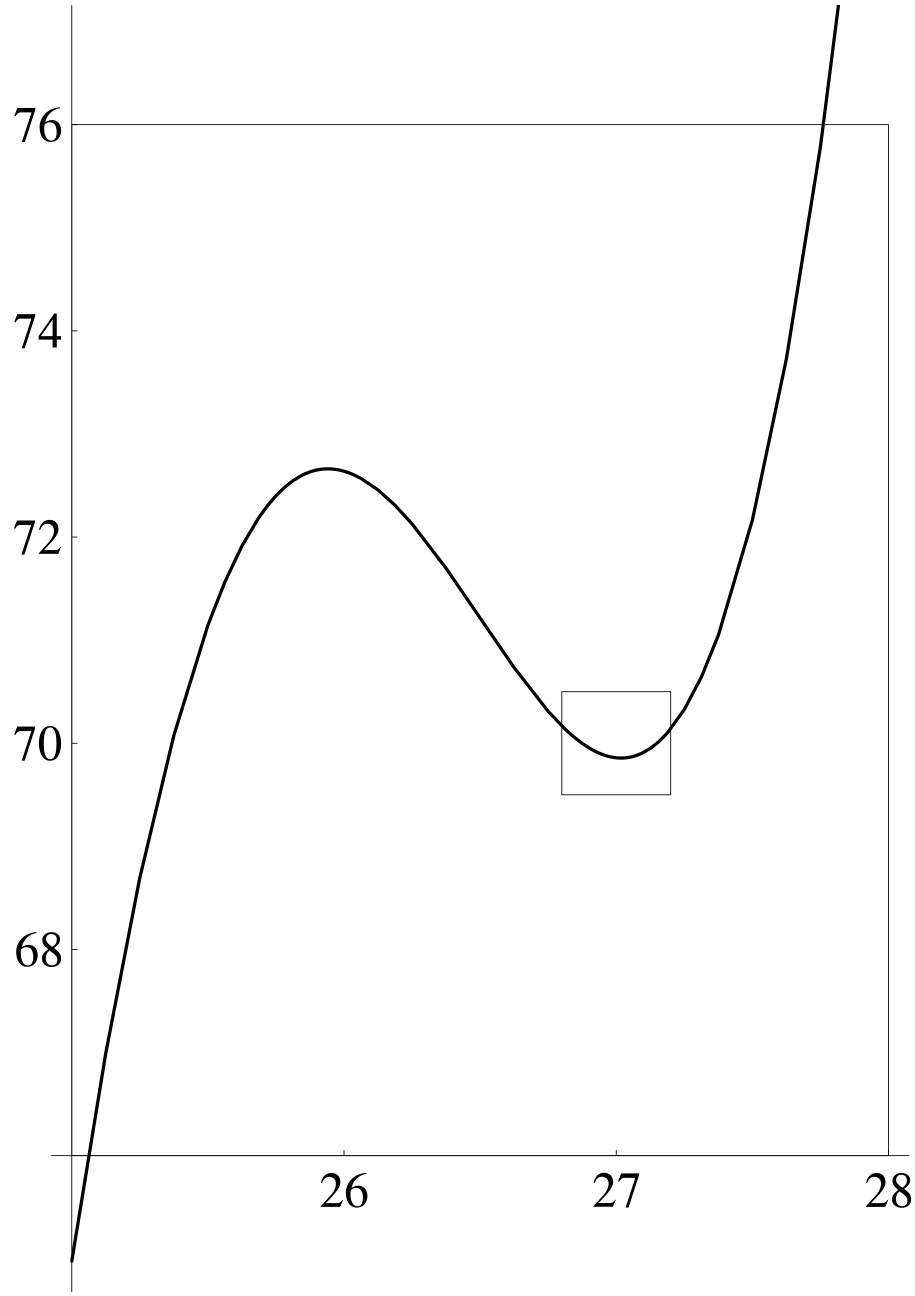


Figure 3.2.4 image

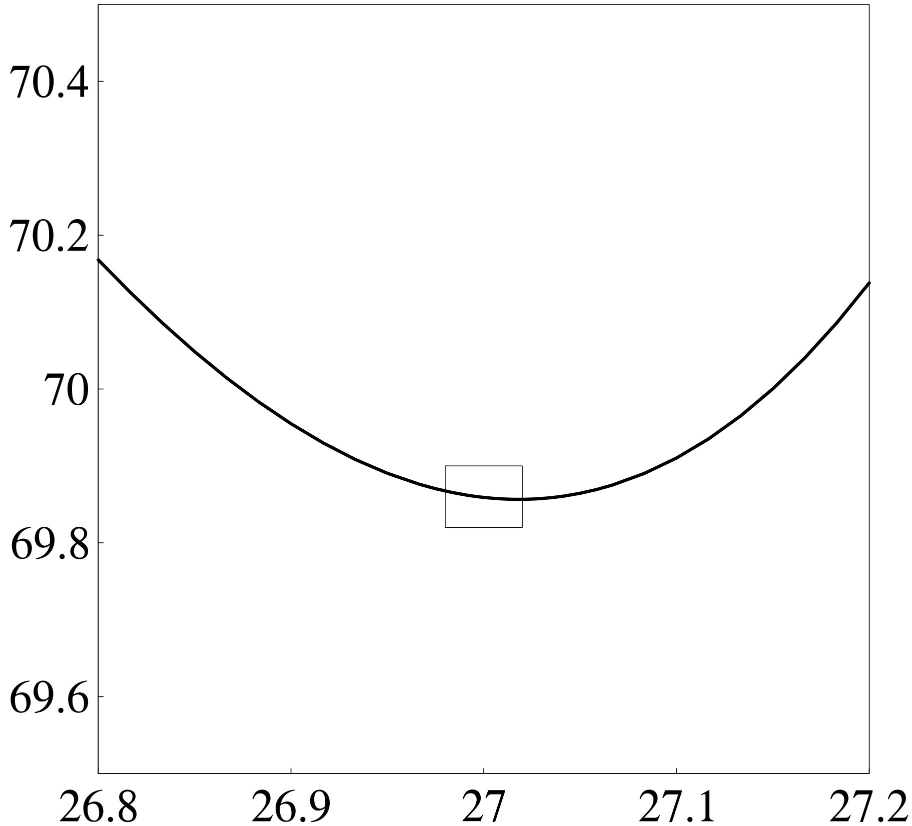


Figure 3.2.5 image

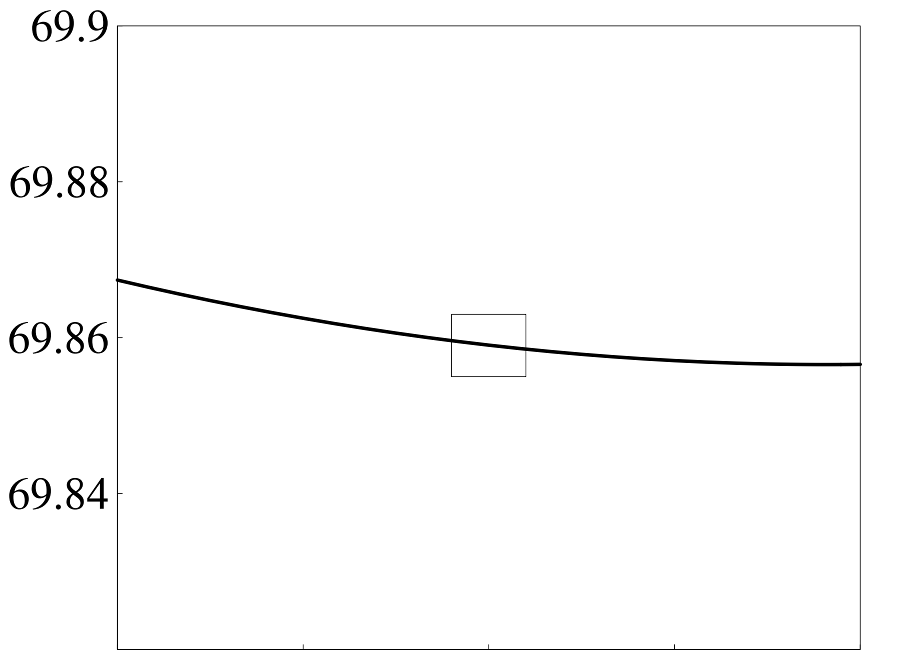


Figure 3.2.6 image

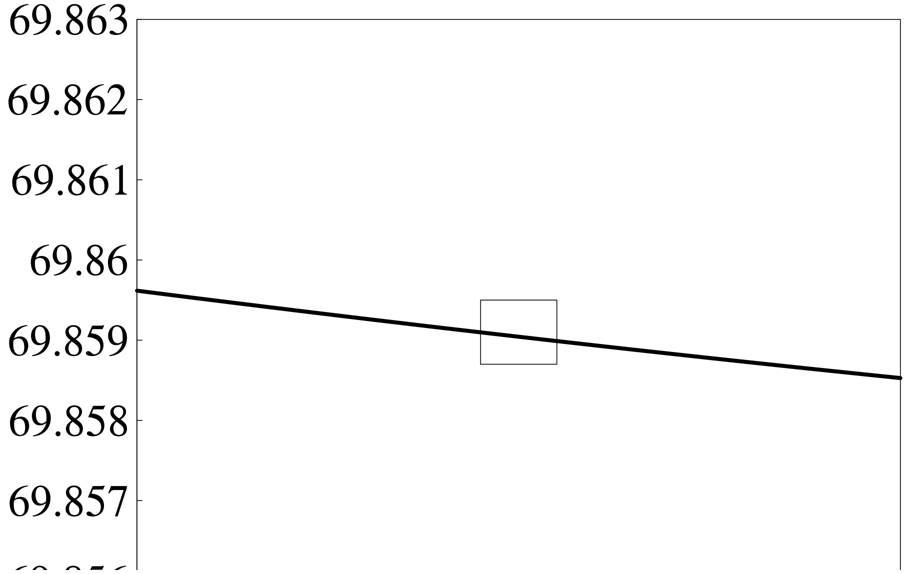


Figure 3.2.7 image

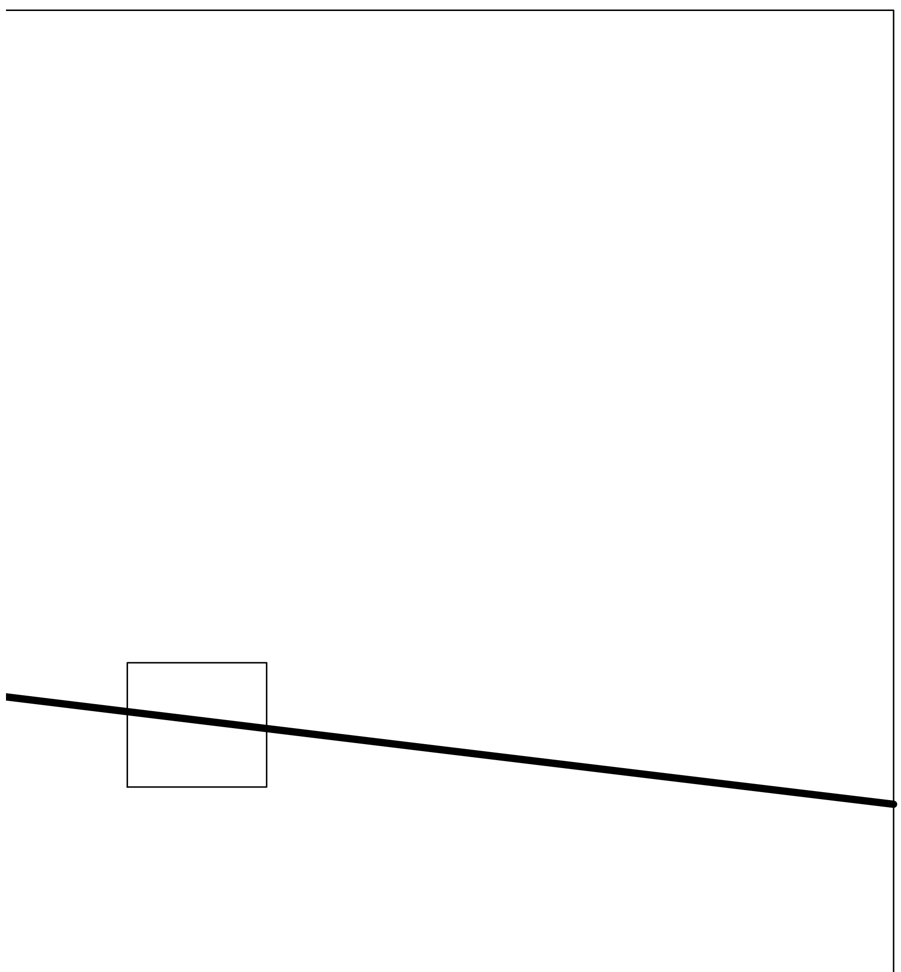


Figure 3.2.8 image

We will need a way to describe the part of the graph that we see in a window. Let's call it the **field of view**.

The field of view of a window

The field of view of each window is only one-tenth as wide as the previous one, and the field of view of the last window is only one-millionth of the first! The last window shows what we would see if we looked at the original graph with a million-power microscope.

The microscope used to study functions is real, but it is different from the one a biologist uses to study micro-organisms. Our microscope is a computer graphing program that can "zoom in" on any point on a graph. The computer screen is the window you look through, and you determine the field of view when you set the size of the interval over which the graph is plotted.

Here is our point of departure: *the rate $f'(27)$ is the slope of the graph of $f(x)$ at $x = 27$ when we magnify the graph enough to make it look straight.* But how much is enough? Which window should we use? The following table gives the slope $\Delta y/\Delta x$ of the line that appears in each of the last four windows in the sequence. For Δx we take the difference between the x-coordinates of the points at the ends of the line, and for Δy we take the difference between the y-coordinates. In particular, the width of the field of view in each case is Δx.

Δx	Δy	$\Delta y/\Delta x$
.04	$-1.081\,508\,24 \times 10^{-2}$	$-.270\,377\,066$
.004	$-1.089\,338\,27 \times 10^{-3}$	$-.272\,334\,556$
.0004	$-1.089\,416\,49 \times 10^{-4}$	$-.272\,354\,131$
.00004	$-1.089\,417\,28 \times 10^{-5}$	$-.272\,354\,327$

As you can see, it *does* matter how much we magnify. The slopes $\Delta y/\Delta x$ in the table are not quite the same, so we don't yet have a definite value for $f'(27)$. The table gives us an idea how we *can* get a definite value, though. Notice that the slopes get more and more alike, the more we magnify. In fact, under successive magnifications the first five digits of $\Delta y/\Delta x$ have **stabilized**. We saw in chapter 2 how to think about a sequence of numbers whose digits stabilize one by one. We should treat the values of $\Delta y/\Delta x$ as

The slope is a limit

successive approximations to the slope of the graph. The exact value of the slope is then the **limit** of these approximations as the width of the field of view shrinks to zero:

$$f'(27) = \text{the slope of the graph} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}.$$

In the limit process we take $\Delta x \to 0$ because Δx is the width of the field of view. Since five digits of $\Delta y/\Delta x$ have stabilized, we can write

$$f'(27) = -.27235\ldots.$$

To find $f'(x)$ at some other point x, proceed the same way. Magnify the graph at that point repeatedly, until the value of the slope stabilizes. The method is very powerful. In the exercises you will have an opportunity to use it with other functions.

By using a microscope of arbitrarily high power, we have obtained further insights about rates and slopes. In fact, with these insights we can now state definitively what we mean by the slope of a curved graph and the rate of change of a function.

Definition. The **slope** of a graph at a point is the *limit* of the slopes seen in a microscope at that point, as the field of view shrinks to zero.

Definition. The **rate of change** of a function at a point is the slope of its graph at that point. Thus the rate of change is also a limit.

To calculate the value of the slope of the graph of $f(x)$ when $x = a$, we have to carry out a limit process. We can break down the process into these four steps:1.Magnify the graph at the point $(a, f(a))$ until it appears straight.2.Calculate the slope of the magnified segment.3.Repeat steps 1 and 2 with successively higher magnifications.4.Take the limit of the succession of slopes produced in step 3.

3.2.3 Local Linearity

The crucial property of a microscope

A microscope gives a local view

is that it allows us to look at a graph **locally**, that is, in a small neighborhood of a particular point. The two functions we have been studying in this section have curved graphs—like most functions. But *locally*, their graphs are straight—or nearly so. This is a remarkable property, and we give it a name. We say these functions are **locally linear**. In other words, a locally linear function looks like a linear function, locally.

The graph of a linear function has a definite slope at every point, and so does a *locally* linear function. For a linear function, the slope is easy to calculate, and it has the same value at every point. For a locally linear function, the slope is harder to calculate; it involves a limit process. The slope also varies from point to point.

How common is local linearity?

All the standard functions are locally linear at almost all points

All the standard functions you already deal with are locally linear almost everywhere. To see why we use the qualifying phrase "almost everywhere," look at what happens when we view the graph of $y = f(x) = x^{2/3}$ with a microscope. At any point other than the origin, the graph is locally linear. For instance, if we view this graph over the interval from 0 to 2 and then zoom in on the point $(1, 1)$ by two successive powers of 10, here's what we see:

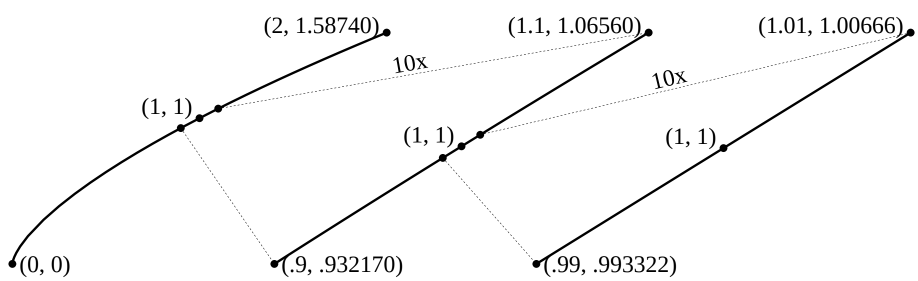


Figure 3.2.9 image

As the field of view shrinks, the graph looks more and more like a straight line. Using the highest magnification given, we estimate the slope of the graph—and hence the rate of change of the function—to be

$$f'(1) \approx \frac{\Delta y}{\Delta x} = \frac{1.006656 - .993322}{1.01 - .99} = \frac{.013334}{.02} = .6667.$$

Similarly, if we zoom in on the point $(.001, .01)$ we get:

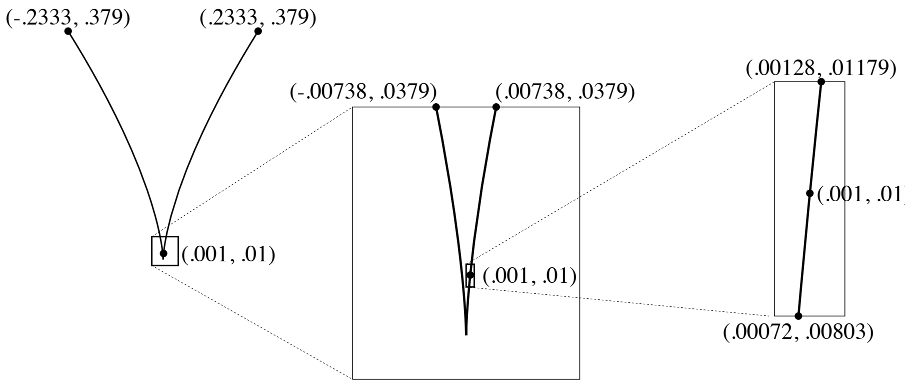


Figure 3.2.10 image

In the last window the graph looks like a line of slope

$$f'(.001) \approx \frac{.0118 - .0082}{.00128181 - .00074254} = \frac{.0036}{.00053937} = 6.674.$$

At the origin, though, something quite different happens:

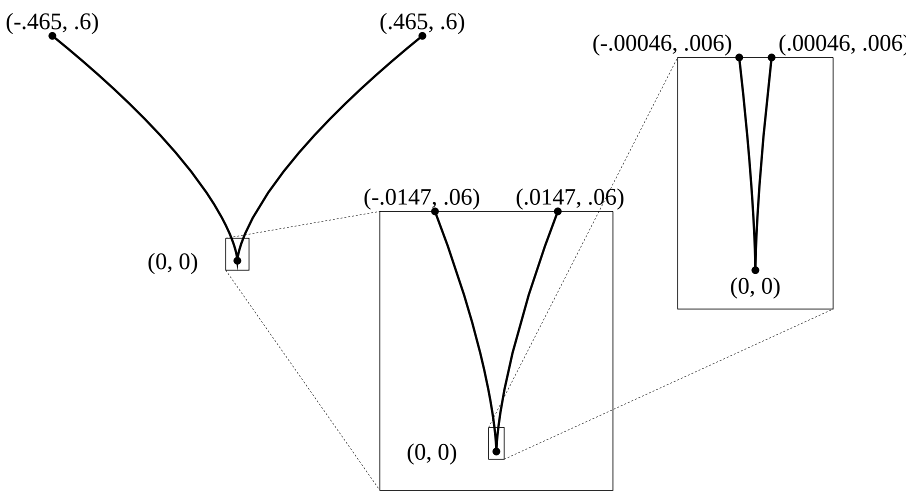


Figure 3.2.11 image

The graph simply looks more and more sharply pointed the closer we zoom in to the origin—it never looks like a straight line. However, the origin turns out to be the only point where the graph does not eventually look like a straight line.

In spite of these examples, it is important to realize that local linearity is a very special property. There are some functions that fail to be locally linear anywhere! Such functions are called **fractals**.

Fractals are locally *non*-linear objects

No matter how much you magnify the graph of a fractal at any point, it continues to look non-linear—bent and "pointy" in various ways. In recent years fractals have been used in problems where the more common (locally linear) functions are inadequate. For instance, they describe irregular shapes like coastlines and clouds, and they model the way molecules are knocked about in a fluid (this is called *Brownian motion*). However, calculus does not deal with such functions. On the contrary:

Calculus studies functions that are locally linear almost everywhere.

3.2.4 Exercises

Using a microscope.

1. Use a computer microscope to do the following. (A suggestion: first look at each graph over a fairly large interval.)

(a) With a window of size $\Delta x = .002$, estimate the rate $f'(1)$ where $f(x) = x^4 - 8x$.

(b) With a window of size $\Delta x = .0002$, estimate the rate $g'(0)$ where $g(x) = 10^x$.

(c) With a window of size $\Delta t = .05$, estimate the slope of the graph of $y = t + 2^{-t}$ at $t = 7$.

(d) With a window of size $\Delta z = .0004$, estimate the slope of the graph of $w = \sin z$ at $z = 0$.

2. Use a computer microscope to determine the following values, correct to one decimal place. Obtain estimates using a *sequence* of windows, and shrink the field of view until the first two decimal places stabilize. Show *all* the estimates you constructed in each sequence.

(a) $f'(1)$ where $f(x) = x^4 - 8x$.

(b) $h'(0)$ where $h(s) = 3^s$.

(c) The slope of the graph of $w = \sin z$ at $z = \pi/4$.

(d) The slope of the graph of $y = t + 2^{-t}$ at $t = 7$.

(e) The slope of the graph of $y = x^{2/3}$ at $x = -5$.

3. For each of the following functions, magnify its graph at the indicated point until the graph appears straight. Determine the equation of that straight line. Then verify that your equation is correct by plotting it as a second function in the same window you are viewing the given function. (The two graphs should "share phosphor"!)

(a) $f(x) = \sin x$ at $x = 0$;

(b) $\varphi(t) = t + 2^{-t}$ at $t = 7$;

(c) $H(x) = x^{2/3}$ at $x = -5$.

4. Consider the function that we investigated in the text:

$$f(x) = \frac{2 + x^3 \cos x + 1.5^x}{2 + x^2}.$$

(a) Determine $f(0)$.

(b) Make a sketch of the graph of f on the interval $-1 \le x \le 1$. Use the same scale on the horizontal and vertical axes so your graph shows slopes accurately.

(c) Sketch what happens when you magnify the previous graph so the field of view is only $-.001 \le x \le .001$.

(d) Estimate the slope of the line you drew in the part (c).

(e) Estimate $f'(0)$. How many decimal places of accuracy does your estimate have?

(f) What is the equation of the line in part (c)?

5. A function that occurs in several different contexts in physical problems is

$$g(x) = \frac{\sin x}{x}.$$

Use a graphing program to answer the following questions.

(a) Estimate the rate of change of g at the following points to two decimal place accuracy:

$$g'(1), \qquad g'(2.79), \qquad g'(\pi), \qquad g'(3.1).$$

(b) Find three values of x where $g'(x) = 0$.

(c) In the interval from 0 to 2π, where is g decreasing the most rapidly? At what rate is it decreasing there?

(d) Find a value of x for which $g'(x) = -\,0.25$.

(e) Although $g(0)$ is not defined, the function $g(x)$ seems to behave nicely in a neighborhood of 0. What seems to be true about $g(x)$ and $g'(x)$ when x is near 0?

(f) According to your graphs, what value does $g(x)$ approach as $x \to 0$? What value does $g'(x)$ approach as $x \to 0$?

Rates from graphs; graphs from rates.

6.)Sketch the graph of a function f that has $f(1) = 1$ and $f'(1) = 2$.

(a) Sketch the graph of a function f that has $f(1) = 1$, $f'(1) = 2$, and $f(1.1) = -5$.

7. A and B start off at the same time, run to a point 50 feet away, and return, all in 10 seconds. A graph of distance from the starting point as a function of time for each runner appears below. It tells where each runner is during this time interval.

(a) Who is in the lead during the race?

(b) At what time(s) is A farthest ahead of B? At what time(s) is B farthest ahead of A?

(c) Estimate how fast A and B are going after one second.

(d) Estimate the velocities of A and B during each of the ten seconds. Be sure to assign *negative* velocities to times when the distance to the starting point is *shrinking*. Use these estimates to sketch graphs of the velocities of A and B versus time. (Although the velocity of B changes rapidly around $t = 5$, assume that the graph of B's distance *is* locally linear at $t = 5$.)

(e) Use your graphs in (d) to answer the following questions. When is A going faster than B? When is B going faster than A? Around what time is A running at -5 feet/second (i.e., running 5 feet/second *toward* the starting point)?

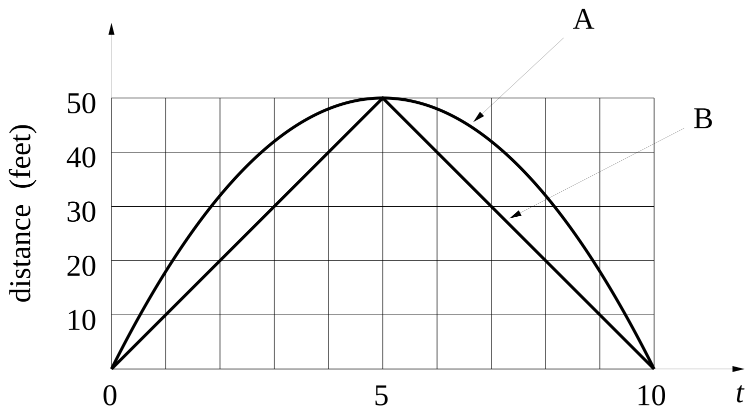


Figure 3.2.12 image

8. For each of the following functions draw a graph that reflects the given information. Restate the given information in the language and notation of rate of change, paying particular attention to the units in which any rate of change is expressed.

(a) A woman's height h (in inches) depends on her age t (in years). Babies grow very rapidly for the first two years, then more slowly until the adolescent growth spurt; much later, many women actually become shorter because of loss of cartilage and bone mass in the spinal column.

(b) The number R of rabbits in a meadow varies with time t (in years). In the early years food is abundant and the rabbit population grows rapidly. However, as the population of rabbits approaches the "carrying capacity" of the meadow environment, the growth rate slows, and the population never exceeds the carrying capacity. Each year, during the harsh conditions of winter, the population dies back slightly, although it never gets quite as low as its value the previous year.

(c) In a fixed population of couples who use a contraceptive, the average number N of children per couple depends on the effectiveness E (in percent) of the contraceptive. If the couples are using a contraceptive of low effectiveness, a small increase in effectiveness has a small effect on the value of N. As we look at contraceptives of greater and greater effectiveness, small additional increases in effectiveness have larger and larger effects on N.

9. If we graph the distance travelled by a parachutist in freefall as a function of the length of time spent falling, we would get a picture something like the following:

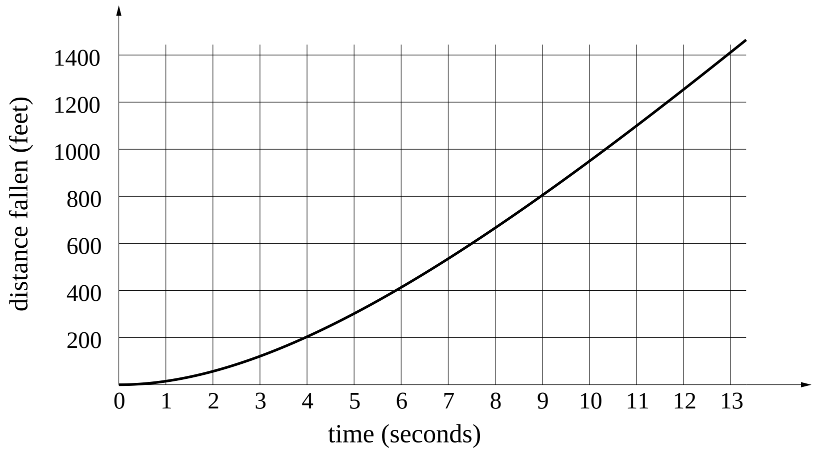


Figure 3.2.13 image

(a) Use this graph to make estimates of the parachutist's velocity at the end of each second.

(b) Describe what happens to the velocity as time passes.

(c) How far do you think the parachutist would have fallen by the end of 15 seconds?

10. True or false. If you think a statement is true, give your reason; if you think a statement is false, give a counterexample—i.e., an example that shows why it must be false.

(a) If $g'(t)$ is positive for all t, we can conclude that $g(214)$ is positive.

(b) If $g'(t)$ is positive for all t, we can conclude that $g(214) > g(17)$.

(c) Bill and Samantha are driving separate cars in the same direction along the same road. At the start Samantha is 1 mile in front of Bill. If their speeds are the same at every moment thereafter, at the end of 20 minutes Samantha will be 1 mile in front of Bill.

(d) Bill and Samantha are driving separate cars in the same direction along the same road. They start from the same point at 10 am and arrive at the same destination at 2 pm the same afternoon. At some time during the four hours their speeds must have been exactly the same.

When local linearity fails.

11. The absolute value function $f(x) = |x|$ is *not* locally linear at $x = 0$. Explore this fact by zooming in on the graph at $(0, 0)$. Describe what you see in successively smaller windows. Is there any change?

12. Find three points where the function $f(x) = |\cos x|$ fails to be locally linear. Sketch the graph of f to demonstrate what is happening.

13. Zoom in on the graph of $y = x^{4/5}$ at $(0, 0)$. In order to get an accurate picture, be sure that you use the same scales on the horizontal and vertical axes. Sketch what you see happening in successive windows. Is the function $x^{4/5}$ locally linear at $x = 0$?

14. Is the function $x^{4/5}$ locally linear at $x = 1$? Explain your answer.

15. This question concerns the function $K(x) = x^{10/9}$.

(a) Sketch the graph of K on the interval $-1 \le x \le 1$. Compare K to the absolute value function $|x|$. Are they similar or dissimilar? In what ways? Would you say K is locally linear at the origin, or not?

(b) Magnify the graph of K at the origin repeatedly, until the field of view is no bigger than $\Delta x \le 10^{-10}$. As you magnify, be sure the scales on the horizontal and vertical axes remain the same, so you get a true picture of the slopes. Sketch what you see in the final window.

(c) After using the microscope do you change your opinion about the local linearity of K at the origin? Explain your response.

3.3 The Derivative

3.3.1 Definition

One of our main goals in this chapter is to make precise the notion of the rate of change of a function. In fact, we have already done that in the last section. We defined the rate of change of a function at a point to be the slope of its graph at that point; we defined the slope, in turn, by a four-step limit process. Thus, the precise definition of a rate of change involves a limit, and it involves geometric visualization—we think of a rate as a slope. We introduce a new word—*derivative*—to embrace both of these concepts as we now understand them.

Definition. The **derivative** of the function $f(x)$ at $x = a$ is its rate of change a
$x = a$, which is the same as the slope of its graph at $(a, f(a))$. The derivative of j
at a is denoted $\boldsymbol{f'(a)}$.

Figure 3.3.1

Later in this section we will extend our interpretation of the derivative to include the idea of a *multiplier*, as well as a rate and a slope. Besides providing us with a single word to describe rates, slopes, and multipliers, the term "derivative" also reminds us that the quantity $f'(a)$ is *derived* from information about the function f in a particular way. It is worth repeating here the four steps by which we derive $f'(a)$:1.Magnify the graph at the point $(a, f(a))$ until it appears straight.2.Calculate the slope of the magnified segment.3.Repeat steps 1 and 2 with successively higher magnifications.4.Take the limit of the succession of slopes produced in step 3.We can express this limit in analytic form in the following way:

$$f'(a) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{h \to 0} \frac{f(a+h) - f(a-h)}{2h}.$$

The **difference quotient**

$$\frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a-h)}{2h}$$

is the usual way we estimate the slope of the magnified graph of f at the point $(a, f(a))$. As the following figure shows, the calculation involves two points equally spaced on either side of $(a, f(a))$.

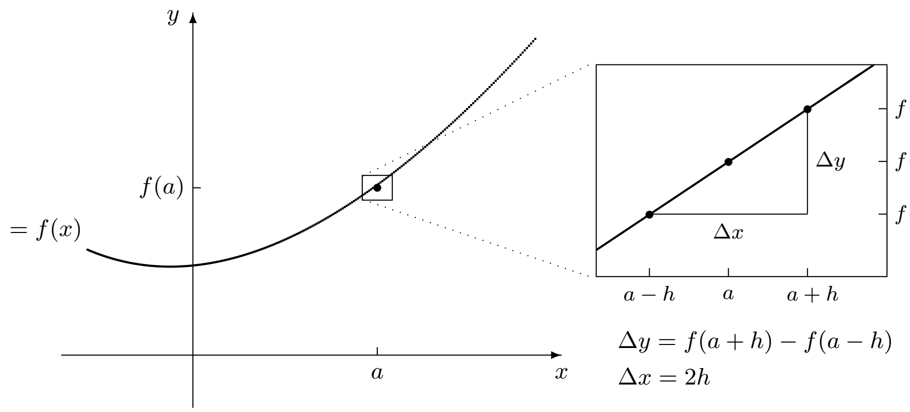


Figure 3.3.2

Estimating the value of the derivative $f'(a)$

Choosing points in a window. To estimate the slope in the window above, we chose two particular points, $(a-h, f(a-h))$ and $(a+h, f(a+h))$. However, *any* two points in the window would give us a valid estimate. Our choice depends on the situation. For example, if we are working with formulas, we want simple expressions. In that case we would probably replace $(a - h, f(a-h))$ by $(a, f(a))$. We do that in the window on the left. The resulting slope is

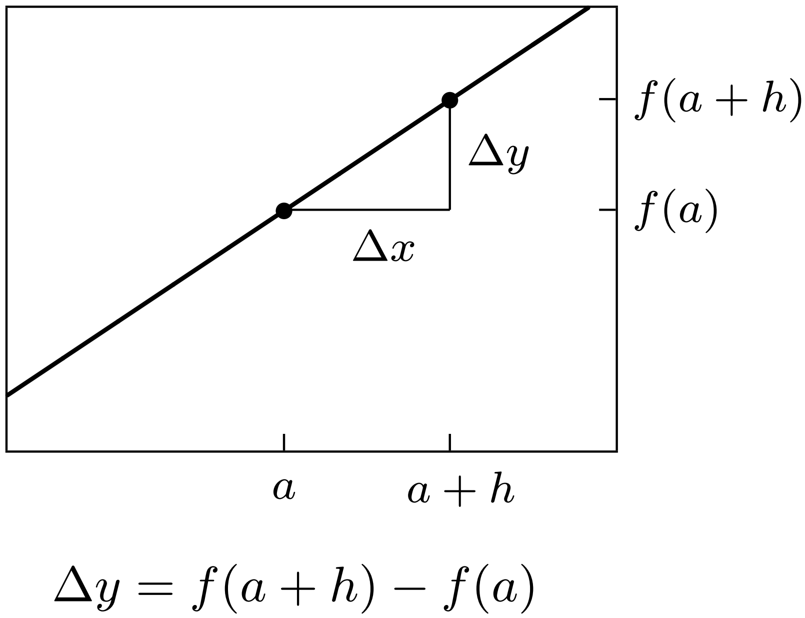


Figure 3.3.3

$$\frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a)}{h}.$$

While the limiting value of $\Delta y/\Delta x$ doesn't depend on the choices you make, the *estimates* you produce with a fixed Δx can be closer to or farther from the true value. The exercises will explore this.

Data versus formulas. The derivative is a limit. To find that limit we have to be able to zoom in arbitrarily close, to make Δx arbitrarily small. For functions given by data, that is usually impossible; we can't use any Δx smaller than the spacing between the data points.

A data function might not have a derivative

Thus, a data function of this sort does not have a derivative, strictly speaking. However, by zooming in as much as the data allow, we get the most precise description possible for the rate of change of the function. In these circumstances it makes a difference which points we choose in a window to calculate $\Delta y/\Delta x$. In the exercises you will have a chance to see how the precision of your estimate depends on which points you choose to calculate the slope.

For a function given by a formula, it is possible to find the value of the derivative exactly. In fact, the derivative of a function given by a formula is itself given by a formula.

There are rules for finding derivatives

Later in this chapter we will describe some general rules which will allow us to produce the formula for the derivative without going through the successive approximation process each time. In chapter 5 we will discuss these rules more fully.

Practical considerations.. The derivative is a limit, and there are always practical considerations to raise when we discuss limits. As we saw in chapter 2, we cannot expect to construct the entire decimal expansion of a limit. In most cases all we can get are a specified number of digits. For example, in section 2 we found that

$$\text{if } f(x) = \frac{2 + x^3 \cos x + 1.5^x}{2 + x^2}, \text{ then } f'(27) = -.27235\ldots.$$

The same digits without the "…" give us **approximations**. Thus we can write $f'(27) \approx -.27235$; we also have $f'(27) \approx -.2723$ and $f'(27) \approx -.272$. Which approximation is the right one to use depends on the context. For example, if f appears in a problem in which all the other quantities are known only to one or two decimal places, we probably don't need a very precise value for $f'(27)$. In that case we don't have to carry the sequence of slopes $\Delta y/\Delta x$ very far. For instance, if we want to know $f'(27)$ to three decimal places, and so justify writing $f'(27) \approx -.272$, we only need to continue the zooming process until the slopes $\Delta y/\Delta x$ all have values that begin $-.2723\ldots$. By the table on page , $\Delta x = .0004$ is sufficient.

3.3.2 Language and Notation

• If f has a derivative at a, we also say f **is differentiable at** a. If f is differentiable at every point a in its domain, we say f **is differentiable**.

• Do *locally linear* and *differentiable* mean the same thing? The awkward case is a function whose graph is vertical at a point (for example, $y = \sqrt[3]{x}$ at the origin). On the one hand, it makes sense to say that the function is locally linear at such a point, because the graph looks straight under a microscope. On the other hand, the derivative itself is undefined, because the line is vertical. So the function is locally linear, but not differentiable, at that point.

There is another way to view the matter. We can say, instead, that a vertical line *does* have a slope, and its value is *infinity* (∞). From this point of view, if the graph of f is vertical at $x = a$, then $f'(a) = \infty$. In other words, f *does* have a derivative at $x = a$; its value just happens to be ∞.

Which view is "right"? Neither; we can choose either. Our choice is a matter of convention.

Convention: if the graph of f is vertical at a, write $f'(a) = \infty$

(In some countries cars travel on the left; in others, on the right. That's a convention, too.) However, we will follow the second alternative. One advantage is that we will be able to use the derivative to indicate where the graph of a function is vertical. Another is that *locally linear* and *differentiable* then mean exactly the same thing.

• Suppose $y = f(x)$ and the quantities x and y appear in a context in which they have units. Then the derivative of $f'(x)$ *also* has units, because it is the rate of change of y with respect to x. The units for the derivative must be

$$\text{units for } f' = \frac{\text{units for output } y}{\text{units for input } x}.$$

We have already seen several examples—persons per day, miles per hour, milligrams per pound, dollars of profit per dollar change in price—and we will see many more.

• There are several notations for the derivative. You should be aware of them because they are all in common use and because they reflect different ways of viewing the derivative. We have been writing the derivative of $y = f(x)$ as $f'(x)$.

Leibniz's notation

Leibniz wrote it as dy/dx. This notation has several advantages. It resembles the quotient $\Delta y/\Delta x$ that we use to approximate the derivative. Also, because dy/dx looks like a rate, it helps remind us that a derivative is a rate. Later on, when we consider the chain rule to find derivatives, you'll see that it can be stated very vividly using Leibniz's notation.

The German philosopher Gottfried Wilhelm Leibniz (1646-1716) developed calculus about the same time Newton did. While Newton dealt with derivatives in more or less the way we do, Leibniz introduced a related idea which he called a *differential*—'infinitesimally small' numbers which he would write as dx and dy.

The other notation still encountered is due to Newton.

Newton's notation

It occurs primarily in physics and is used to denote rates with respect to time. If a quantity y is changing over time, then the Newton notation expresses the derivative of y as $\dot{y}$ (that's the variable y with a dot over it).

3.3.3 The Microscope Equation

A Context: Driving Time. If you make a 400 mile trip at an average speed of 50 miles per hour, then the trip takes 8 hours. Suppose you increase the average speed by 2 miles per hour. How much time does that cut off the trip?

One way to approach this question is to start with the basic formula

$$\text{speed} \times \text{time} = \text{distance}.$$

The distance is known to be 400 miles,

Travel time depends on speed

and we really want to understand how *time* T depends on *speed* s. We get T as a function of s by rewriting the last equation:

$$T \text{ hours} = \frac{400 \text{ miles}}{s \text{ miles per hour}}.$$

To answer the question, just set $s = 52$ miles per hour in this equation. Then $T = 7.6923$ hours, or about 7 hours, 42 minutes. Thus, compared to the original 8 hours, the higher speed cuts 18 minutes off your driving time.

What happens to the driving time if you increase your speed by 4 miles per hour, or 5, instead of 2? What happens if you go slower, say 2 or 3 miles per hour slower? We could make a fresh start with each of these questions and answer them, one by one, the same way we did the first. But taking the questions one at a time misses the point. What we really want to know is the general pattern:

If I'm travelling at 50 miles per hour,

How does travel time respond to changes in speed?

how much does anygiven increase in speed decrease my travel time?

We already know how T and s are related: $T = 400/s$ hours. This question, however, is about the connection between a *change* in speed of

$$\Delta s = s - 50 \text{ miles per hour}$$

and a *change* in arrival time of

$$\Delta T = T - 8 \text{ hours.}$$

To answer it we should change our point of view slightly. It is not the relation between s and T, but between Δs and ΔT, that we want to understand.

Since we are considering speeds s that are only slightly above or below 50 miles per hour, Δs will be small. Consequently, the arrival time T will be only slightly different from 8 hours, so ΔT will also be small. Thus we want to study small changes in the function $T = 400/s$ near $(s, T) = (50, 8)$. The natural tool to use is a microscope.

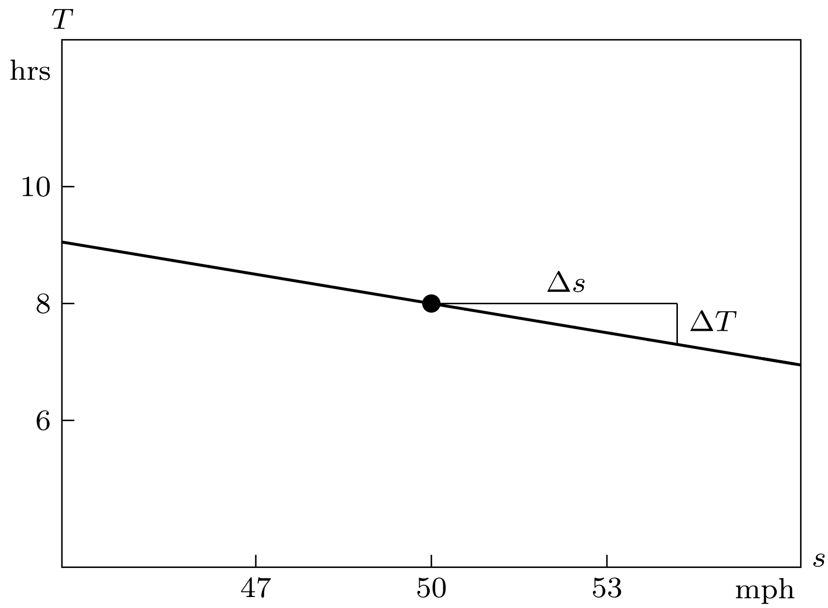


Figure 3.3.4

How travel time changes with speed around 50 miles per hour

In the microscope window above we see the graph of $T = 400/s$, magnified at the point $(s, T) = (50, 8)$. The field of view was chosen so that s can take values about 6 mph above or below 50 mph.

The slope of the graph in the microscope window

The graph looks straight, and its slope is $T'(50)$. In the exercises you are asked to determine the value of $T'(50)$; you should find $T'(50) \approx -.16$. (Later on, when we have rules for finding the derivative of a formula, you will see that $T'(50) = -.16$ *exactly*.) Since the quotient $\Delta T/\Delta s$ is also an estimate for the slope of the line in the window, we can write

$$\frac{\Delta T}{\Delta s} \approx -.16 \text{ hours } \textit{per} \text{ mile per hour.}$$

If we multiply both sides of this approximate equation by Δs miles per hours, we get

$$\Delta T \approx -.16\,\Delta s \text{ hours.}$$

This equation answers our question

How travel time changes with speed

about the general pattern relating changes in travel time to changes in speed. It says that the changes are *proportional*. For every mile per hour increase in speed, travel time decreases by about .16 hours, or about $9\frac{1}{2}$ minutes. Thus, if the speed is 1 mph over 50 mph, travel time is cut by about $9\frac{1}{2}$ minutes. If we double the increase in speed, that doubles the savings in time: if the speed is 2 mph over 50 mph, travel time is cut by about 19 minutes. Compare this with a value of about 18 minutes that we got with the exact equation $T = 400/s$.

Notice that we are using Δs and ΔT

Δs and ΔT now have a special meaning

in a slightly more restricted way than we have previously. Up to now, Δs measured the horizontal distance between *any* two points on a graph. Now, however, Δs just measures the horizontal distance from the fixed point $(s, T) = (50, 8)$ (marked with a large dot) that sits at the center of the window. Likewise, ΔT just measures the vertical distance from this point. The central point therefore plays the role of an origin, and Δs and ΔT are the *coordinates* of a point measured from that origin. To underscore the fact that Δs and ΔT are really coordinates, we have added a Δs-axis and a ΔT-axis in the window below. Notice that these coordinate axes have their own labels and scales.

Every point in the window can therefore

Δs and ΔT are coordinates in the window

be described in two different coordinate systems. The two different sets of coordinates of the point labelled P, for instance, are $(s, T) = (53, 7.52)$ and $(\Delta s, \Delta T) = (3, -.48)$. The first pair says "When your speed is 53 mph, the trip will take 7.52 hrs." The second pair says "When you increase your speed by 3 mph, you will decrease travel time by .48 hrs." Each statement can be translated into the other, but each statement has its own point of reference.

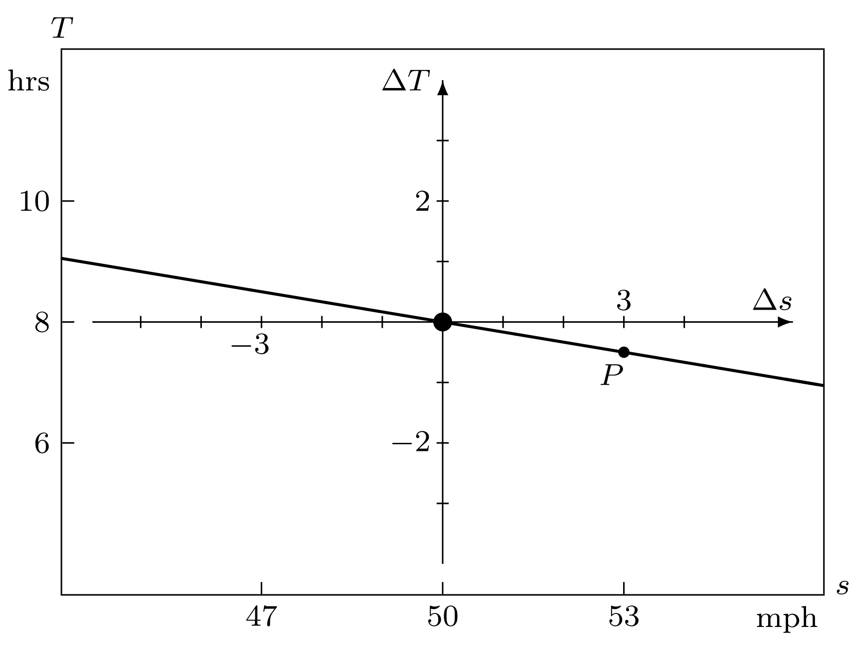


Figure 3.3.5

The microscope equation: $\Delta T \approx -.16\,\Delta s$ hours

We call $\Delta T \approx -.16\,\Delta s$ the **microscope equation** because it tells us how the microscope coordinates Δs and ΔT are related.

In fact, we now have two different ways to describe how travel time is related to speed. They can be compared in the following table.

Table 3.3.6

	global
coordinates:	s,T
equation:	$T = 400/s$

Table 3.3.7 **Table**

	exact
properties:	non-linear

We say the microscope equation is *local*

Global vs. local descriptions

because it is intended to deal only with speeds near 50 miles per hour. There is a different microscope equation for speeds near 40 miles per hour, for instance. By contrast, the original equation is *global*, because it works for all speeds. While the global equation is exact it is also non-linear; this can make it more difficult to compute. The microscope equation is approximate but linear; it is easy to compute. It is also easy to put into words:

At 50 miles per hour, the travel time of a 400 mile journeydecreases $9\frac{1}{2}$ minutes for each mile per hour increase in speed.

The connection between the global equation and the microscope equation is shown in the following illustration.

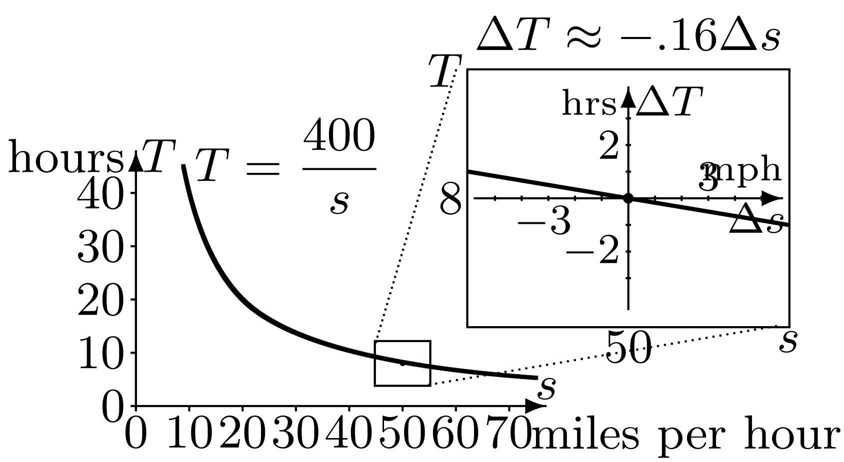


Figure 3.3.9

Local Linearity and Multipliers. The reasoning that led us to a microscope equation for travel time can be applied to any locally linear function. If $y = f(x)$ is locally linear, then at $x = a$ we can write

The microscope equation: $\boldsymbol{\Delta y \approx f'(a) \cdot \Delta x}$

Figure 3.3.10

We know an equation of the form $\Delta y = m \cdot \Delta x$ tells us that y is a linear function of x, in which m plays the role of slope, rate, and multiplier. The microscope equation therefore tells us that y **is a linear function of** x **when** x **is near** a—at least approximately. In this almost-linear relation, the derivative $f'(a)$ plays the role of slope, rate, and multiplier.

The microscope equation

The microscope equation is the analytic form of local linearity

is just the idea of local linearity expressed analytically rather than geometrically—that is, by a formula rather than by a picture. Here is a chart that shows how the two descriptions of local linearity fit together.

$y = f(x)$ **is locally linear at** $x = a$**:**

Table 3.3.11

geometrically
When magnified at $(a, f(a))$,
the graph of f is almost straight,
and the slope of the line is $f'(a)$.

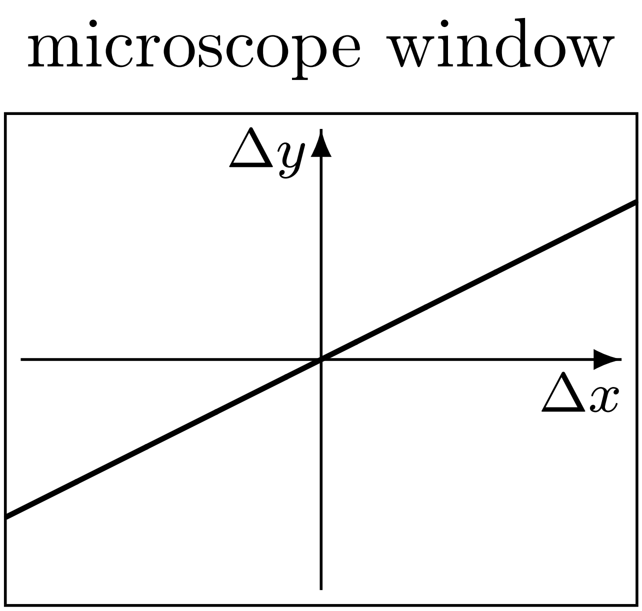


Figure 3.3.12

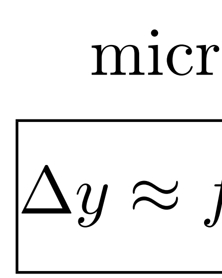


Figure 3.3.13

Of course, the graph in a microscope window is not *quite* straight. The analytic counterpart of this statement is that the microscope equation is not *quite* exact—the two sides of the equation are only approximately equal. We write "≈" instead of "=". However, we can make the graph look even straighter by increasing the magnification—or, what is the same thing, by decreasing the field of view. Analytically, this increases the exactness of the microscope equation. Like a laboratory microscope, our microscope is most accurate at the center of the field of view, with increasing aberration toward the periphery!

In the microscope equation $\Delta y \approx f'(a) \cdot \Delta x$,

the derivative is the multiplier that tells how y responds to changes in x. In particular, a small increase in x produces a change in y that depends on the sign and magnitude of $f'(a)$ in the following way:

- $f'(a)$ is large and positive $\Rightarrow$ large increase in y,
- $f'(a)$ is small and positive $\Rightarrow$ small increase in y,
- $f'(a)$ is large and negative $\Rightarrow$ large decrease in y,
- $f'(a)$ is small and negative $\Rightarrow$ small decrease in y.

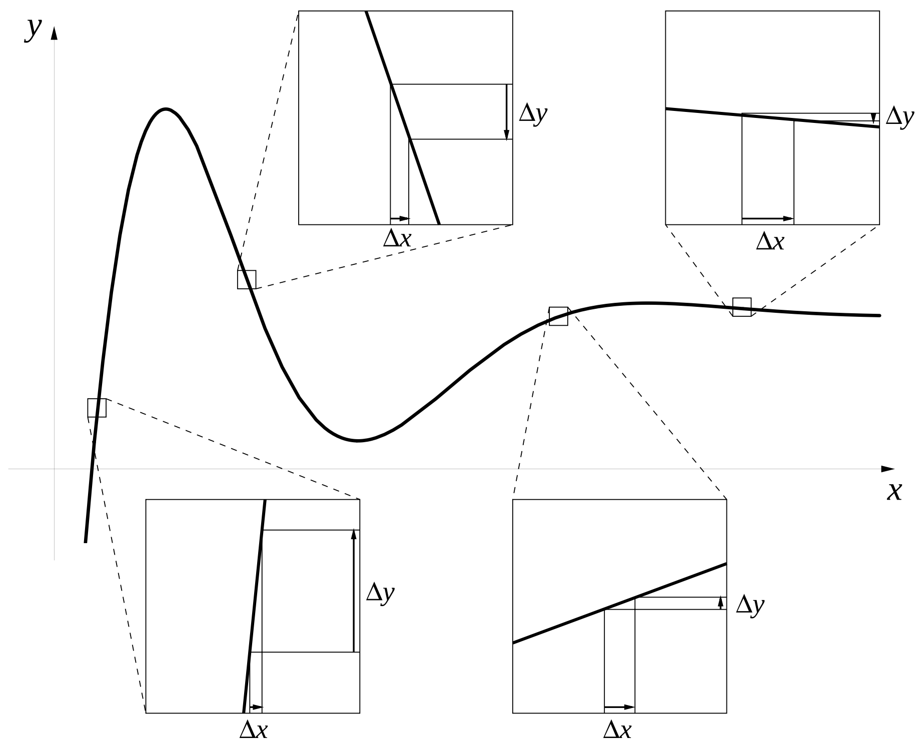


Figure 3.3.14 image

For example, suppose we are told the value of the derivative is 2. Then any small change in x induces a change in y approximately twice as large. If, instead, the derivative is $-1/5$, then a small change in x produces a change in y only one fifth as large, and in the opposite direction. That is, if x increases, then y decreases, and vice-versa.

The microscope equation should look familiar to you.

The microscope equation is the recipe for building solutions to rate equations

It has been with us from the beginning of the course. Our "recipe" $\Delta S = S' \cdot \Delta t$ for predicting future values of S in the S-I-R model is just the microscope equation for the function $S(t)$. (Although we wrote it with an "=" instead of an "≈" in the first chapter, we noted that ΔS would provide us only an *estimate* for the new value of S.) The success of Euler's method in producing solutions to rate equations depends fundamentally on the fact that the functions we are trying to find are locally linear.

The derivative is one of the fundamental concepts of the calculus, and one of its most important roles is in the microscope equation. Besides giving us a tool for building solutions to rate equations, the microscope equation helps us do estimation and error analysis, the subject of the next section.

We conclude with a summary that compares linear and locally linear functions. Note there are two differences, but only two: 1) the equation for local linearity is only an approximation; 2) it holds only locally—i.e., near a given point.

If $y = f(x)$ is linear, then $\Delta y = m \cdot \Delta x$; the constant m is rate, slope, and multiplier for all x.	**If $y = f(x)$ is locally linear, then $\Delta y \approx f'(a) \cdot \Delta x$; the derivative $f'(a)$ is rate, slope, and multiplier for x near a.**

Figure 3.3.15

3.3.4 Exercises

Computing Derivatives.

1. Sketch graphs of the following functions and use these graphs to determine which function has a derivative that is always positive (except at $x = 0$, where neither the function nor its derivative is defined).

$$y = \frac{1}{x} \quad y = \frac{-1}{x} \quad y = \frac{1}{x^2} \quad y = \frac{-1}{x^2}$$

What feature of the graph told you whether the derivative was positive?

2. For each of the functions f below, approximate its derivative at the given value $x = a$ in two different ways. First, use a computer microscope (i.e., a graphing program) to view the graph of f near $x = a$. Zoom in until the graph looks straight and find its slope. Second, use a calculator to find the value of the quotient

$$\frac{f(a+h) - f(a-h)}{2h}$$

for $h = .1, .01, .001, \ldots, .000001$. Based on these values of the quotients, give your best estimate for $f'(a)$, and say how many decimal places of accuracy it has.

(a) $f(x) = 1/x$ at $x = 2$.

(b) $f(x) = \sin(7x)$ at $x = 3$.

(c) $f(x) = x^3$ at $x = 200$.

(d) $f(x) = 2^x$ at $x = 5$.

3. In a later section we will establish that the derivative of $f(x) = x^3$ at $x = 1$ is exactly 3: $f'(1) = 3$. This question concerns the freedom we have to choose points in a window to estimate $f'(1)$ (see page). Its purpose is to compare two quotients, to see which gets closer to the exact value of $f'(1)$ for a fixed “field of view” Δx. The two quotients are

$$Q_1 = \frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a-h)}{2h} \quad \text{and} \quad Q_2 = \frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a)}{h}.$$

In this problem $a = 1$.

(a) Construct a table that shows the values of Q_1 and Q_2 for each $h = 1/2^k$, where $k = 0, 1, 2, \ldots, 8$. If you wish, you can use this program to compute the values:

```
a = 1
```

(b) How many digits of Q_1 stabilize in this table? How many digits of Q_2?

(c) Which is a better estimator—Q_1 or Q_2? To indicate how much better, give the value of h for which the *better* estimator provides an estimate that is as close as the best estimate provided by the *poorer* estimator.

4. Repeat all the steps of the last question for the function $f(x) = \sqrt{x}$ at $x = 9$. The exact value of $f'(9)$ is $1/6$.

Comment:. Note that, in section 1, we estimated the rate of change of the sunrise function using an expression like Q_1 rather than one like Q_2. The previous exercises should persuade you this was deliberate. We were trying to get the most representative estimates, given the fact that we could not reduce the size of Δx arbitrarily.

5. At this point you will find it convenient to write a more general derivative-finding program. You can modify the program in problem 3. to do this by having a `DEF` command at the beginning of the program to specify the function you are currently interested in. For instance, if you insert the command `DEF fnf (x) = x ^ 3` at the beginning of the program, how could you simplify the lines specifying `q1` and `q2`? If you then wanted to calculate the derivative of another function at a different x-value, you would only need to change the `DEF` specification and, depending on the point you were interested in, the `a = 1` line.

6. Use one of the methods of problem to estimate the value of the derivative of each of the following functions at $x = 0$:

$$y = 2^x, \quad y = 3^x, \quad y = 10^x, \quad \text{and} \quad y = (1/2)^x.$$

These are called **exponential** functions, because the input variable x appears in the exponent. How many decimal places accuracy do your approximations to the derivatives have?

7. In this problem we look again at the exponential function $f(x) = 2^x$ from the previous problem.

(a) Use the rules for exponents to put the quotient

$$\frac{f(a+h) - f(a)}{h}$$

in the simplest form you can.

(b) We know that

$$f'(0) = \lim_{h \to 0} \frac{f(h) - f(0)}{h}.$$

Use this fact, along with the algebraic result of part (a), to explain why $f'(a) = f'(0) \cdot 2^a$.

8. Apply all the steps of the previous question to the exponential function $f(x) = b^x$ with an arbitrary base b. Show that $f'(x) = f'(0) \cdot b^x$.

9. For which values of x is the absolute value function $y = |x|$ differentiable?

(a) At each point where $y = |x|$ is differentiable, find the value of the derivative.

The microscope equation.

10. Write the microscope equation for each of the following functions at the indicated point. (To find the necessary derivative, consult problem .)

(a) $f(x) = 1/x$ at $x = 2$.

(b) $f(x) = \sin(7x)$ at $x = 3$.

(c) $f(x) = x^3$ at $x = 200$.

(d) $f(x) = 2^x$ at $x = 5$.

11. This question uses the microscope equation for $f(x) = 1/x$ at $x = 2$ that you constructed in the previous question.

(a) Draw the graph of what you would see in the microscope if the field of view is .2 units wide.

(b) If we take $x = 2.05$, what is Δx in the microscope equation? What estimate does the microscope equation give for Δy? What estimate does the microscope equation then give for $f(2.05) = 1/2.05$? Calculate the true value of $1/2.05$ and compare the two values; how far is the microscope estimate from the true value?

(c) What estimate does the microscope equation give for $1/2.02$? How far is this from the true value?

(d) What estimate does the microscope equation give for $1/1.995$? How far is this from the true value?

12. This question concerns the travel time function $T = 400/s$ hours, discussed in the text.

(a) How many hours does a 400-mile trip take at an average speed of 40 miles per hour?